UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☑ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2020

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from _____ to _____

Commission file number 001-14905

BERKSHIRE HATHAWAY INC.

(Exact name of Registrant as specified in its charter)

Delaware	47-0813844
State or other jurisdiction of incorporation or organization	(I.R.S. Employer Identification No.)
3555 Farnam Street, Omaha, Nebraska	68131
(Address of principal executive office)	(Zip Code)

Registrant's telephone number, including area code (402) 346-1400

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbols	Name of each exchange on which registered
Class A Common Stock	BRK.A	New York Stock Exchange
Class B Common Stock	BRK.B	New York Stock Exchange
0.750% Senior Notes due 2023	BRK23	New York Stock Exchange
1.125% Senior Notes due 2027	BRK27	New York Stock Exchange
1.625% Senior Notes due 2035	BRK35	New York Stock Exchange
1.300% Senior Notes due 2024	BRK24	New York Stock Exchange
2.150% Senior Notes due 2028	BRK28	New York Stock Exchange
0.625% Senior Notes due 2023	BRK23A	New York Stock Exchange
0.000% Senior Notes due 2025	BRK25	New York Stock Exchange
2.375% Senior Notes due 2039	BRK39	New York Stock Exchange
0.500% Senior Notes due 2041	BRK41	New York Stock Exchange
2.625% Senior Notes due 2059	BRK59	New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act: NONE

Indicate by check mark if the Registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☑ No ☐

Indicate by check mark if the Registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☑

Indicate by check mark whether the Registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☑ No ☐

Indicate by check mark whether the Registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☑ No ☐

Indicate by check mark whether the Registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act. Large accelerated filer ☑ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company ☐

If an emerging growth company, indicate by check mark if the Registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☑

Indicate by check mark whether the Registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☑

State the aggregate market value of the voting stock held by non-affiliates of the Registrant as of June 30, 2020: $336,500,000,000*

Indicate the number of shares outstanding of each of the Registrant's classes of common stock:

February 16, 2021—Class A common stock, $5 par value	640,586 shares
February 16, 2021—Class B common stock, $0.0033 par value	1,336,348,609 shares

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the Proxy Statement for the Registrant's Annual Meeting to be held May 1, 2021 are incorporated in Part III.

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* This aggregate value is computed at the last sale price of the common stock as reported on the New York Stock Exchange on June 30, 2020. It does not include the value of Class A common stock and Class B common stock held by Directors and Executive Officers of the Registrant and members of their immediate families, some of whom may not constitute "affiliates" for purpose of the Securities Exchange Act of 1934.

Table of Contents

		Page No.
	Part I	
Item 1.	**Business Description**	K-1
Item 1A.	**Risk Factors**	K-22
Item 1B.	**Unresolved Staff Comments**	K-26
Item 2.	**Description of Properties**	K-26
Item 3.	**Legal Proceedings**	K-28
Item 4.	**Mine Safety Disclosures**	K-29
	Part II	
Item 5.	**Market for Registrant's Common Equity, Related Security Holder Matters and Issuer Purchases of Equity Securities**	K-29
Item 6.	**Selected Financial Data**	K-32
Item 7.	**Management's Discussion and Analysis of Financial Condition and Results of Operations**	K-33
Item 7A.	**Quantitative and Qualitative Disclosures About Market Risk**	K-66
Item 8.	**Financial Statements and Supplementary Data**	K-67
	Consolidated Balance Sheets— December 31, 2020 and December 31, 2019	K-70
	Consolidated Statements of Earnings— Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-72
	Consolidated Statements of Comprehensive Income— Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-73
	Consolidated Statements of Changes in Shareholders' Equity— Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-73
	Consolidated Statements of Cash Flows— Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-74
	Notes to Consolidated Financial Statements	K-75
Item 9.	**Changes in and Disagreements with Accountants on Accounting and Financial Disclosure**	K-116
Item 9A.	**Controls and Procedures**	K-116
Item 9B.	**Other Information**	K-116
	Part III	
Item 10.	**Directors, Executive Officers and Corporate Governance**	K-116
Item 11.	**Executive Compensation**	K-116
Item 12.	**Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters**	K-116
Item 13.	**Certain Relationships and Related Transactions and Director Independence**	K-116
Item 14.	**Principal Accountant Fees and Services**	K-116
	Part IV	
Item 15.	**Exhibits and Financial Statement Schedules**	K-116
	Exhibit Index	K-120
	Signatures	K-122

Part I

Item 1. Business Description

Berkshire Hathaway Inc. ("Berkshire," "Company" or "Registrant") is a holding company owning subsidiaries engaged in a large number of diverse business activities. The most important of these are insurance businesses conducted on both a primary basis and a reinsurance basis, a freight rail transportation business and a group of utility and energy generation and distribution businesses. Berkshire also owns and operates numerous other businesses engaged in a variety of activities, as identified herein. Berkshire is domiciled in the state of Delaware, and its corporate headquarters is in Omaha, Nebraska.

Berkshire's operating businesses are managed on an unusually decentralized basis. There are few centralized or integrated business functions. Berkshire's corporate senior management team participates in and is ultimately responsible for significant capital allocation decisions, investment activities and the selection of the Chief Executive to head each of the operating businesses.

Berkshire's senior management is also responsible for establishing and monitoring Berkshire's corporate governance practices, including monitoring governance efforts, including those at the operating businesses, and participating in the resolution of governance-related issues as needed. Berkshire's Board of Directors is responsible for assuring an appropriate successor to the Chief Executive Officer. The Berkshire Code of Business Conduct and Ethics emphasizes, among other things, the commitment to ethics and compliance with the law and provides basic standards for ethical and legal behavior of its employees.

Berkshire and its consolidated subsidiaries employed approximately 360,000 people worldwide at the end of 2020. Human capital and resources are an integral and essential component of Berkshire's businesses. Consistent with Berkshire's decentralized management philosophy, Berkshire's operating businesses establish specific policies and practices for their businesses concerning the attraction and retention of personnel within the organizations. Such policies and practices generally address, among other things: maintaining a safe work environment for employees, customers and other business partners, offering competitive compensation to employees, including health insurance and retirement benefits and incentives, providing learning and career development opportunities, and hiring practices intended to identify qualified candidates and promote diversity and inclusion in the workforce.

Insurance and Reinsurance Businesses

Berkshire's insurance and reinsurance business activities are conducted through numerous domestic and foreign-based insurance entities. Berkshire's insurance businesses provide insurance and reinsurance of property and casualty and life, accident and health risks worldwide. Berkshire's insurance businesses employed approximately 51,000 people at the end of 2020.

In direct or primary insurance activities, the insurer assumes the risk of loss from persons or organizations that are directly subject to the risks. Such risks may relate to property, casualty (or liability), life, accident, health, financial or other perils that may arise from an insurable event. In reinsurance activities, the reinsurer assumes defined portions of risks that other direct insurers or reinsurers assumed in their own insuring activities.

Reinsurance contracts are normally classified as treaty or facultative contracts. Treaty reinsurance refers to reinsurance coverage for all or a portion of a specified group or class of risks ceded by the direct insurer, while facultative reinsurance involves coverage of specific individual underlying risks. Reinsurance contracts are further classified as quota-share or excess. Under quota-share (proportional or pro-rata) reinsurance, the reinsurer shares proportionally in the original premiums and losses of the direct insurer or reinsurer. Excess (or non-proportional) reinsurance provides for the indemnification of the direct insurer or reinsurer for all or a portion of the loss in excess of an agreed upon amount or "retention." Both quota-share and excess reinsurance contracts may provide for aggregate limits of indemnification.

Insurance and reinsurance are generally subject to regulatory oversight throughout the world. Except for regulatory considerations, there are virtually no barriers to entry into the insurance and reinsurance industry. Competitors may be domestic or foreign, as well as licensed or unlicensed. The number of competitors within the industry is not known. Insurers and reinsurers compete on the basis of reliability, financial strength and stability, financial ratings, underwriting consistency, service, business ethics, price, performance, capacity, policy terms and coverage conditions.

Insurers based in the United States ("U.S.") are subject to regulation by their states of domicile and by those states in which they are licensed to write policies on an admitted basis. The primary focus of regulation is to assure that insurers are financially solvent and that policyholder interests are otherwise protected. States establish minimum capital levels for insurance companies and establish guidelines for permissible business and investment activities. States have the authority to suspend or revoke a company's authority to do business as conditions warrant. States regulate the payment of dividends by insurance companies to their shareholders and other transactions with affiliates. Dividends, capital distributions and other transactions of extraordinary amounts are subject to prior regulatory approval.

Insurers may market, sell and service insurance policies in the states where they are licensed. These insurers are referred to as admitted insurers. Admitted insurers are generally required to obtain regulatory approval of their policy forms and premium rates. Non-admitted insurance markets have developed to provide insurance that is otherwise unavailable through admitted insurers. Non-admitted insurance, often referred to as "excess and surplus" lines, is procured by either state-licensed surplus lines brokers who place risks with insurers not licensed in that state or by the insured party's direct procurement from non-admitted insurers. Non-admitted insurance is subject to considerably less regulation with respect to policy rates and forms. Reinsurers are normally not required to obtain regulatory approval of premium rates or reinsurance contracts.

The insurance regulators of every state participate in the National Association of Insurance Commissioners ("NAIC"). The NAIC adopts forms, instructions and accounting procedures for use by U.S. insurers and reinsurers in preparing and filing annual statutory financial statements. However, an insurer's state of domicile has ultimate authority over these matters. In addition to its activities relating to the annual statement, the NAIC develops or adopts statutory accounting principles, model laws, regulations and programs for use by its members. Such matters deal with regulatory oversight of solvency, risk management, compliance with financial regulation standards and risk-based capital reporting requirements.

U.S. states, through the NAIC, and international insurance regulators through the International Association of Insurance Supervisors ("IAIS") have been developing standards and best practices focused on establishing a common set of principles ("Insurance Core Principles") and framework ("ComFrame") for the regulation of large multi-national insurance groups. The standards address a variety of topics regarding supervision, coordination of regulators, insurance capital standards, risk management and governance. While the IAIS standards do not have legal effect, the states and the NAIC are implementing various regulatory tools and mandates that are responsive to certain IAIS standards. For example, the U.S. state regulators now require insurance groups to file an annual report, called an Own Risk Solvency Assessment or ORSA, with the group's lead regulator. U.S. state regulators formed supervisory colleges intended to promote communication and cooperation amongst the various domestic international insurance regulators. The Nebraska Department of Insurance acts as the lead group wide supervisor for our group of insurance companies and chairs the Berkshire supervisory college. The NAIC is also developing further tools, including a group capital calculation tool and various liquidity assessments, that could be imposed on insurance groups in the future.

Berkshire's insurance companies maintain capital strength at exceptionally high levels, which differentiates them from their competitors. Collectively, the combined statutory surplus of Berkshire's U.S.-based insurers was approximately $237 billion at December 31, 2020. Berkshire's major insurance subsidiaries are rated AA+ by Standard & Poor's and A++ (superior) by A.M. Best with respect to their financial condition and claims paying ability.

The Terrorism Risk Insurance Act of 2002 established within the Department of the Treasury a Terrorism Insurance Program ("Program") for commercial property and casualty insurers by providing federal reinsurance of insured terrorism losses. The Program currently extends to December 31, 2027 through other Acts, most recently the Terrorism Risk Insurance Program Reauthorization Act of 2019 (the "2019 TRIA Reauthorization"). Hereinafter these Acts are collectively referred to as TRIA. Under TRIA, the Department of the Treasury is charged with certifying "acts of terrorism." Coverage under TRIA occurs if the industry insured loss for certified events occurring during the calendar year exceeds $200 million in 2020, or any calendar year thereafter.

To be eligible for federal reinsurance, insurers must make available insurance coverage for acts of terrorism, by providing policyholders with clear and conspicuous notice of the amount of premium that will be charged for this coverage and of the federal share of any insured losses resulting from any act of terrorism. Assumed reinsurance is specifically excluded from TRIA participation. TRIA currently also excludes certain forms of direct insurance (such as personal and commercial auto, burglary, theft, surety and certain professional liability lines). Reinsurers are not required to offer terrorism coverage and are not eligible for federal reinsurance of terrorism losses.

During 2020 and thereafter, in the event of a certified act of terrorism, the federal government will reimburse insurers (conditioned on their satisfaction of policyholder notification requirements) for 80% of their insured losses in excess of an insurance group's deductible. Under the Program, the deductible is 20% of the aggregate direct subject earned premium for relevant commercial lines of business in the immediately preceding calendar year. The aggregate deductible in 2021 for Berkshire's insurance group is expected to approximate $1.4 billion. There is also an aggregate program limit of $100 billion on the amount of the federal government coverage for each TRIA year.

The extent of insurance regulation varies significantly among the countries in which our non-U.S. operations conduct business. While each country imposes licensing, solvency, auditing, and financial reporting requirements, the type and extent of the requirements differ substantially. For example:

- in some countries, insurers are required to prepare and file monthly and/or quarterly financial reports, and in others, only annual reports;
- some regulators require intermediaries to be involved in the sale of insurance products, whereas other regulators permit direct sales contact between the insurer and the customer;
- the extent of restrictions imposed upon an insurer's use of local and offshore reinsurance vary;
- policy form filing and rate regulation vary by country;
- the frequency of contact and periodic on-site examinations by insurance authorities differ by country;
- the scope and prescriptive requirements of an insurer's risk management and governance framework vary significantly by country; and
- regulatory requirements relating to insurer dividend policies vary by country.

Significant variations can also be found in the size, structure, and resources of the local regulatory departments that oversee insurance activities. Certain regulators prefer close relationships with all subject insurers and others operate a risk-based approach.

Berkshire's insurance group operates in some countries through subsidiaries and in some countries through branches of subsidiaries. Berkshire insurance subsidiaries are located in several countries, including Germany, the United Kingdom ("UK"), Ireland, Australia and South Africa, and also maintain branches in other countries, including Canada, various members of the European Union ("EU"), Australia, New Zealand, Singapore, Hong Kong, Macau and Dubai. Most of these foreign jurisdictions impose local capital requirements. Other legal requirements include discretionary licensing procedures, local retention of funds and records, and data privacy and protection program requirements. Berkshire's international insurance companies are also subject to multinational application of certain U.S. laws.

There are various regulatory bodies and initiatives that impact Berkshire in multiple international jurisdictions and the potential for significant effect on the Berkshire insurance group could be heightened as a result of recent industry and economic developments.

On June 23, 2016, the UK voted in a national referendum to withdraw from the EU ("Brexit"), which resulted in the UK's withdrawal from the EU on January 31, 2020. In anticipation of the UK leaving the EU, Berkshire Hathaway European Insurance DAC in Ireland was established to permit property and casualty insurance and reinsurance businesses to continue to operate in the EU following Brexit. Following the withdrawal of the UK from the EU as a result of Brexit, Berkshire expects to continue to maintain a substantial presence in London.

Berkshire's insurance underwriting operations include the following groups: (1) GEICO, (2) Berkshire Hathaway Primary Group and (3) Berkshire Hathaway Reinsurance Group. Except for retroactive reinsurance and periodic payment annuity products that generate significant amounts of up-front premiums along with estimated claims expected to be paid over very long time periods (creating "float," see Investments section below), Berkshire expects to achieve a net underwriting profit over time and to reject inadequately priced risks. Underwriting profit is defined as earned premiums less associated incurred losses, loss adjustment expenses and underwriting and policy acquisition expenses. Underwriting profit does not include income earned from investments. Additional information related to each of Berkshire's underwriting groups follows.

GEICO—GEICO is headquartered in Chevy Chase, Maryland. GEICO's insurance subsidiaries consist of Government Employees Insurance Company, GEICO General Insurance Company, GEICO Indemnity Company, GEICO Casualty Company, GEICO Advantage Insurance Company, GEICO Choice Insurance Company, GEICO Secure Insurance Company, GEICO County Mutual Insurance Company and GEICO Marine Insurance Company. The GEICO companies primarily offer private passenger automobile insurance to individuals in all 50 states and the District of Columbia. GEICO also provides insurance for motorcycles, all-terrain vehicles, recreational vehicles, boats and small commercial fleets and acts as an agent for other insurers who offer homeowners, renters, life and identity management insurance to individuals who desire insurance coverages other than those offered by GEICO.

GEICO's marketing is primarily through direct response methods in which applications for insurance are submitted directly to the companies via the Internet or by telephone, and to a lesser extent, through captive agents. GEICO conducts business through regional service centers and claims adjustment and other facilities in 39 states.

The automobile insurance business is highly competitive in the areas of price and service. GEICO competes for private passenger automobile insurance customers in the preferred, standard and non-standard risk markets with other companies that sell directly to the customer as well as with companies that use agency sales forces, including State Farm, Allstate, Progressive and USAA. GEICO's advertising campaigns and competitive rates contributed to a cumulative increase in voluntary policies-in-force of approximately 36% over the past five years. According to the most recently published A.M. Best data for 2019, the five largest automobile insurers had a combined market share in 2019 of approximately 58% based on written premiums, with GEICO's market share being second largest at approximately 13.8%. Since that data was published, GEICO's management estimates its current market share may have declined, depending on how the effects of pandemic-related premium credit programs will be reflected in A.M. Best's measurements. Seasonal variations in GEICO's insurance business are not significant. However, extraordinary weather conditions or other factors may have a significant effect upon the frequency or severity of automobile claims.

State insurance departments stringently regulate private passenger auto insurance. As a result, it is difficult for insurance companies to differentiate their products. Competition for private passenger automobile insurance, which is substantial, tends to focus on price and level of customer service provided. GEICO's cost-efficient direct response marketing methods and emphasis on customer satisfaction enable it to offer competitive rates and value to its customers. GEICO primarily uses its own claims staff to manage and settle claims. The name and reputation of GEICO are material assets and management protects it and other service marks through appropriate registrations.

Berkshire Hathaway Primary Group—The Berkshire Hathaway Primary Group ("BH Primary") is a collection of independently managed insurers that provide a wide variety of insurance coverages to policyholders located principally in the United States. These various operations are discussed below.

NICO and certain affiliates ("NICO Primary") underwrite commercial motor vehicle and general liability insurance on an admitted basis and on an excess and surplus basis. Insurance coverages are offered nationwide primarily through insurance agents and brokers.

The Berkshire Hathaway Homestate Companies ("BHHC") is a group of insurers offering workers' compensation, commercial auto and commercial property coverages to a diverse client base. BHHC has a national reach, with the ability to provide first-dollar and small to large deductible workers' compensation coverage to employers in all states, except those where coverage is available only through state-operated workers' compensation funds. NICO Primary and BHHC are each based in Omaha, Nebraska.

Berkshire Hathaway Specialty Insurance ("BH Specialty") offers commercial property, casualty, healthcare professional liability, executive and professional, surety, travel, medical stop loss and homeowner's insurance through Berkshire Hathaway Specialty Insurance Company and other Berkshire insurance affiliates. BH Specialty writes primary and excess policies on an admitted and surplus basis in the U.S., and on a local or foreign non-admitted basis outside the U.S. BH Specialty is based in Boston, Massachusetts, with regional offices currently in several U.S. cities. BH Specialty also maintains international offices located in Australia, New Zealand, Canada and several countries in Asia, Europe and the Middle East. BH Specialty writes business through wholesale and retail insurance brokers, as well as managing general agents.

MedPro Group ("MedPro") is a leading provider of healthcare liability ("HCL") insurance in the United States. MedPro provides customized HCL insurance, claims, patient safety and risk solutions to physicians, surgeons, dentists and other healthcare professionals, as well as hospitals, senior care and other healthcare facilities. Additionally, MedPro provides HCL insurance solutions to the international markets through other Berkshire insurance affiliates, delivers liability insurance to other professionals, and offers specialized accident and health insurance solutions to colleges and other customers through its subsidiaries and other Berkshire affiliates. MedPro is based in Fort Wayne, Indiana.

U.S. Liability Insurance Company ("USLI") includes a group of five specialty insurers that underwrite commercial, professional and personal lines insurance on an admitted basis, as well as an excess and surplus basis. USLI markets policies in all 50 states and the District of Columbia and Canada through wholesale and retail insurance agents. USLI companies also underwrite and market a wide variety of specialty insurance products. USLI is based in Wayne, Pennsylvania.

The Berkshire Hathaway GUARD Insurance Companies ("GUARD") is a group of five insurance companies that provide workers' compensation, business owners', commercial auto, commercial package and homeowners' products to over 350,000 small and mid-sized businesses. GUARD also offers complementary professional liability and umbrella products nationwide. Policies are offered through independent agents and retail and wholesale brokers. GUARD is based in Wilkes-Barre, Pennsylvania. Central States Indemnity Company of Omaha, based in Omaha, Nebraska, primarily writes credit card credit insurance, Medicare Supplement insurance and agricultural equipment insurance.

On October 1, 2018, NICO acquired MLMIC Insurance Company ("MLMIC"). MLMIC has been the leading writer of medical professional liability insurance in New York State for over 40 years. MLMIC distributes its policies mostly on a

direct basis to medical and dental professionals, health care providers and hospitals. In October 2019, Berkshire sold its 81% interest in Applied Underwriters, Inc.

Berkshire Hathaway Reinsurance Group—Berkshire's combined global reinsurance business, referred to as the Berkshire Hathaway Reinsurance Group ("BHRG"), offers a wide range of coverages on property, casualty, life and health risks to insurers and reinsurers worldwide. Reinsurance business is written through National Indemnity Company ("NICO"), domiciled in Nebraska, its subsidiaries and various other insurance subsidiaries wholly owned by Berkshire (collectively, the "NICO Group") and General Re Corporation, domiciled in Delaware, and its subsidiaries (collectively the "General Re Group"). BHRG's underwriting operations in the U.S. are based in Stamford, Connecticut. BHRG also conducts business activities globally in 23 countries.

The type and volume of business written is dependent on market conditions, including prevailing premium rates and coverage terms. The level of underwriting activities often fluctuates significantly from year to year depending on the perceived level of price adequacy in specific insurance and reinsurance markets as well as from the timing of particularly large reinsurance transactions.

Property/casualty

The NICO Group offers traditional property/casualty reinsurance on both an excess-of-loss and a quota-share basis, catastrophe excess-of-loss treaty and facultative reinsurance, and primary insurance on an excess-of-loss basis for large or unusual risks for clients worldwide.

The type and volume of business written by the NICO Group may vary significantly from period to period resulting from changes in perceived premium rate adequacy and from unique or large transactions. A significant portion of NICO Group's annual reinsurance premium volume currently derives from a 20% quota-share agreement with Insurance Australia Group Limited ("IAG") that expires July 1, 2025. IAG is a multi-line insurer in Australia, New Zealand and other Asia-Pacific countries. The General Re Group conducts a global property and casualty reinsurance business. Reinsurance contracts are written on both a quota-share and excess basis for multiple lines of business. Contracts are primarily in the form of treaties, and to a lesser degree, on a facultative basis.

General Re Group conducts business in North America primarily through General Reinsurance Corporation ("GRC"), which is licensed in the District of Columbia and all states, except Hawaii, where it is an accredited reinsurer. GRC conducts operations in North America from its headquarters in Stamford, Connecticut and through 13 branch offices in the U.S. and Canada.

In North America, the General Re Group includes General Star National Insurance Company, General Star Indemnity Company and Genesis Insurance Company, which offer a broad array of specialty and surplus lines and property, casualty and professional liability coverages. Such business is marketed through a select group of wholesale brokers, managing general underwriters and program administrators, and offer solutions for the unique needs of public entity, commercial and captive customers.

General Re Group's international reinsurance business is conducted on a direct basis through General Reinsurance AG ("GRAG"), based in Cologne, Germany, and through several other subsidiaries and branches in 22 countries. International business is also written through brokers, including Faraday Underwriting Limited ("Faraday"), a wholly-owned subsidiary. Faraday owns the managing agent of Syndicate 435 at Lloyd's of London and provides capacity and participates in 100% of the results of Syndicate 435.

Life/health

The General Re Group also conducts a global life and health reinsurance business. In the U.S. and internationally, the General Re Group writes life, disability, supplemental health, critical illness and long-term care coverages. The life/health business is marketed on a direct basis. Approximately 35% of the aggregate life/health net premiums written by the General Re Group were in the Asia Pacific compared to 26% in the United States, 22% in Western Europe and 17% throughout the rest of the world.

Berkshire Hathaway Life Insurance Company of Nebraska ("BHLN"), a subsidiary of NICO, and its affiliates write reinsurance covering various forms of traditional life insurance exposures and, on a limited basis, health insurance exposures. BHLN and its affiliates have also periodically reinsured certain guaranteed minimum death, income, and similar benefit risks on closed-blocks of variable annuity reinsurance contracts.

Retroactive reinsurance

NICO also periodically writes retroactive reinsurance contracts. Retroactive reinsurance contracts indemnify ceding companies against the adverse development of claims arising from loss events that have already occurred under property and casualty policies issued in prior years. Coverages under such contracts are provided on an excess basis (above a stated retention) or for losses payable after the inception of the contract with no additional ceding company retention. Contracts are normally subject to aggregate limits of indemnification, which can be exceptionally large in amount. For instance, an excess contract written in January 2017 provides indemnification for 80% of up to $25 billion in excess of $25 billion retained by the ceding company. Significant amounts of asbestos, environmental and latent injury claims may arise under these contracts.

The concept of time-value-of-money is an important element in establishing retroactive reinsurance contract prices and terms since loss payments may occur over decades. Normally, expected ultimate losses payable under these policies are expected to exceed premiums, thus producing underwriting losses. Nevertheless, this business is written, in part, because of the large amounts of policyholder funds generated for investment, the economic benefit of which will be reflected through investment results in future periods.

Periodic payment annuity

BHLN writes periodic payment annuity insurance policies and reinsures existing annuity-like obligations. Under these policies, BHLN receives upfront premiums and agrees in the future to make periodic payments that often extend for decades. These policies, generally relate to the settlement of underlying personal injury or workers' compensation cases of other insurers, known as structured settlements. Consistent with retroactive reinsurance contracts, time-value-of-money concepts are an important factor in establishing annuity premiums and underwriting losses are expected from the periodic accretion of time-value discounted liabilities.

Investments of insurance businesses—Berkshire's insurance subsidiaries hold significant levels of invested assets. Investment portfolios are primarily managed by Berkshire's Chief Executive Officer. Investments include a very large portfolio of publicly traded equity securities, which are concentrated in relatively few issuers, as well as fixed maturity securities and cash and short-term investments. Generally, there are no targeted allocations by investment type or attempts to match investment asset and insurance liability durations. However, investment portfolios have historically included a much greater proportion of equity securities than is customary in the insurance industry.

Invested assets derive from shareholder capital as well as funds provided from policyholders through insurance and reinsurance business ("float"). Float is the approximate amount of net policyholder funds generated through underwriting activities that is available for investment. The major components of float are unpaid losses and loss adjustment expenses, life, annuity and health benefit liabilities, unearned premiums and other policyholder liabilities less premium and reinsurance receivables, deferred policy acquisition costs and deferred charges on reinsurance contracts. On a consolidated basis, float has grown from approximately $88 billion at the end of 2015 to approximately $138 billion at the end of 2020. The cost of float can be measured as the net pre-tax underwriting loss as a percentage of average float. In four of the past five years, Berkshire's cost of float was negative, as its insurance businesses produced net underwriting gains.

Railroad Business—Burlington Northern Santa Fe

Burlington Northern Santa Fe, LLC ("BNSF") is based in Fort Worth, Texas, and through BNSF Railway Company ("BNSF Railway") operates one of the largest railroad systems in North America. BNSF Railway had approximately 35,000 employees at the end of 2020. BNSF also operates a relatively smaller third-party logistics services business.

In serving the Midwest, Pacific Northwest, Western, Southwestern and Southeastern regions and ports of the United States, BNSF transports a range of products and commodities derived from manufacturing, agricultural and natural resource industries. Freight revenues are covered by contractual agreements of varying durations or common carrier published prices or company quotations. BNSF's financial performance is influenced by, among other things, general and industry economic conditions at the international, national and regional levels.

BNSF's primary routes, including trackage rights, allow it to access major cities and ports in the western and southern United States as well as parts of Canada and Mexico. In addition to major cities and ports, BNSF Railway efficiently serves many smaller markets by working closely with approximately 200 shortline railroads. BNSF Railway has also entered into marketing agreements with other rail carriers, expanding the marketing reach for each railroad and their customers. For the year ending December 31, 2020, approximately 37% of freight revenues were derived from consumer products, 26% from industrial products, 24% from agricultural products and 13% from coal.

Regulatory Matters

BNSF is subject to federal, state and local laws and regulations generally applicable to its businesses. Rail operations are subject to the regulatory jurisdiction of the Surface Transportation Board ("STB"), the Federal Railroad Administration of the United States Department of Transportation ("DOT"), the Occupational Safety and Health Administration ("OSHA"), as well as other federal and state regulatory agencies and Canadian regulatory agencies for operations in Canada. The STB has jurisdiction over disputes and complaints involving certain rates, routes and services, the sale or abandonment of rail lines, applications for line extensions and construction, and the merger with or acquisition of control of rail common carriers. The outcome of STB proceedings can affect the profitability of BNSF Railway's business.

The DOT and OSHA have jurisdiction under several federal statutes over a number of safety and health aspects of rail operations, including the transportation of hazardous materials. BNSF Railway is required to transport these materials to the extent of its common carrier obligation. State agencies regulate some aspects of rail operations with respect to health and safety in areas not otherwise preempted by federal law.

Environmental Matters

BNSF's rail operations, as well as those of its competitors, are also subject to extensive federal, state and local environmental regulation covering discharges to the ground or waters, air emissions, toxic substances and the generation, handling, storage, transportation and disposal of waste and hazardous materials. Such regulations effectively increase the costs and liabilities associated with rail operations. Environmental risks are also inherent in rail operations, which frequently involve transporting chemicals and other hazardous materials.

Many of BNSF's land holdings are or have been used for industrial or transportation-related purposes or leased to commercial or industrial companies whose activities may have resulted in discharges onto the property. Under federal (in particular, the Comprehensive Environmental Response, Compensation and Liability Act) and state statutes, BNSF may be held jointly and severally liable for cleanup and enforcement costs associated with a particular site without regard to fault or the legality of the original conduct. BNSF may also be subject to claims by third parties for investigation, cleanup, restoration or other environmental costs under environmental statutes or common law with respect to properties they own that have been impacted by BNSF operations.

Competition

The business environment in which BNSF operates is highly competitive. Depending on the specific market, deregulated motor carriers and other railroads, as well as river barges, ships and pipelines, may exert pressure on price and service levels. The presence of advanced, high service truck lines with expedited delivery, subsidized infrastructure and minimal empty mileage continues to affect the market for non-bulk, time-sensitive freight. The potential expansion of longer combination vehicles could further encroach upon markets traditionally served by railroads. In order to remain competitive, BNSF Railway and other railroads seek to develop and implement operating efficiencies to improve productivity.

As railroads streamline, rationalize and otherwise enhance their franchises, competition among rail carriers intensifies. BNSF Railway's primary rail competitor in the Western region of the United States is the Union Pacific Railroad Company. Other Class I railroads and numerous regional railroads and motor carriers also operate in parts of the same territories served by BNSF Railway.

Utilities and Energy Businesses—Berkshire Hathaway Energy

Berkshire currently owns 91.1% of the outstanding common stock of Berkshire Hathaway Energy Company ("BHE"). BHE is a global energy company with subsidiaries that generate, transmit, store, distribute and supply energy. BHE's locally managed businesses are organized as separate operating units. BHE's domestic regulated energy interests are comprised of four regulated utility companies serving approximately 5.2 million retail customers, five interstate natural gas pipeline companies with approximately 21,300 miles of operated pipeline having a design capacity of approximately 21 billion cubic feet of natural gas per day and ownership interests in electricity transmission businesses. BHE's Great Britain electricity distribution subsidiaries serve about 3.9 million electricity end-users and its electricity transmission-only business in Alberta, Canada serves approximately 85% of Alberta's population. BHE's interests also include a diversified portfolio of independent power projects, a liquefied natural gas export, import and storage facility, the largest residential real estate brokerage firm in the United States, and one of the largest residential real estate brokerage franchise networks in the United States. BHE employs approximately 24,000 people in connection with its various operations.

General Matters

PacifiCorp is a regulated electric utility company headquartered in Oregon, serving electric customers in portions of Utah, Oregon, Wyoming, Washington, Idaho and California. The combined service territory's diverse regional economy ranges from rural, agricultural and mining areas to urban, manufacturing and government service centers. No single segment of the economy dominates the combined service territory, which helps mitigate PacifiCorp's exposure to economic fluctuations. In addition to retail sales, PacifiCorp sells electricity on a wholesale basis.

MidAmerican Energy Company ("MEC") is a regulated electric and natural gas utility company headquartered in Iowa, serving electric and natural gas customers primarily in Iowa and also in portions of Illinois, South Dakota and Nebraska. MEC has a diverse retail customer base consisting of urban and rural residential customers and a variety of commercial and industrial customers. In addition to retail sales and natural gas transportation, MEC sells electricity principally to markets operated by regional transmission organizations and natural gas on a wholesale basis.

NV Energy, Inc. ("NV Energy") is an energy holding company headquartered in Nevada, primarily consisting of two regulated utility subsidiaries, Nevada Power Company ("Nevada Power") and Sierra Pacific Power Company ("Sierra Pacific") (collectively, the "Nevada Utilities"). Nevada Power serves retail electric customers in southern Nevada and Sierra Pacific serves retail electric and natural gas customers in northern Nevada. The Nevada Utilities' combined service territory's economy includes gaming, mining, recreation, warehousing, manufacturing and governmental services. In addition to retail sales and natural gas transportation, the Nevada Utilities sell electricity and natural gas on a wholesale basis.

As vertically integrated utilities, BHE's domestic utilities own approximately 29,000 net megawatts of generation capacity in operation and under construction. The domestic utilities business is subject to seasonal variations principally related to the use of electricity for air conditioning and natural gas for heating. Typically, regulated electric revenues are higher in the summer months, while regulated natural gas revenues are higher in the winter months.

The Great Britain distribution companies consist of Northern Powergrid (Northeast) plc and Northern Powergrid (Yorkshire) plc, which own a substantial electricity distribution network that delivers electricity to end-users in northeast England in an area covering approximately 10,000 square miles. The distribution companies primarily charge supply companies regulated tariffs for the use of their distribution systems.

AltaLink L.P. ("AltaLink") is a regulated electric transmission-only utility company headquartered in Calgary, Alberta. AltaLink's high voltage transmission lines and related facilities transmit electricity from generating facilities to major load centers, cities and large industrial plants throughout its 87,000 square mile service territory.

The natural gas pipelines consist of BHE GT&S, LLC ("BHE GT&S"), Northern Natural Gas Company ("Northern Natural") and Kern River Gas Transmission Company ("Kern River"). BHE GT&S was acquired on November 1, 2020.

BHE GT&S, based in Virginia, operates three interstate natural gas pipeline systems that consist of approximately 5,400 miles of natural gas transmission, gathering and storage pipelines and operates seventeen underground natural gas storage fields in the eastern region of the United States. BHE GT&S's large underground natural gas storage assets and pipeline systems are part of an interconnected gas transmission network that provides transportation services to utilities and numerous other customers. BHE GT&S is also an industry leader in liquefied natural gas solutions through its investments in and ownership of several liquefied natural gas facilities located throughout the eastern region of the United States.

Northern Natural, based in Nebraska, operates the largest interstate natural gas pipeline system in the United States, as measured by pipeline miles, reaching from west Texas to Michigan's Upper Peninsula. Northern Natural's pipeline system consists of approximately 14,500 miles of natural gas pipelines. Northern Natural's extensive pipeline system, which is interconnected with many interstate and intrastate pipelines in the national grid system, has access to supplies from multiple major supply basins and provides transportation services to utilities and numerous other customers. Northern Natural also operates three underground natural gas storage facilities and two liquefied natural gas storage peaking units. Northern Natural's pipeline system experiences significant seasonal swings in demand and revenue, with the highest demand typically occurring during the months of November through March.

Kern River, based in Utah, operates an interstate natural gas pipeline system that consists of approximately 1,400 miles and extends from supply areas in the Rocky Mountains to consuming markets in Utah, Nevada and California. Kern River transports natural gas for electric and natural gas distribution utilities, major oil and natural gas companies or affiliates of such companies, electric generating companies, energy marketing and trading companies, and financial institutions.

BHE Renewables, based in Iowa, owns interests in independent power projects having approximately 4,700 net megawatts of generation capacity that are in service in California, Texas, Illinois, Nebraska, New York, Arizona, Minnesota, Kansas, Hawaii and the Philippines. These independent power projects sell power generated primarily from wind, solar, geothermal and hydro sources under long-term contracts. Additionally, BHE Renewables has invested over $6 billion in 32 wind projects sponsored by third parties, commonly referred to as tax equity investments.

Regulatory Matters

PacifiCorp, MEC and the Nevada Utilities are subject to comprehensive regulation by various federal, state and local agencies. The Federal Energy Regulatory Commission ("FERC") is an independent agency with broad authority to implement provisions of the Federal Power Act, the Natural Gas Act, the Energy Policy Act of 2005 and other federal statutes. The FERC regulates rates for wholesale sales of electricity; transmission of electricity, including pricing and regional planning for the expansion of transmission systems; electric system reliability; utility holding companies; accounting and records retention; securities issuances; construction and operation of hydroelectric facilities; and other matters. The FERC also has the enforcement authority to assess civil penalties of up to $1.3 million per day per violation of rules, regulations and orders issued under the Federal Power Act. MEC is also subject to regulation by the Nuclear Regulatory Commission pursuant to the Atomic Energy Act of 1954, as amended, with respect to its 25% ownership of the Quad Cities Nuclear Station.

With certain limited exceptions, BHE's domestic utilities have an exclusive right to serve retail customers within their service territories and, in turn, have an obligation to provide service to those customers. In some jurisdictions, certain classes of customers may choose to purchase all or a portion of their energy from alternative energy suppliers, and in some jurisdictions retail customers can generate all or a portion of their own energy. Historically, state regulatory commissions have established retail electric and natural gas rates on a cost-of-service basis, designed to allow a utility the opportunity to recover what each state regulatory commission deems to be the utility's reasonable costs of providing services, including a fair opportunity to earn a reasonable return on its investments based on its cost of debt and equity. The retail electric rates of PacifiCorp, MEC and the Nevada Utilities are generally based on the cost of providing traditional bundled services, including generation, transmission and distribution services; however, rates are available for transmission and distribution-only services.

Northern Powergrid (Northeast) and Northern Powergrid (Yorkshire) each charge fees for the use of their distribution systems that are controlled by a formula prescribed by the British electricity regulatory body, the Gas and Electricity Markets Authority. The current eight-year price control period runs from April 1, 2015 through March 31, 2023.

AltaLink is regulated by the Alberta Utilities Commission ("AUC"), pursuant to the Electric Utilities Act (Alberta), the Public Utilities Act (Alberta), the Alberta Utilities Commission Act (Alberta) and the Hydro and Electric Energy Act (Alberta). The AUC is an independent quasi-judicial agency, which regulates and oversees Alberta's electricity transmission sector with broad authority that may impact many of AltaLink's activities, including its tariffs, rates, construction, operations and financing. Under the Electric Utilities Act, AltaLink prepares and files applications with the AUC for approval of tariffs to be paid by the Alberta Electric System Operator ("AESO") for the use of its transmission facilities, and the terms and conditions governing the use of those facilities. The AESO is an independent system operator in Alberta, Canada that oversees Alberta's integrated electrical system ("AIES") and wholesale electricity market. The AESO is responsible for directing the safe, reliable and economic operation of the AIES, including long-term transmission system planning.

The natural gas pipelines are subject to regulation by various federal and state agencies. The natural gas pipeline and storage operations of BHE GT&S, Northern Natural and Kern River are regulated by the FERC pursuant to the Natural Gas Act and the Natural Gas Policy Act of 1978. Under this authority, the FERC regulates, among other items, (a) rates, charges, terms and conditions of service, (b) the construction and operation of interstate pipelines, storage and related facilities, including the extension, expansion or abandonment of such facilities and (c) the construction and operation of liquefied natural gas import/export facilities. Interstate natural gas pipeline companies are also subject to regulations administered by the Office of Pipeline Safety within the Pipeline and Hazardous Materials Safety Administration, an agency of the DOT. Federal pipeline safety regulations are issued pursuant to the Natural Gas Pipeline Safety Act of 1968, as amended, which establishes safety requirements in the design, construction, operation and maintenance of interstate natural gas pipeline facilities.

Environmental Matters

BHE and its energy businesses are subject to federal, state, local and foreign laws and regulations regarding climate change, renewable portfolio standards, air and water quality, emissions performance standards, coal combustion byproduct disposal, hazardous and solid waste disposal, protected species and other environmental matters that have the potential to impact current and future operations. In addition to imposing continuing compliance obligations, these laws and regulations, such as the Federal Clean Air Act, provide regulators with the authority to levy substantial penalties for noncompliance, including fines, injunctive relief and other sanctions.

The Federal Clean Air Act, as well as state laws and regulations impacting air emissions, provides a framework for protecting and improving the nation's air quality and controlling sources of air emissions. These laws and regulations continue to be promulgated and implemented and will impact the operation of BHE's generating facilities and require them to reduce emissions at those facilities to comply with the requirements.

Renewable portfolio standards have been established by certain state governments and generally require electricity providers to obtain a minimum percentage of their power from renewable energy resources by a certain date. Utah, Oregon, Washington, California, Iowa and Nevada have adopted renewable portfolio standards. In addition, the potential adoption of state or federal clean energy standards, which include low-carbon, non-carbon and renewable electricity generating resources, may also impact electricity generators and natural gas providers.

In December 2015, an international agreement was negotiated by 195 nations to create a universal framework for coordinated action on climate change in what is referred to as the Paris Agreement. The Paris Agreement reaffirms the goal of limiting global temperature increase well below 2 degrees Celsius, while urging efforts to limit the increase to 1.5 degrees Celsius; establishes commitments by all parties to make nationally determined contributions and pursue domestic measures aimed at achieving the commitments; commits all countries to submit emissions inventories and report regularly on their emissions and progress made in implementing and achieving their nationally determined commitments; and commits all countries to submit new commitments every five years, with the expectation that the commitments will get more aggressive. In the context of the Paris Agreement, the United States agreed to reduce greenhouse gas emissions 26% to 28% by 2025 from 2005 levels. The Paris Agreement formally entered into force November 4, 2016. The United States completed its withdrawal from the Paris Agreement on November 4, 2020. President Biden accepted the terms of the climate agreement on January 21, 2021, and the United States completed its reentry on February 19, 2021.

On October 10, 2017, the Environmental Protection Agency ("EPA") issued a proposal to repeal the Clean Power Plan, which was intended to achieve an overall reduction in carbon dioxide emissions from existing fossil-fueled electric generating units of 32% below 2005 levels. On June 19, 2019, the EPA repealed the Clean Power Plan and issued the Affordable Clean Energy rule, which fully replaced the Clean Power Plan. In the Affordable Clean Energy rule, the EPA determined that the best system of emissions reduction for existing coal fueled power plants is heat rate improvements and identified a set of candidate technologies and measures that could improve heat rates. Measures taken to meet the standards of performance must be achieved at the source itself.

The EPA's repeal and replacement of the Clean Power Plan is not expected to have a material impact on BHE and its energy subsidiaries. Increasingly, states are adopting legislation and regulations to reduce greenhouse gas emissions, and local governments and consumers are seeking increasing amounts of clean and renewable energy.

BHE and its energy subsidiaries continue to focus on delivering reliable, affordable, safe and clean energy to its customers and on actions to mitigate greenhouse gas emissions. For example, through December 31, 2020, BHE's cumulative investment in wind, solar, geothermal and biomass generation is approximately $34 billion.

Non-Energy Businesses

HomeServices of America, Inc. ("HomeServices") is the largest residential real estate brokerage firm in the United States. In addition to providing traditional residential real estate brokerage services, HomeServices offers other integrated real estate services, including mortgage originations and mortgage banking, title and closing services, property and casualty insurance, home warranties, relocation services and other home-related services. It operates under 46 brand names with over 43,000 real estate agents in nearly 900 brokerage offices in 30 states and the District of Columbia.

HomeServices' franchise network currently includes approximately 370 franchisees in over 1,600 brokerage offices throughout the United States and Europe with over 53,000 real estate agents under two brand names. In exchange for certain fees, HomeServices provides the right to use the Berkshire Hathaway HomeServices or Real Living brand names and other related service marks, as well as providing orientation programs, training and consultation services, advertising programs and other services.

HomeServices' principal sources of revenue are dependent on residential real estate sales, which are generally higher in the second and third quarters of each year. This business is highly competitive and subject to general real estate market conditions.

Manufacturing Businesses

Berkshire's numerous and diverse manufacturing subsidiaries are grouped into three categories: (1) industrial products, (2) building products and (3) consumer products. Berkshire's industrial products businesses manufacture specialty chemicals, metal cutting tools, components for aerospace and power generation applications, and a variety of other products primarily for industrial use. The building products group produces prefabricated and site-built residential homes, flooring products, insulation, roofing and engineered products, building and engineered components, paint and coatings and bricks and masonry products. The consumer products group manufactures recreational vehicles, alkaline batteries, various apparel products, jewelry and custom picture framing products. Information concerning the major activities of these three groups follows. Berkshire's manufacturing businesses employed approximately 179,000 people at the end of 2020.

Industrial products

Precision Castparts

Precision Castparts Corp. ("PCC") manufactures complex metal components and products, provides high-quality investment castings, forgings, fasteners/fastener systems and aerostructures for critical aerospace and power and energy applications. PCC also manufactures seamless pipe for coal-fired, industrial gas turbine ("IGT") and nuclear power plants; downhole casing and tubing, fittings and various mill forms in a variety of nickel and steel alloys for severe-service oil and gas environments; investment castings and forgings for general industrial, armament, medical and other applications; nickel and titanium alloys in all standard mill forms from large ingots and billets to plate, foil, sheet, strip, tubing, bar, rod, extruded shapes, rod-in-coil, wire and welding consumables, as well as cobalt alloys, for the aerospace, chemical processing, oil and gas, pollution control and other industries; revert management solutions; fasteners for automotive and general industrial markets; specialty alloys for the investment casting and forging industries; heat treating and destructive testing services for the investment cast products and forging industries; grinder pumps and affiliated components for low-pressure sewer systems; critical auxiliary equipment and gas monitoring systems for the power generation industry; and metalworking tools for the fastener market and other applications.

Investment casting technology involves a multi-step process that uses ceramic molds in the manufacture of metal components with more complex shapes, closer tolerances and finer surface finishes than parts manufactured using other methods. PCC uses this process to manufacture products for aircraft engines, IGT and other aeroderivative engines, airframes, medical implants, armament, unmanned aerial vehicles and other industrial applications. PCC also manufactures high temperature carbon and ceramic composite components, including ceramic matrix composites, for use in next-generation aerospace engines.

PCC uses forging processes to manufacture components for the aerospace and power generation markets, including seamless pipe for coal-fired, industrial gas turbine and nuclear power plants, and downhole casings and tubing pipe for severe service oil and gas markets. PCC manufactures high-performance, nickel-based alloys used to produce forged components for aerospace and non-aerospace applications in such markets as oil and gas, chemical processing and pollution control. These titanium products are used to manufacture components for the commercial and military aerospace, power generation, energy and other industrial end markets.

PCC is also a leading developer and manufacturer of highly engineered fasteners, fastener systems, aerostructures and precision components, primarily for critical aerospace applications. These products are produced for the aerospace and power and energy markets, as well as for construction, automotive, heavy truck, farm machinery, mining and construction equipment, shipbuilding, machine tools, medical equipment, appliance and recreation markets. PCC has several significant customers, including aerospace original equipment manufacturers (Boeing and Airbus) and aircraft engine manufacturer suppliers (General Electric, Rolls Royce and Pratt &Whitney).

The majority of PCC's sales are from customer orders or demand schedules pursuant to long-term agreements. Contractual terms may provide for termination by the customer, subject to payment for work performed. PCC typically does not experience significant order cancellations, although periodically it receives requests for delays in delivery schedules. In 2020, delay requests increased due to the COVID-19 pandemic.

The effects of the COVID-19 pandemic and the grounding of the Boeing 737 MAX produced significant adverse effects on the PCC aerospace business in 2020. The sudden and material reductions in air travel led to aircraft build rate reductions and customer destocking at extraordinary rates. Aircraft build rates have not yet begun to recover in any meaningful way. During 2020, PCC significantly reduced its worldwide workforce by about 40% since the end of 2019 to help align operations to reduced aircraft build rates. The restructuring actions taken began to improve margins in late 2020 from the low margins experienced earlier in the year and further margin improvements are expected going forward.

PCC is subject to substantial competition in all of its markets. Components and similar products may be produced by competitors, who use either the same types of manufacturing processes as PCC or other processes. Although PCC believes its manufacturing processes, technology and experience provide advantages to its customers, such as high quality, competitive prices and physical properties that often meet more stringent demands, alternative forms of manufacturing can be used to produce many of the same components and products. Despite intense competition, PCC is a leading supplier in most of its principal markets. Several factors, including long-standing customer relationships, technical expertise, state-of-the-art facilities and dedicated employees, aid PCC in maintaining competitive advantages.

Several raw materials used in PCC products, including certain metals such as nickel, titanium, cobalt, tantalum and molybdenum, are found in only a few parts of the world. These metals are required for the alloys used in manufactured products. The availability and costs of these metals may be influenced by private or governmental cartels, changes in world politics, labor relations between the metal producers and their workforces and inflation.

Lubrizol Corporation

The Lubrizol Corporation ("Lubrizol") is a specialty chemical and performance materials company that produces and supplies technologies for the global transportation, industrial and consumer markets. Lubrizol currently operates two businesses: (1) Lubrizol Additives, which includes engine lubricant additives, driveline lubricant additives and industrial specialties products; and (2) Lubrizol Advanced Materials, which includes Engineered Materials (engineered polymers and performance coatings) and Life Sciences (beauty and personal care, health and home care solutions).

Lubrizol Additives products are used in a broad range of applications including engine oils, transmission fluids, gear oils, specialty driveline lubricants, fuels, metalworking fluids, compressor lubricants and greases for transportation and industrial applications. Lubrizol Advanced Materials products are used in many different types of applications including over-the-counter pharmaceutical products, medical devices, performance coatings, personal care products, sporting goods and plumbing and fire sprinkler systems. Lubrizol is an industry leader in many of the markets in which it competes. Lubrizol's principal lubricant additives competitors are Infineum International Ltd., Chevron Oronite Company and Afton Chemical Corporation. Advanced Materials competes in many markets with a variety of competitors in each product line.

With its considerable patent portfolio, Lubrizol uses its technological leadership position in product development and applies its science capabilities and formulation and market expertise to improve the quality and value of its products. Lubrizol leverages its scientific and applications knowledge to meet and exceed customer performance and sustainability requirements. While Lubrizol typically has patents that expire each year, it invests resources to protect its intellectual property and to develop or acquire innovative products for the markets it serves. Lubrizol uses many specialty and commodity chemical raw materials in its manufacturing processes. Raw materials are primarily feedstocks derived from petroleum and petrochemicals and, generally, are obtainable from several sources. The materials that Lubrizol chooses to purchase from a single source typically are subject to long-term supply contracts to ensure supply reliability.

Lubrizol operates its business on a global basis through more than 100 offices, laboratories, production facilities and warehouses on six continents, the most significant of which are North America, Europe, Asia and South America. Lubrizol markets its products worldwide through a direct sales organization and sales agents and distributors. Lubrizol's customers principally consist of major global and regional oil companies and industrial and consumer products companies that are located throughout the world. Some of its largest customers also may be suppliers. During 2020, no single customer accounted for more than 10% of Lubrizol's consolidated revenues. In 2020, the global pandemic had an adverse effect on many of the markets that Lubrizol serves, including the transportation and industrial markets. This was offset in part by strong demand for Lubrizol's technology that is used in personal care applications, such as hand sanitizers.

Lubrizol continues to expend necessary capital to upgrade and optimize operations, ensure compliance with health, safety and environmental requirements, and increase global manufacturing capacity, while reducing the environmental footprint of its operations. Lubrizol also makes a significant investment in its human capital to ensure that it attracts, develops and retains a talented and diverse employee workforce.

Lubrizol is subject to foreign, federal, state and local laws to protect the environment, limit manufacturing waste and emissions, ensure product and employee safety and regulate trade. The company believes that its policies, practices and procedures are designed to limit the risks of non-compliance with laws and consequent financial liability. Nevertheless, the operation of manufacturing plants entails ongoing environmental and other risks, and significant costs or liabilities could be incurred in the future.

IMC International Metalworking Companies

IMC International Metalworking Companies ("IMC") is one of the world's three largest multinational manufacturers of consumable precision carbide metal cutting tools for applications in a broad range of industrial end markets. IMC's principal brand names include ISCAR®, TaeguTec®, Ingersoll®, Tungaloy®, Unitac®, UOP®, It.te.di®, Qutiltec®, Tool—Flo®, PCT® and IMCO®. IMC's primary manufacturing facilities are located in Israel, the United States, South Korea, Japan, Germany, Italy, Switzerland, India and China.

IMC has five primary product lines: milling tools, gripping tools, turning/thread tools, drilling tools and tooling. The main products are split within each product line between consumable cemented tungsten carbide inserts and steel tool holders. Inserts comprise the vast majority of sales and earnings. Metal cutting inserts are used by industrial manufacturers to cut metals and are consumed during their use in cutting applications. IMC manufactures hundreds of types of highly engineered inserts within each product line that are tailored to maximize productivity and meet the technical requirements of customers. IMC's staff of scientists and engineers continuously develop and innovate products that address end user needs and requirements.

IMC's global sales and marketing network operates in virtually every major manufacturing center around the world, staffed with highly skilled engineers and technical personnel. IMC's customer base is very diverse, with its primary customers being large, multinational businesses in the automotive, aerospace, engineering and machinery industries. IMC operates a regional central warehouse system with locations in Israel, the United States, Belgium, Korea, Japan, China and Brazil. Additional small quantities of products are maintained at local IMC offices to provide on-time customer support and inventory management.

IMC competes in the metal cutting tools segment of the global metalworking tools market. The segment includes hundreds of participants who range from small, private manufacturers of specialized products for niche applications and markets to larger, global multinational businesses (such as Sandvik and Kennametal, Inc.) with a wide assortment of products and extensive distribution networks. Other manufacturing companies such as Kyocera, Mitsubishi, Sumitomo, Ceratizit and Korloy also play a significant role in the cutting tool market.

Marmon Holdings

Marmon Holdings, Inc. ("Marmon") is a global industrial organization comprising 11 diverse business sectors and more than 100 autonomous manufacturing and service businesses. Marmon's manufacturing and service operations are conducted at approximately 400 manufacturing, distribution and service facilities located primarily in the United States, as well as 22 other countries worldwide. Marmon's business sectors are described as follows.

Foodservice Technologies manufactures beverage dispensing and cooling equipment, hot and cold food preparation and holding equipment and related products for restaurants, global brand owners and other foodservice providers. Operations are based in the U.S. with manufacturing in the U.S., Mexico, China, the U.K., Germany and Italy. Products are sold primarily throughout the U.S., Europe and Asia.

Water Technologies manufactures water treatment equipment for residential, commercial and industrial applications worldwide. Operations are based primarily in the U.S., Canada, China, Singapore, India and Mexico with business centers located in Belgium, France, Poland, Germany, the U.K., Italy, Switzerland and U.A.E.

Transportation Products serves the automotive, heavy-duty highway transportation, and aerospace industries with precision-molded plastic components; fastener thread solutions; metal tubing; auto aftermarket transmission and chassis products; platform and lowbed trailers; and truck and trailer components. Operations and business are conducted primarily in the U.S., Mexico, Canada, Europe and Asia.

Retail Solutions provides retail environment design services; in-store digital merchandising, dispensing and display fixtures; shopping, material handling and security carts. Operations and business are conducted in the U.S., U.K. and Czech Republic.

Metal Services provides specialty metal pipe, tubing and related value-added services to customers across a broad range of industries. Operations are based in the U.S., Canada and Mexico and business is conducted primarily in those countries.

Electrical produces electrical wire for use in residential and commercial buildings, and specialty wire and cable for use in energy, transit, aerospace, defense, communication and other industrial applications. Operations are based in the U.S., Canada, India and England. Business is conducted globally and primarily in the U.S., Canada, India, the U.K., U.A.E. and China.

Plumbing & Refrigeration supplies copper tubing and copper, brass, aluminum and stainless-steel fittings and components for the plumbing, HVAC and refrigeration markets; custom coils for the HVAC market; and aluminum and brass forgings for many commercial and industrial applications. Business and operations are conducted primarily in the U.S.

Industrial Products supplies construction fasteners; gloves and other protective wear; gear drives, gearboxes, fan drives and pump drives for various markets; wind machines for agricultural use; and wheels, axles, and gears for rail, mining and other applications. Operations are primarily based in the U.S., Canada and China and business is conducted in those countries.

Rail & Leasing manufactures, leases and maintains railcars; leases intermodal tank containers; manufactures mobile railcar movers; provides in-plant rail switching and loading services; performs track construction and maintenance; and manufactures steel tank heads and cylinders.

Union Tank Car Company ("UTLX") is the largest component of Rail & Leasing and is a leading designer, builder and full-service lessor of railroad tank cars and other specialized railcars. Together with its Canadian affiliate Procor, UTLX owns a fleet of approximately 124,000 railcars for lease to customers in chemical, petrochemical, energy and agricultural/food industries. UTLX manufactures tank cars in the U.S. and performs railcar maintenance services at more than 100 locations across North America.

UTLX has a diversified customer base, both geographically and across industries. UTLX, while subject to cyclicality and significant competition in most of its markets, competes by offering a broad range of high-quality products and services targeted at its niche markets. Railcars are typically leased for multiple-year terms and most of the leases are renewed upon expiration. Due to selective ongoing capital investment, utilization rates (the number of railcars on lease as a percentage of the total fleet) of the railcar fleet are generally high.

Intermodal tank containers are leased through EXSIF Worldwide. EXSIF is a leading international lessor of intermodal tank containers with a fleet of approximately 69,000 units, primarily serving chemical producers and logistics operators.

Crane Services is a provider of mobile cranes and operators in North America and Australia. Sterling Crane, Joyce Crane, Freo Group, and WGC Cranes operate a combined fleet of approximately 1,200 cranes primarily serving the energy, mining, petrochemical and infrastructure markets.

Medical (formed in 2019 through the acquisition of the Colson Medical Companies) develops, manufactures and distributes a wide range of innovative medical devices in the extremities fixation, craniomaxillofacial surgery, neurosurgery, biologics, aesthetics and powered instruments markets. The sector's leading-edge medical technology and products are used globally to help improve patient care and outcomes. Operations are based in the U.S., Europe and China. Business is conducted primarily in North and South America, Europe, Asia and Australia.

Other industrial products

CTB International Corp. ("CTB"), headquartered in Milford, Indiana, is a leading global designer, manufacturer and marketer of a wide range of agricultural systems and solutions for preserving grain, producing poultry, pigs and eggs, and for processing poultry, fish, vegetables and other foods. CTB operates from facilities located around the globe and supports customers through a worldwide network of independent distributors and dealers.

CTB competes with a variety of manufacturers and suppliers, many of which offer only a limited number of the products offered by CTB and two of which offer products across many of CTB's product lines. Competition is based on the price, value, reputation, quality and design of the products offered and the customer service provided by distributors, dealers and manufacturers of the products. CTB's leading brand names, distribution network, diversified product line, product support and high-quality products enable it to compete effectively. CTB manufactures its products primarily from galvanized steel, steel wire, stainless steel and polymer materials and supplies of these materials have been sufficient in recent years.

LiquidPower Specialty Products Inc. ("LSPI"), headquartered in Houston, Texas, is a global leader in the science of drag reduction application ("DRA') technology by maximizing the flow potential of pipelines, increasing operational flexibility and throughput capacity, and efficiencies for customers. LSPI develops innovative flow improver solutions with customers in over 40 countries on six continents, treating over 50 million barrels of hydrocarbon liquids per day. LSPI's DRA offering is part of a comprehensive, full-service solution that encompasses industry-leading technology, quality manufacturing, technical support and consulting, a reliable supply chain, injection equipment and field service. The Scott Fetzer companies are a group of businesses that manufacture, distribute, service and finance a wide variety of products for residential, industrial and institutional use.

Building Products

Clayton Homes

Clayton Homes, Inc. ("Clayton"), headquartered near Knoxville, Tennessee, is a vertically integrated housing company offering traditional site-built homes and off-site built housing – including modular homes, manufactured homes, CrossMod™ homes and tiny homes. In 2020, Clayton delivered 46,765 off-site built and 9,475 site-built homes. Clayton also offers home financing and insurance products and competes on price, service, location and delivery capabilities.

All Clayton Built® off-site homes are designed, engineered and assembled in the United States. As of December 2020, off-site backlog was $1.3 billion, up 237% from prior year. Clayton sells its homes through independent and company owned home centers, realtors and subdivision channels. Clayton considers its ability to make financing available to retail purchasers, a factor affecting the market acceptance of its off-site built homes. Clayton's financing programs utilize proprietary loan underwriting guidelines, which include ability to repay calculations, including debt to income limits, consideration of residual income and credit score requirements, which are considered in evaluating loan applicants.

Since 2015, Clayton's site-built division, Clayton Properties Group, has expanded through the acquisition of nine builders across 14 states with a total of 312 subdivisions, supplementing the portfolio of housing products offered to customers. Clayton's site-builders currently own and control a total of 62,514 homesites, with a home order backlog of approximately $2.2 billion.

Shaw Industries

Shaw Industries Group, Inc. (“Shaw”), headquartered in Dalton, Georgia, is a leading manufacturer and distributor of carpet and flooring products. Shaw designs and manufactures over 3,800 styles of tufted carpet, wood and resilient flooring for residential and commercial use under about 30 brand and trade names and under certain private labels. Shaw also provides project management and installation services. Shaw’s manufacturing operations are fully integrated from the processing of raw materials used to make fiber through the finishing of carpet. In 2018, Shaw acquired Sanquahar Tile Services in Scotland, which manufactures and distributes carpet tile throughout Europe. Shaw also manufactures or distributes a variety of hardwood, wood plastic composite (WPC), stone plastic composite (SPC) and vinyl and laminate floor products (“hard surfaces”). Shaw’s soft and hard surface products are sold in a broad range of patterns, colors and textures. Shaw operates Shaw Sports Turf and Southwest Greens International, LLC, which provide synthetic sports turf, golf greens and landscape turf products.

Shaw products are sold wholesale to over 47,000 retailers, distributors and commercial users throughout the United States, Canada and Mexico and are also exported to various overseas markets. Shaw’s wholesale products are marketed domestically by over 2,100 salaried and commissioned sales personnel directly to retailers and distributors and to large national accounts. Shaw’s seven carpet, nine hard surface, one sample full-service distribution facility, three sample satellite locations and 30 redistribution centers, along with centralized management information systems, enable it to provide prompt and efficient delivery of its products to both its retail customers and wholesale distributors.

Substantially all carpet manufactured by Shaw is tufted carpet made from nylon, polypropylene and polyester. In the tufting process, yarn is inserted by multiple needles into a synthetic backing, forming loops, which may be cut or left uncut, depending on the desired texture or construction. During 2020, Shaw processed approximately 97% of its requirements for carpet yarn in its own yarn processing facilities. The availability of raw materials is adequate but costs are impacted by petro-chemical and natural gas price changes. Raw material cost changes are periodically factored into selling prices to customers.

The soft floor covering industry is highly competitive with only a handful of key players domestically where the majority of Shaw’s business occurs. There are numerous manufacturers, domestically and internationally, that are engaged in hard surface floor covering production, distribution and sales. According to industry estimates, carpet accounts for approximately 44% of the total United States consumption of all flooring types. The principal competitive measures within the floor covering industry are quality, style, price and service.

Johns Manville

Johns Manville (“JM”), headquartered in Denver, Colorado, is a leading manufacturer and marketer of premium-quality products for building insulation, mechanical and industrial insulation, commercial roofing and roof insulation, as well as fibers and nonwovens for commercial, industrial and residential applications. JM serves markets that include aerospace, automotive and transportation, air handling, appliance, HVAC, pipe and equipment, filtration, waterproofing, building, flooring, interiors and wind energy. Fiberglass is the basic material in a majority of JM’s products, although JM also manufactures a significant portion of its products with other materials to satisfy the broader needs of its customers. Raw materials are readily available in sufficient quantities from various sources for JM to maintain and expand its current production levels. JM regards its patents and licenses as valuable, however it does not consider any of its businesses to be materially dependent on any single patent or license. JM operates over 40 manufacturing facilities in North America, Europe and China and conducts research and development at its technical center in Littleton, Colorado and at other facilities in the U.S. and Europe.

Fiberglass is made from earthen raw materials and recycled glass, together with proprietary agents to bind many of its glass fibers. JM’s products also contain materials other than fiberglass, including various chemical and petrochemical-based materials used in roofing and other specialized products. JM uses recycled material when available and suitable to satisfy the broader needs of its customers. The raw materials used in these various products are readily available in sufficient quantities from various sources to maintain and expand its current production levels.

JM’s operations are subject to a variety of federal, state and local environmental laws and regulations, which regulate or impose liability for the discharge of materials into the air, land and water and govern the use and disposal of hazardous substances and use of chemical substances generally. The most relevant of the federal laws are the Federal Clean Air Act, the Clean Water Act, the Toxic Substances Control Act, the Resource Conservation and Recovery Act and the Comprehensive Environmental Response, Compensation and Liability Act, which are administered by the EPA. Canadian, European and Asian regulatory authorities have also adopted their own environmental laws and regulations. JM continually monitors new and pending regulations and assesses their potential impact on the business.

JM sells its products through a wide variety of channels including contractors, distributors, retailers, manufacturers and fabricators. JM operates in highly competitive markets, with competitors comprising primarily several large global and national manufacturers and smaller regional manufacturers. JM holds leadership positions in the key markets that it serves. JM's products compete primarily on value, differentiation and customization, and breadth of product line. Sales of JM's products are moderately seasonal due to increases in construction activity that typically occur in the second and third quarters of the calendar year. JM sees a marketplace trend in customer purchasing decisions being influenced by the sustainable and energy efficient attributes of its products, services and operations.

MiTek Industries, Inc.

MiTek Industries, Inc. ("MiTek"), based in Chesterfield, Missouri, operates in two separate markets: residential and commercial. MiTek operates worldwide with sales in over 100 countries and with manufacturing facilities and/or sales/engineering offices located in 21 countries.

In the residential segment, MiTek is a leading supplier of engineered connector products, construction hardware, engineering software and services and computer-driven manufacturing machinery to the truss component market of the building components industry. MiTek's primary customers are component manufacturers who manufacture prefabricated roof and floor trusses and wall panels for the residential building market. MiTek also sells construction hardware to commercial distributors and do-it-yourself retail stores.

MiTek's commercial businesses provide products and services sold to the commercial construction industry. Commercial products include curtain wall systems, masonry and stone anchoring systems, light gauge steel framing products, engineering services for a proprietary high-performance steel frame connection and a comprehensive range of ductwork for the ventilation market, customized air handling systems for commercial, institutional and industrial markets, design and supply of nuclear safety related HVAC systems and components, energy recovery and dehumidification systems for commercial applications and pre-engineered and pre-fabricated custom structural mezzanines and platforms for distribution and manufacturing facilities.

A significant raw material used by MiTek is hot dipped galvanized sheet steel. While supplies are presently adequate, variations in supply have historically occurred, producing significant variations in cost and availability.

Benjamin Moore

Benjamin Moore & Co. ("Benjamin Moore"), headquartered in Montvale, New Jersey, is one of North America's leading manufacturers of premium quality residential, commercial and industrial maintenance coatings. Benjamin Moore is committed to innovation and sustainable manufacturing practices. The Benjamin Moore premium portfolio spans the brand's flagship paint lines including Aura®, Regal® Select, Ultra Spec®, ben®, ADVANCE®, ARBORCOAT® and more. The Benjamin Moore diversified brands include specialty and architectural paints from Coronado®, Insl-x® and Lenmar®. Benjamin Moore coatings are available from its more than 7,500 independently owned and operated paint, decorating and hardware retailers throughout the United States and Canada as well as 75 countries globally. In July 2019, Benjamin Moore announced the expansion of its relationship with Ace Hardware ("Ace"), through which Benjamin Moore has become the preferred paint supplier for approximately 3,300 Ace Hardware stores, which are included in the count above. Through this agreement, these Ace stores are afforded the opportunity to carry a full line premium assortment of Benjamin Moore products or a streamlined offering of Regal® Select and ben®, or ben® only branded products. As part of the expansion, Benjamin Moore assumed responsibility for manufacturing Clark+Kensington® and Royal®, as well as the balance of Ace's private label paint brands.

In addition, Benjamin Moore operates an online "pick up in store" program, which allows consumers to place orders via an e-commerce site or, for national accounts and government agencies, via its customer information center. These orders may be picked up at the customer's nearest retailer or delivered. For national accounts, drop-ship orders can be fulfilled by Benjamin Moore if a minimum gallon threshold is met.

Benjamin Moore competes with numerous manufacturers, distributors and paint, coatings and related products retailers. Product quality, product innovation, breadth of product line, technical expertise, service and price determine the competitive advantage. Competitors include other paint and decorating stores, mass merchandisers, home centers, independent hardware stores, hardware chains and manufacturer-operated direct outlets, such as Sherwin-Williams Company, PPG Industries, Inc., The Valspar Corporation, The Home Depot, Inc. and Lowe's Companies, Inc.

The most significant raw materials in Benjamin Moore products are titanium dioxide, monomers, polymers and pigments. Historically, these materials have been generally available, with pricing and availability subject to fluctuation.

Acme Brick

Acme Brick Company ("Acme"), headquartered in Fort Worth, Texas, manufactures and distributes clay bricks (Acme Brick®) and concrete block (Featherlite). In addition, Acme distributes numerous other building products of other manufacturers, including floor and wall tile, wood flooring and other masonry products. Products are sold primarily in the South Central and South Eastern United States through company-operated sales offices. Acme distributes products primarily to homebuilders and masonry and general contractors.

In 2018 and 2019, Acme closed multiple underperforming manufacturing and sales facilities. Acme operates 12 clay brick manufacturing sites located in four states, three concrete block facilities and a quarrying operation all located in Texas. The demand for Acme's products is seasonal, with higher sales in the warmer weather months, and is subject to the level of construction activity, which is cyclical. Acme also owns and leases properties and mineral rights that supply raw materials used in many of its manufactured products. Acme's raw materials supply is believed to be adequate.

The brick industry is subject to the EPA Maximum Achievable Control Technology Standards ("MACT"). As required under the 1990 Clean Air Act, the EPA developed a list of source categories that require the development of National Emission Standards for Hazardous Air Pollutants, which are also referred to as MACT Standards ("Rule"). Key elements of the MACT Rule include emission limits established for certain hazardous air pollutants and acidic gases. Acme's brick plants are in compliance with the current Rule.

Consumer Products

Apparel

Fruit of the Loom ("FOL"), headquartered in Bowling Green, Kentucky, is primarily a manufacturer and distributor of basic apparel, underwear, casualwear, athletic apparel and sports equipment. Products under the Fruit of the Loom® and JERZEES® labels are primarily sold in the mass merchandise, mid-tier chains and wholesale markets. In the Vanity Fair Brands product line, Vassarette®, Curvation® and Radiant® by Vanity Fair are sold in the mass merchandise market, while Vanity Fair® and Lily of France® products are sold to mid-tier chains and department stores. FOL also markets and sells apparel, sports equipment and balls to team dealers and athletic apparel, sports equipment and balls to sporting goods retailers under the Russell Athletic® and Spalding® brands. Additionally, Spalding® markets and sells balls and sports equipment in the mass merchandise market and dollar store channels. In 2020, approximately 58% of FOL's sales were to five customers.

FOL generally performs its own knitting, cloth finishing, cutting, sewing and packaging for apparel. For the North American market, which is FOL's predominant sales region, the majority of FOL's cloth manufacturing is performed in Honduras. Labor-intensive cutting, sewing and packaging operations are located in Central America, the Caribbean and Vietnam. For the European market, products are either sourced from third-party contractors in Europe or Asia or sewn in Morocco from textiles internally produced in Morocco. Manufacturing of bras, athletic equipment, sporting goods and other athletic apparel lines are generally sourced from third-party contractors located primarily in Asia.

U.S. grown cotton and polyester fibers are the main raw materials used in the manufacturing of FOL's apparel products and are purchased from a limited number of third-party suppliers. In 2015, FOL entered into an eight-year agreement with one key supplier to provide the majority of FOL's yarn. Management currently believes there are readily available alternative sources of raw materials and yarn. However, if relationships with suppliers cannot be maintained or delays occur in obtaining alternative sources of supply, production could be adversely affected, which could have a corresponding adverse effect on results of operations. Additionally, raw materials are subject to price volatility caused by weather, supply conditions, government regulations, economic climate and other unpredictable factors. FOL has secured contracts to purchase cotton, either directly or through the yarn suppliers, to meet a large percentage of its production plans for 2021. FOL's markets are highly competitive, consisting of many domestic and foreign manufacturers and distributors. Competition is generally based upon product features, quality, customer service and price.

Garan, headquartered in New York, New York designs, manufactures, imports and sells apparel primarily for children, including boys, girls, toddlers and infants. Products are sold under its own trademark Garanimals® and customer private label brands. Garan conducts its business through operating subsidiaries located in the United States, Central America and Asia. Garan's products are sold through its distribution centers in the United States. Fechheimer Brothers manufactures, distributes and sells uniforms, principally for the public service and safety markets, including police, fire, postal and military markets. Fechheimer Brothers is based in Cincinnati, Ohio.

The BH Shoe Holdings Group, headquartered in Greenwich, Connecticut, manufactures and distributes work, rugged outdoor and casual shoes and western-style footwear under a number of brand names, including Justin, Tony Lama®, Chippewa®, BØRN®, B•Ø•C®, Carolina®, EuroSofft, Söfft, Double-H Boots®, Nursemates® and Comfortiva®. Brooks Sports, headquartered in Seattle, Washington, markets and sells performance running footwear and apparel to specialty and national retailers and directly to consumers under the Brooks® brand. A significant volume of the shoes sold by Berkshire's shoe businesses are manufactured or purchased from sources located outside the United States. Products are sold worldwide through a variety of channels including department stores, footwear chains, specialty stores, catalogs and the Internet, as well as through company-owned retail stores.

Other consumer products

Forest River, Inc. ("Forest River") is a manufacturer of recreational vehicles ("RV"), utility cargo trailers, buses and pontoon boats, headquartered in Elkhart, Indiana with products sold in the United States and Canada through an independent dealer network. Forest River has numerous manufacturing facilities located in seven states. Forest River is a leading manufacturer of RVs with numerous brand names, including Forest River, Coachmen RV and Prime Time. Utility cargo trailers are sold under a variety of brand names. Buses are sold under several brand names, including Starcraft Bus. Pontoon boats are sold under the Berkshire, South Bay and Trifecta brand names. The RV industry is very competitive. Competition is based primarily on price, design, quality and service. The industry has consolidated over the past several years and is currently concentrated in a few companies, the largest of which had a market share of approximately 42% based on industry data as of December 2020. Forest River held a market share of approximately 37% at that time.

The Duracell Company ("Duracell'), headquartered in Chicago, Illinois, is a leading manufacturer of high-performance alkaline batteries. Duracell manufactures batteries in the U.S., Europe and China and provides a network of worldwide sales and distribution centers. Costco and Walmart are significant customers, representing approximately 23% of Duracell's annual revenue. There are several competitors in the battery manufacturing market with Duracell holding an approximately 31% market share of the global alkaline battery market. Management believes there are currently sufficient sources of raw materials available, which are primarily steel, zinc and manganese.

Albecca Inc. ("Albecca"), headquartered in Norcross, Georgia, operates in the U.S., Canada and 12 other countries, with products primarily under the Larson-Juhl® name. Albecca designs, manufactures and distributes a complete line of high quality, branded custom framing products, including wood and metal moulding, matboard, foamboard, glass and framing supplies. Complementary to its framing products, Albecca offers art printing and fulfillment services.

Richline Group, Inc., headquartered in New York, New York, operates five strategic business units: Richline Jewelry, Richline Digital, LeachGarner, Rio Grande and Inverness. Each business unit is a manufacturer and/or distributor of precious metal and non-precious metal products to specific target markets including large jewelry chains, department stores, shopping networks, mass merchandisers, e-commerce retailers and artisans plus worldwide manufacturers and wholesalers and the medical, electronics and aerospace industries.

Service and Retailing Businesses

Service Businesses

Berkshire's service businesses provide grocery and foodservice distribution, professional aviation training programs, shared aircraft ownership programs and distribution of electronic components. Other service businesses include franchising and servicing of quick service restaurants, media businesses (television and information distribution), as well as logistics businesses. Berkshire's service businesses employed approximately 45,000 people at the end of 2020. Information concerning these activities follows.

McLane Company

McLane Company, Inc. ("McLane") provides wholesale distribution services in all 50 states to customers that include convenience stores, discount retailers, wholesale clubs, drug stores, military bases, quick service restaurants and casual dining restaurants. McLane provides wholesale distribution services to Walmart, which accounted for approximately 18% of McLane's revenues in 2020. McLane's other significant customers include 7-Eleven (approximately 13% of revenues) and Yum! Brands, (approximately 11% of revenues). McLane's business model is based on a high volume of sales, rapid inventory turnover and stringent expense controls. Operations are currently divided into three business units: grocery distribution, foodservice distribution and beverage distribution.

McLane's grocery distribution unit, based in Temple, Texas, maintains a dominant market share within the convenience store industry and serves most of the national convenience store chains and major oil company retail outlets. Grocery operations provide products to approximately 50,000 retail locations nationwide, including Walmart. McLane's grocery distribution unit operates 25 distribution facilities in 20 states.

McLane's foodservice distribution unit, based in Carrollton, Texas, focuses on serving the quick service and casual dining restaurant industry with high quality, timely-delivered products. Operations are conducted through 46 facilities in 22 states. The foodservice distribution unit services approximately 33,200 restaurants nationwide.

Through its subsidiaries, McLane also operates wholesale distributors of distilled spirits, wine and beer. The beverage unit operates as Empire Distributors and operations are conducted through 14 distribution centers in Georgia, North Carolina, Tennessee and Colorado. Empire Distributors services approximately 25,600 retail locations in the southeastern United States and Colorado.

FlightSafety International

FlightSafety International Inc. ("FlightSafety") is an industry leading provider of professional aviation training services and flight simulation products. FlightSafety and FlightSafety Textron Aviation Training, a joint venture with Textron which began operations in 2019, provide high technology training to pilots, aircraft maintenance technicians, flight attendants and dispatchers who operate and support a wide variety of business, commercial and military aircraft. The training is provided using a large fleet of advanced full flight simulators at learning centers and training locations in the United States, Australia, Brazil, Canada, France, Hong Kong, Japan, Norway, South Africa and the United Kingdom. The vast majority of the instructors, training programs and flight simulators are qualified by the United States Federal Aviation Administration and other aviation regulatory agencies around the world.

FlightSafety is also a leader in the design and manufacture of full flight simulators, visual systems, displays and other advanced technology training devices. This equipment is used to support FlightSafety training programs and is offered for sale to airlines and government and military organizations around the world. Manufacturing facilities are located in Oklahoma, Missouri and Texas. FlightSafety strives to maintain and manufacture simulators and develop courseware using state-of-the-art technology and invests in research and development as it builds new equipment and training programs.

NetJets

NetJets Inc. ("NetJets") is the world's leading provider of shared ownership programs for general aviation aircraft. NetJets' global headquarters is located in Columbus, Ohio, with most of its logistical and flight operations based at John Glenn Columbus International Airport. NetJets' European operations are based in Lisbon, Portugal. The shared ownership concept is designed to meet the travel needs of customers who require the scale, flexibility and access of a large fleet that whole aircraft ownership cannot deliver. In addition, shared ownership programs are available for corporate flight departments seeking to outsource their general aviation needs or add capacity for peak periods and for others that previously chartered aircraft.

With a focus on safety and service, NetJets' programs are designed to offer customers guaranteed availability of aircraft, predictable operating costs and increased liquidity. NetJets' shared aircraft ownership programs permit customers to acquire a specific percentage of a certain aircraft type and allows customers to utilize the aircraft for a specified number of flight hours annually. In addition, NetJets offers prepaid flight cards and other aviation solutions and services for aircraft management, customized aircraft sales and acquisition, ground support and flight operation services under a number of programs including NetJets Shares™, NetJets Leases™ and the Marquis Jet Card®.

NetJets is subject to the rules and regulations of the United States Federal Aviation Administration, the Portuguese Civil Aviation Authority and the European Union Aviation Safety Agency. Regulations address aircraft registration, maintenance requirements, pilot qualifications and airport operations, including flight planning and scheduling as well as security issues and other matters. NetJets maintains a comprehensive training and development program in compliance with regulatory requirements for pilots, flight attendants, maintenance mechanics, and other flight operations specialists.

TTI, Inc.

TTI, Inc. ("TTI"), headquartered in Fort Worth, Texas, is a global specialty distributor of passive, interconnect, electromechanical, discrete, and semiconductor components used by customers in the manufacturing and assembling of electronic products. TTI's customer base includes original equipment manufacturers, electronic manufacturing services, original design manufacturers and military and commercial customers, as well as design and system engineers. TTI's distribution agreements with the industry's leading suppliers allow it to uniquely leverage its product cost and to expand its business by providing new lines and products to its customers. TTI operates sales offices and distribution centers from more than 100 locations throughout North America, Europe, Asia and Israel.

TTI services a variety of industries including telecommunications, medical devices, computers and office equipment, military/aerospace, automotive and industrial electronics. TTI's core customers include businesses in the design through production stages in the electronic component supply chain, which supports its high-volume business, and its Mouser subsidiary, which supports a broader base of customers with lower volume purchases through internet-based marketing.

Other services

XTRA Corporation ("XTRA"), headquartered in St. Louis, Missouri, is a leading transportation equipment lessor operating under the XTRA Lease® brand name. XTRA manages a diverse fleet of approximately 86,000 units located at 48 facilities throughout the United States. The fleet includes over-the-road and storage trailers, chassis, temperature-controlled vans and flatbed trailers. XTRA is one of the largest lessors (in terms of units available) of over-the-road trailers in North America. Transportation equipment customers lease equipment to cover cyclical, seasonal and geographic needs and as a substitute for purchasing equipment. Therefore, as a provider of marginal capacity to its customers, XTRA's utilization rates and operating results tend to be cyclical. In addition, transportation providers often use leasing to maximize their asset utilization and reduce capital expenditures. By maintaining a large fleet, XTRA is able to provide customers with a broad selection of equipment and quick response times.

International Dairy Queen develops and services a worldwide system of over 7,000 franchised restaurants operating primarily under the names DQ Grill and Chill®, Dairy Queen® and Orange Julius® that offer various dairy desserts, beverages, prepared foods and blended fruit drinks. Business Wire provides electronic dissemination of full-text news releases to the media, online services and databases and the global investment community in 150 countries and in 45 languages. Approximately 93% of Business Wire's revenues derive from its core news distribution business. CORT Business Services Corporation is a leading national provider of rental furniture and related services in the "rent-to-rent" segment of the furniture rental industry. CORT's primary revenue streams include furniture rental to individuals, businesses, government agencies, the trade show and events industry and retail sales of used furniture. WPLG, Inc. is an ABC affiliate broadcast station in Miami, Florida and Charter Brokerage is a leading non-asset based third party logistics provider to the petroleum and chemical industries. Until March 2020, other services included the newspaper publishing businesses conducted through The Buffalo News and BH Media Group, Inc. These operations were sold in 2020.

Retailing Businesses

Berkshire's retailing businesses include automotive, home furnishings and several other operations that sell various consumer products to consumers. Information regarding each of these operations follows. Berkshire's retailing businesses employed approximately 25,000 people at the end of 2020.

Berkshire Hathaway Automotive

The Berkshire Hathaway Automotive Group, Inc. ("BHA") is one of the largest automotive retailers in the United States, currently operating 104 new vehicle franchises through 81 dealerships located primarily in major metropolitan markets in the United States. The dealerships sell new and used vehicles, vehicle maintenance and repair services, extended service contracts, vehicle protection products and other aftermarket products. BHA also arranges financing for its customers through third-party lenders. BHA operates 29 collision centers directly connected to the dealerships' operations and owns and operates two auto auctions and a fluid maintenance products distribution company.

Dealership operations are highly concentrated in the Arizona and Texas markets, with approximately 70% of dealership-related revenues derived from sales in these markets. BHA currently maintains franchise agreements with 27 different vehicle manufacturers, although it derives a significant portion of its revenue from the Toyota/Lexus, General Motors, Ford/Lincoln, Nissan/Infiniti and Honda/Acura brands. Approximately 90% of BHA's annual revenues are from dealerships representing these manufacturers.

The retail automotive industry is highly competitive. BHA faces competition from other large public and private dealership groups, as well as individual franchised dealerships and competition via the Internet. Given the pricing transparency available via the Internet, and the fact that franchised dealers acquire vehicles from the manufacturers on the same terms irrespective of volume, the location and quality of the dealership facility, customer service and transaction speed are key differentiators in attracting customers.

BHA's overall relationships with the automobile manufacturers are governed by framework agreements. The framework agreements contain provisions relating to the management, operation, acquisition and the ownership structure of BHA's dealerships. Failure to meet the terms of these agreements could adversely impact BHA's ability to acquire additional dealerships representing those manufacturers. Additionally, these agreements contain limitations on the number of dealerships from a specific manufacturer that may be owned by BHA.

Individual dealerships operate under franchise agreements with the manufacturer, which grants the dealership entity a non-exclusive right to sell the manufacturer's brand of vehicles and offer related parts and service within a specified market area, as well as the right to use the manufacturer's trademarks. The agreements contain various requirements and restrictions related to the management and operation of the franchised dealership and provide for termination of the agreement by the manufacturer or non-renewal for a variety of causes. The states generally have automotive dealership franchise laws that provide substantial protection to the franchisee, and it is difficult for a manufacturer to terminate or not renew a franchise agreement outside of bankruptcy or with "good cause" under the applicable state franchise law.

BHA also develops, underwrites and administers various vehicle protection plans as well as life and accident and health insurance plans sold to consumers through BHA's dealerships and third-party dealerships. BHA also develops proprietary training programs and materials and provides ongoing monitoring and training of the dealership's finance and insurance personnel.

Home furnishings retailing

The home furnishings businesses consist of Nebraska Furniture Mart ("NFM"), R.C. Willey Home Furnishings ("R.C. Willey"), Star Furniture Company ("Star") and Jordan's Furniture, Inc. ("Jordan's"). These businesses offer a wide selection of furniture, bedding and accessories. In addition, NFM and R.C. Willey sell a full line of major household appliances, electronics, computers and other home furnishings and offer customer financing to complement their retail operations. An important feature of each of these businesses is their ability to control costs and to produce high business volume by offering significant value to their customers.

NFM operates its business from four retail complexes with almost 4.5 million square feet of retail, warehouse and administrative facilities located in Omaha, Nebraska, Clive, Iowa, Kansas City, Kansas and The Colony, Texas. NFM also owns Homemakers Furniture located in Urbandale, Iowa, which has approximately 600,000 square feet of retail, warehouse and administrative space. NFM is the largest furniture retailer in each of these markets. R.C. Willey, based in Salt Lake City, Utah, currently operates twelve full-line retail home furnishings stores and three distribution centers. These facilities include approximately 1.5 million square feet of retail space with six stores located in Utah, one store in Meridian, Idaho, three stores in Nevada (Las Vegas and Reno) and two stores in the Sacramento, California area.

Jordan's operates a retail furniture business from seven locations with approximately 890,000 square feet of retail space in stores located in Massachusetts, New Hampshire, Rhode Island, Maine and Connecticut. The retail stores are supported by an 800,000 square foot distribution center in Taunton, Massachusetts. Jordan's is the largest furniture retailer, as measured by sales, in Massachusetts and New Hampshire. Jordan's is well known in its markets for its unique store arrangements and advertising campaigns. Star has operated home furnishings retail stores in Texas for many years. Star's retail facilities currently include about 700,000 square feet of retail space in 11 locations in Texas, including eight in Houston.

Other retailing

Borsheim Jewelry Company, Inc. ("Borsheims") operates from a single store in Omaha, Nebraska. Borsheims is a high-volume retailer of fine jewelry, watches, crystal, china, stemware, flatware, gifts and collectibles. Helzberg's Diamond Shops, LLC. ("Helzberg") is based in North Kansas City, Missouri, and operates a chain of 213 retail jewelry stores in 36 states, which includes approximately 500,000 square feet of retail space. Helzberg's stores are located in malls, lifestyle centers, power strip centers and outlet malls, and all stores operate under the name Helzberg Diamonds® or Helzberg Diamonds Outlet®. The Ben Bridge Corporation ("Ben Bridge Jeweler"), based in Seattle, Washington, operates 75 retail jewelry stores under three different brand names, located primarily in major shopping malls in 10 western states and in British Columbia, Canada. Thirty-six of its retail locations are upscale jewelry stores selling loose diamonds, finished jewelry and high-end timepieces. Thirty-eight of its retail locations are concept stores operating under a franchise agreement that sell only Pandora jewelry. One store is a Breitling concept store, selling only Breitling timepieces.

See's Candies ("See's") produces boxed chocolates and other confectionery products with an emphasis on quality and distinctiveness in two large kitchens in Los Angeles and San Francisco and one smaller facility in Burlingame, California. See's operates approximately 250 retail and quantity discount stores located mainly in California and other Western states, as well as over 110 seasonal in-line locations. See's revenues are highly seasonal with approximately half of its annual revenues earned in the fourth quarter.

The Pampered Chef, Ltd. ("Pampered Chef") is a premier direct seller of distinctive high-quality kitchenware products with sales and operations in the United States, Canada, Germany, Austria and France and operations in China. Pampered Chef's product portfolio consists of approximately 650 Pampered Chef® branded kitchenware items in categories ranging from stoneware and cutlery to grilling and entertaining. Pampered Chef's products are available through its sales force of independent cooking consultants and online.

Oriental Trading Company ("OTC") is a leading multi-channel retailer and online destination for value-priced party supplies, arts and crafts, toys and novelties, school supplies, educational games, patient giveaways and personalized products. OTC, headquartered in Omaha, Nebraska, serves a broad base of nearly four million customers annually, including consumers, schools, churches, non-profit organizations, medical and dental offices and other businesses. OTC offers a unique assortment of over 50,000 products and utilizes sophisticated digital and print marketing efforts to drive significant traffic and industry leading customer satisfaction.

Detlev Louis Motorrad ("Louis"), headquartered in Hamburg, Germany, is a leading retailer of motorcycle apparel and equipment in Europe. Louis carries over 32,000 different products from more than 600 manufacturers, primarily covering the clothing, technical equipment and leisure markets. Louis has over 80 stores in Germany, Austria, Switzerland and the Netherlands and also sells through catalogs and via the Internet throughout most of Europe.

Additional information with respect to Berkshire's businesses

Revenue, earnings before taxes and identifiable assets attributable to Berkshire's reportable business segments are included in Note 27 to Berkshire's Consolidated Financial Statements contained in Item 8, Financial Statements and Supplementary Data. Additional information regarding Berkshire's investments in fixed maturity and equity securities is included in Notes 3 and 4, respectively, to Berkshire's Consolidated Financial Statements.

Berkshire owns 26.6% of the outstanding common stock of The Kraft Heinz Company ("Kraft Heinz"). Kraft Heinz is one of the largest food and beverage companies in the world, with sales in numerous countries within developed and emerging markets and territories. Kraft Heinz manufactures and markets food and beverage products, including condiments and sauces, cheese and dairy meals, meats, refreshment beverages, coffee and other grocery products, throughout the world, under a diverse mix of iconic and emerging brands. Berkshire subsidiaries also own a 50% joint venture interest in Berkadia Commercial Mortgage LLC ("Berkadia"), a 38.6% interest in Pilot Travel Centers LLC ("Pilot") and a 50% joint venture interest in Electric Transmission Texas, LLC ("ETT"). Information concerning these investments is included in Note 5 to Berkshire's Consolidated Financial Statements.

Berkshire maintains a website (*http://www.berkshirehathaway.com*) where its annual reports, certain corporate governance documents, press releases, interim shareholder reports and links to its subsidiaries' websites can be found. Berkshire's periodic reports filed with the SEC, which include Form 10-K, Form 10-Q, Form 8-K and amendments thereto, may be accessed by the public free of charge from the SEC and through Berkshire. Electronic copies of these reports can be accessed at the SEC's website (*http://www.sec.gov*) and indirectly through Berkshire's website (*http://www.berkshirehathaway.com*). Copies of these reports may also be obtained, free of charge, upon written request to: Berkshire Hathaway Inc., 3555 Farnam Street, Omaha, NE 68131, Attn: Corporate Secretary.

Item 1A. Risk Factors

Berkshire and its subsidiaries (referred to herein as "we," "us," "our" or similar expressions) are subject to certain risks and uncertainties in its business operations which are described below. The risks and uncertainties described below are not the only risks we face. Additional risks and uncertainties that are presently unknown or are currently deemed immaterial may also impair our business operations.

General Business Risks

Terrorist acts could hurt our operating businesses.

A cyber, biological, nuclear or chemical attack could produce significant losses to our worldwide operations. Our business operations could be adversely affected from such acts through the loss of human resources or destruction of production facilities and information systems. We share the risk with all businesses.

Cyber security risks

We rely on technology in virtually all aspects of our business. Like those of many large businesses, certain of our information systems have been subject to computer viruses, malicious codes, unauthorized access, phishing efforts, denial-of-service attacks and other cyber-attacks and we expect to be subject to similar attacks in the future as such attacks become more sophisticated and frequent. A significant disruption or failure of our technology systems could result in service interruptions, safety failures, security events, regulatory compliance failures, an inability to protect information and assets against unauthorized users and other operational difficulties. Attacks perpetrated against our systems could result in loss of assets and critical information and expose us to remediation costs and reputational damage.

Although we have taken steps intended to mitigate these risks, including business continuity planning, disaster recovery planning and business impact analysis, a significant disruption or cyber intrusion could adversely affect our results of operations, financial condition and liquidity. Additionally, if we are unable to acquire, develop, implement, adopt or protect rights around new technology, we may suffer a competitive disadvantage, which could also have an adverse effect on our results of operations, financial condition and/or liquidity.

Cyber-attacks could further adversely affect our ability to operate facilities, information technology and business systems or compromise confidential customer and employee information. Political, economic, social or financial market instability or damage to or interference with our operating assets, customers or suppliers from cyber-attacks may result in business interruptions, lost revenues, higher commodity prices, disruption in fuel supplies, lower energy consumption, unstable markets, increased security, repair or other costs, or may materially adversely affect us in ways that cannot be predicted at this time. Any of these risks could materially affect our consolidated financial results. Furthermore, instability in the financial markets resulting from terrorism, sustained or significant cyber-attacks or war could also have a material adverse effect on our ability to raise capital. We share these risks with all businesses.

We are dependent on a few key people for our major investment and capital allocation decisions.

Major investment decisions and all major capital allocation decisions are made by Warren E. Buffett, Chairman of the Board of Directors and Chief Executive Officer, age 90, in consultation with Charles T. Munger, Vice Chairman of the Board of Directors, age 97. If for any reason the services of our key personnel, particularly Mr. Buffett, were to become unavailable, there could be a material adverse effect on our operations. However, Berkshire's Board of Directors has identified certain current Berkshire managers who, in their judgment, are capable of succeeding Mr. Buffett and has agreed on a replacement for Mr. Buffett should a replacement be needed currently. The Board continually monitors this risk and could alter its current view regarding a replacement for Mr. Buffett in the future. We believe that the Board's succession plan, together with the outstanding managers running our numerous and highly diversified operating units helps to mitigate this risk. In 2018, Berkshire's Board of Directors appointed Mr. Gregory Abel as Vice Chairman of Berkshire's non-insurance operations and Mr. Ajit Jain as Vice Chairman of Berkshire's insurance operations. Mr. Abel and Mr. Jain each report directly to Mr. Buffett and Mr. Buffett continues to be responsible for major capital allocation and investment decisions.

We need qualified personnel to manage and operate our various businesses.

In our decentralized business model, we need qualified and competent management to direct day-to-day business activities of our operating subsidiaries and to manage changes in future business operations due to changing business or regulatory environments. Our operating subsidiaries also need qualified and competent personnel in executing their business plans and serving their customers, suppliers and other stakeholders. Our inability to recruit and retain qualified and competent managers and personnel could negatively affect the operating results, financial condition and liquidity of our subsidiaries and Berkshire as a whole.

Investments are unusually concentrated in equity securities and fair values are subject to loss in value.

We concentrate a high percentage of the equity security investments of our insurance subsidiaries in a relatively small number of equity securities. A significant decline in the fair values of our larger investments in equity securities may produce a material decline in our consolidated shareholders' equity and our consolidated earnings.

Since a large percentage of our equity securities are held by our insurance subsidiaries, significant decreases in the fair values of these investments will produce significant declines in the statutory surplus of our insurance business. Our large statutory surplus is a competitive advantage, and a long-term material decline could have an adverse effect on our claims-paying ability ratings and our ability to write new insurance business thus potentially reducing our future underwriting profits.

Over ten years ago, we assumed the risk of potentially significant losses under a number of equity index put option contracts, which contain equity price risks. Most of the contracts remaining at year end 2020 will expire by February 2023. Risks of losses under these contracts are based on declines in equity prices of stocks comprising certain major U.S. and international stock indexes. We received considerable cash premiums as compensation for accepting these risks. Absent major reductions in future equity securities prices, our ultimate payment obligations are not likely to be significant. Nevertheless, there can be no assurance that equity securities prices will not decline significantly resulting in significant settlement payments upon contract expirations.

Competition and technology may erode our business franchises and result in lower earnings.

Each of our operating businesses face intense competition within markets in which they operate. While we manage our businesses with the objective of achieving long-term sustainable growth by developing and strengthening competitive advantages, many factors, including technological changes, may erode or prevent the strengthening of competitive advantages. Accordingly, our future operating results will depend to some degree on our operating units successfully protecting and enhancing their competitive advantages. If our operating businesses are unsuccessful in these efforts, our periodic operating results in the future may decline.

Unfavorable general economic conditions may significantly reduce our operating earnings and impair our ability to access capital markets at a reasonable cost.

Our operating businesses are subject to normal economic cycles affecting the general economy or the specific industries in which they operate. Significant deteriorations of economic conditions over a prolonged period could produce a material adverse effect on one or more of our significant operations. In addition, our utilities and energy businesses and our railroad business regularly utilize debt as a component of their capital structures and depend on having access to borrowed funds through the capital markets at reasonable rates. To the extent that access to the capital markets is restricted or the cost of funding increases, these operations could be adversely affected.

Epidemics, pandemics or other outbreaks, including COVID-19, could hurt our operating businesses.

The outbreak of COVID-19 has adversely affected, and in the future it or other epidemics, pandemics or outbreaks may adversely affect, our operations, including our equity securities portfolio. This is or may be due to closures or restrictions requested or mandated by governmental authorities, disruption to supply chains and workforce, reduction of demand for our products and services, credit losses when customers and other counterparties fail to satisfy their obligations to us, and volatility in global equity securities markets, among other factors. We share most of these risks with all businesses.

Regulatory changes may adversely impact our future operating results.

Over time, in response to financial markets crises, global economic recessions, and social and environmental issues, regulatory initiatives were adopted in the United States and elsewhere. Such initiatives addressed for example, the regulation of banks and other major financial institutions, products and environmental and global-warming matters. These initiatives impact all of our businesses, albeit in varying ways. Increased regulatory compliance costs could have a significant negative impact on our operating businesses, as well as on the businesses in which we have a significant, but not controlling economic interests. We cannot predict whether such initiatives will have a material adverse impact on our consolidated financial position, results of operations and/or cash flows.

Data privacy regulations have recently been enacted in various jurisdictions in the U.S. and throughout the world. These regulations address numerous aspects related to the security of personal information that is stored in our information systems, networks and facilities. Failure to comply with these regulations could result in reputational damage and significant penalties.

Risks unique to our regulated businesses

Our tolerance for risk in our insurance businesses may result in significant underwriting losses.

When properly paid for the risk assumed, we have been and will continue to be willing to assume more risk from a single event than any other insurer has knowingly assumed. Accordingly, we could incur a significant loss from a single catastrophe event resulting from a natural disaster or man-made catastrophes such as terrorism or cyber-attacks. We employ various disciplined underwriting practices intended to mitigate potential losses and attempt to take into account all possible correlations and avoid writing groups of policies from which pre-tax losses from a single catastrophe event might aggregate above $10 billion. Currently, we estimate that our aggregate exposure from a single event under outstanding policies is significantly below $10 billion. However, despite our efforts, it is possible that losses could manifest in ways that we do not anticipate and that our risk mitigation strategies are not designed to address. Additionally, various provisions of our policies, such as limitations or exclusions from coverage, negotiated to limit our risks, may not be enforceable in the manner we intend. Our tolerance for significant insurance losses may result in lower reported earnings in a future period.

The degree of estimation error inherent in the process of estimating property and casualty insurance loss reserves may result in significant underwriting losses.

The principal cost associated with the property and casualty insurance business is claims. In writing property and casualty insurance policies, we receive premiums today and promise to pay covered losses in the future. However, it will take decades before all claims that have occurred as of any given balance sheet date will be reported and settled. Although we believe that liabilities for unpaid losses are adequate, we will not know whether these liabilities or the premiums charged for the coverages provided were sufficient until well after the balance sheet date. Estimating insurance claim costs is inherently imprecise. Our estimated unpaid losses arising under contracts covering property and casualty insurance risks are large ($120.8 billion at December 31, 2020), and a small percentage increase to those liabilities can result in materially lower reported earnings.

Changes in regulations and regulatory actions can adversely affect our operating results and our ability to allocate capital.

Our insurance businesses are subject to regulation in the jurisdictions in which we operate. Such regulations may relate to among other things, the types of business that can be written, the rates that can be charged for coverage, the level of capital that must be maintained, and restrictions on the types and size of investments that can be made. Regulations may also restrict the timing and amount of dividend payments to Berkshire by these businesses. U.S. state insurance regulators and international insurance regulators are also actively developing various regulatory mechanisms to address the regulation of large internationally active insurance groups, including regulations concerning group capital, liquidity, governance and risk management. Accordingly, changes in regulations related to these or other matters or regulatory actions imposing restrictions on our insurance businesses may adversely impact our results of operations and restrict our ability to allocate capital.

Our railroad business conducted through BNSF is also subject to a significant number of laws and regulations with respect to rates and practices, taxes, railroad operations and a variety of health, safety, labor, environmental and other matters. Failure to comply with applicable laws and regulations could have a material adverse effect on BNSF's business. Governments may change the legislative and/or regulatory framework within which BNSF operates, without providing any recourse for any adverse effects that the change may have on the business. Complying with legislative and regulatory changes may pose significant operating and implementation risks and require significant capital expenditures.

BNSF derives significant amounts of revenue from the transportation of energy-related commodities, particularly coal. To the extent that changes in government policies limit or restrict the usage of coal as a source of fuel in generating electricity or alternate fuels, such as natural gas, or displace coal on a competitive basis, revenues and earnings could be adversely affected. As a common carrier, BNSF is also required to transport toxic inhalation hazard chemicals and other hazardous materials. A release of hazardous materials could expose BNSF to significant claims, losses, penalties and environmental remediation obligations. Changes in the regulation of the rail industry could negatively impact BNSF's ability to determine prices for rail services and to make capital improvements to its rail network, resulting in an adverse effect on our results of operations, financial condition and/or liquidity.

Our utilities and energy businesses operated under BHE are highly regulated by numerous federal, state, local and foreign governmental authorities in the jurisdictions in which they operate. These laws and regulations are complex, dynamic and subject to new interpretations or change. Regulations affect almost every aspect of our utilities and energy businesses. Regulations broadly apply and may limit management's ability to independently make and implement decisions regarding numerous matters including: acquiring businesses; constructing, acquiring, disposing or retiring of operating assets; operating and maintaining generating facilities and transmission and distribution system assets; complying with pipeline safety and integrity and environmental requirements; setting rates charged to customers; establishing capital structures and issuing debt; transacting between our domestic utilities and our other subsidiaries and affiliates; and paying dividends or similar distributions. Failure to comply with or reinterpretations of existing regulations and new legislation or regulations, such as those relating to air and water quality, renewable portfolio standards, emissions performance standards, climate change, coal combustion byproduct disposal, hazardous and solid waste disposal, protected species and other environmental matters, or changes in the nature of the regulatory process may have a significant adverse impact on our financial results.

Our railroad business requires significant ongoing capital investment to improve and maintain its railroad network so that transportation services can be safely and reliably provided to customers on a timely basis. Our utilities and energy businesses also require significant amounts of capital to construct, operate and maintain generation, transmission and distribution systems to meet their customers' needs and reliability criteria. Additionally, system assets may need to be operational for long periods of time in order to justify the financial investment. The risk of operational or financial failure of capital projects is not necessarily recoverable through rates that are charged to customers. Further, a significant portion of costs of capital improvements may be funded through debt issued by BNSF and BHE and their subsidiaries. Disruptions in debt capital markets that restrict access to funding when needed could adversely affect the results of operations, liquidity and/or capital resources of these businesses.

Item 1B. Unresolved Staff Comments

None.

Item 2. Description of Properties

The properties used by Berkshire's business segments are summarized in this section. Berkshire's railroad and utilities and energy businesses, in particular, utilize considerable physical assets in their businesses.

Railroad Business—Burlington Northern Santa Fe

Through BNSF Railway, BNSF operates approximately 32,500 route miles of track (excluding multiple main tracks, yard tracks and sidings) in 28 states, and also operates in three Canadian provinces. BNSF owns over 23,000 route miles, including easements, and operates over 9,000 route miles of trackage rights that permit BNSF to operate its trains with its crews over other railroads' tracks. As of December 31, 2020, the total BNSF Railway system, including single and multiple main tracks, yard tracks and sidings, consisted of over 50,000 operated miles of track.

BNSF operates various facilities and equipment to support its transportation system, including its infrastructure, locomotives and freight cars. It also owns or leases other equipment to support rail operations, such as vehicles. Support facilities for rail operations include yards and terminals throughout its rail network, system locomotive shops to perform locomotive servicing and maintenance, a centralized network operations center for train dispatching and network operations monitoring and management, computers, telecommunications equipment, signal systems and other support systems. Transfer facilities are maintained for rail-to-rail as well as intermodal transfer of containers, trailers and other freight traffic and include approximately 25 intermodal hubs located across the system. BNSF owns or holds under non-cancelable leases exceeding one year approximately 7,700 locomotives and 66,000 freight cars, in addition to maintenance of way and other equipment.

In the ordinary course of business, BNSF incurs significant costs in repairing and maintaining its properties. In 2020, BNSF recorded approximately $2 billion in repairs and maintenance expense.

Utilities and Energy Businesses—Berkshire Hathaway Energy

BHE's energy properties consist of the physical assets necessary to support its electricity and natural gas businesses. Properties of BHE's electricity businesses include electric generation, transmission and distribution facilities, as well as coal mining assets that support certain of BHE's electric generating facilities. Properties of BHE's natural gas businesses include natural gas distribution facilities, interstate pipelines, storage facilities, liquefied natural gas facilities, compressor stations and meter stations. The transmission and distribution assets are primarily within each of BHE's utility service territories. In addition to these physical assets, BHE has rights-of-way, mineral rights and water rights that enable BHE to utilize its facilities. Pursuant to separate financing agreements, the majority of these properties are pledged or encumbered to support or otherwise provide the security for the related subsidiary debt. BHE or its affiliates own or have interests in the following types of operating electric generating facilities at December 31, 2020:

Energy Source	Entity	Location by Significance	Facility Net Capacity (MW) (1)	Net Owned Capacity (MW) (1)
Natural gas	PacifiCorp, MEC, NV Energy and BHE Renewables	Nevada, Utah, Iowa, Illinois, Washington, Wyoming, Oregon, Texas, New York and Arizona	11,171	10,892
Wind	PacifiCorp, MEC and BHE Renewables	Iowa, Wyoming, Texas, Nebraska, Washington, California, Illinois, Oregon, Kansas and Montana	10,302	10,302
Coal	PacifiCorp, MEC and NV Energy	Wyoming, Iowa, Utah, Nevada, Colorado and Montana	13,249	8,198
Solar	BHE Renewables and NV Energy	California, Texas, Arizona, Minnesota and Nevada	1,699	1,551
Hydroelectric	PacifiCorp, MEC and BHE Renewables	Washington, Oregon, The Philippines, Idaho, California, Utah, Hawaii, Montana, Illinois and Wyoming	1,299	1,277
Nuclear	MEC	Illinois	1,815	454
Geothermal	PacifiCorp and BHE Renewables	California and Utah	377	377
		Total	39,912	33,051

(1) Facility Net Capacity in megawatts (MW) represents the lesser of nominal ratings or any limitations under applicable interconnection, power purchase, or other agreements for intermittent resources and the total net dependable capability available during summer conditions for all other units. An intermittent resource's nominal rating is the manufacturer's contractually specified capability (in MW) under specified conditions. Net Owned Capacity indicates BHE's ownership of Facility Net Capacity.

As of December 31, 2020, BHE's subsidiaries also have electric generating facilities that are under construction in Iowa, Wyoming and Montana having total Facility Net Capacity and Net Owned Capacity of 603 MW.

PacifiCorp, MEC and NV Energy own electric transmission and distribution systems, including approximately 27,600 miles of transmission lines and approximately 1,650 substations and gas distribution facilities, including approximately 27,600 miles of gas mains and service lines.

Northern Powergrid (Northeast) and Northern Powergrid (Yorkshire) operate an electricity distribution network that includes approximately 17,300 miles of overhead lines, approximately 42,800 miles of underground cables and approximately 770 major substations. AltaLink's electricity transmission system includes approximately 8,200 miles of transmission lines and approximately 310 substations.

The BHE GT&S pipeline system consists of approximately 5,400 miles of natural gas transmission, gathering and storage pipelines. BHE GT&S provides natural gas storage and transportation service to on-system customers in Maryland, New York, Ohio, Pennsylvania, South Carolina, Virginia and West Virginia. Additionally, through multiple interconnects with other pipelines, BHE GT&S provides services to off-system customers broadly in the Northeast, Southeast and Mid-Atlantic regions. Storage services are provided through the operation of 17 underground natural gas storage fields located in Pennsylvania, West Virginia and New York. BHE GT&S also operates, as the general partner, and owns a 25% limited partnership interest in one liquefied natural gas export, import and storage facility in Maryland and operates and has ownership interests in three modular liquefied natural gas facilities in Alabama, Florida and Pennsylvania.

Northern Natural's pipeline system consists of approximately 14,500 miles of natural gas pipelines, including approximately 6,000 miles of mainline transmission pipelines and approximately 8,500 miles of branch and lateral pipelines. Northern Natural's end-use and distribution market area includes points in Iowa, Nebraska, Minnesota, Wisconsin, South Dakota, Michigan and Illinois and its natural gas supply and delivery service area includes points in Kansas, Texas, Oklahoma and New Mexico. Storage services are provided through the operation of one underground natural gas storage field in Iowa, two underground natural gas storage facilities in Kansas and two liquefied natural gas storage peaking units, one in Iowa and one in Minnesota.

Kern River's system consists of approximately 1,400 miles of natural gas pipelines, which extends from the system's point of origination in Wyoming through the Central Rocky Mountains into California.

Other Segments

Significant physical properties used by Berkshire's other business segments are summarized below:

Business	Country	Locations	Property/Facility type	Number of Properties Owned	Number of Properties Leased
Insurance:					
GEICO	U.S.		Offices and claims centers	10	122
BHRG	U.S.		Offices	1	30
	Non-U.S.	Locations in 22 countries	Offices	1	37
BH Primary	U.S.		Offices	7	51
	Non-U.S.	Locations in 7 countries	Offices	—	16
Manufacturing	U.S.		Manufacturing facility	485	119
			Offices/Warehouses	207	443
			Retail/Showroom	261	213
			Housing communities	312	—
	Non-U.S.	Locations in 63 countries	Manufacturing facility	199	124
			Offices/Warehouses	88	448
			Retail/Showroom	—	4
Service	U.S.		Training facilities/Hangars	19	94
			Offices/Distribution	15	144
			Production facilities	4	3
			Leasing/Showroom/Retail	31	48
	Non-U.S.	Locations in 18 countries	Training facilities/Hangars	2	12
			Offices/Distribution	—	48
McLane Company	U.S.		Distribution centers	59	26
			Offices	4	1
Retailing	U.S.		Offices/Warehouses	21	26
			Retail/Showroom	142	543
	Non-U.S.	Locations in 6 countries	Offices/Warehouses	1	9
			Retail/Offices	—	93

Item 3. Legal Proceedings

Berkshire and its subsidiaries are parties in a variety of legal actions that routinely arise out of the normal course of business, including legal actions seeking to establish liability directly through insurance contracts or indirectly through reinsurance contracts issued by Berkshire subsidiaries. Plaintiffs occasionally seek punitive or exemplary damages. We do not believe that such normal and routine litigation will have a material effect on our financial condition or results of operations. Berkshire and certain of its subsidiaries are also involved in other kinds of legal actions, some of which assert or may assert claims or seek to impose fines and penalties. We believe that any liability that may arise as a result of other pending legal actions will not have a material effect on our consolidated financial condition or results of operations.

Item 4. Mine Safety Disclosures

Information regarding the Company's mine safety violations and other legal matters disclosed in accordance with Section 1503 (a) of the Dodd-Frank Reform Act is included in Exhibit 95 to this Form 10-K.

Executive Officers of the Registrant

Following is a list of the Registrant's named executive officers:

Name	Age	Position with Registrant	Since
Warren E. Buffett	90	Chairman and Chief Executive Officer	1970
Charles T. Munger	97	Vice Chairman	1978
Gregory E. Abel	58	Vice Chairman – Non-Insurance Operations	2018
Ajit Jain	69	Vice Chairman – Insurance Operations	2018
Marc D. Hamburg	71	Senior Vice-President – Chief Financial Officer	1992

Each executive officer serves, in accordance with the by-laws of the Registrant, until the first meeting of the Board of Directors following the next annual meeting of shareholders and until a successor is chosen and qualified or until such executive officer sooner dies, resigns, is removed or becomes disqualified.

FORWARD-LOOKING STATEMENTS

Investors are cautioned that certain statements contained in this document as well as some statements in periodic press releases and some oral statements of Berkshire officials during presentations about Berkshire or its subsidiaries are "forward-looking" statements within the meaning of the Private Securities Litigation Reform Act of 1995 (the "Act"). Forward-looking statements include statements which are predictive in nature, which depend upon or refer to future events or conditions, which include words such as "expects," "anticipates," "intends," "plans," "believes," "estimates" or similar expressions. In addition, any statements concerning future financial performance (including future revenues, earnings or growth rates), ongoing business strategies or prospects and possible future Berkshire actions, which may be provided by management, are also forward-looking statements as defined by the Act. Forward-looking statements are based on current expectations and projections about future events and are subject to risks, uncertainties and assumptions about Berkshire and its subsidiaries, economic and market factors and the industries in which we do business, among other things. These statements are not guarantees of future performance and we have no specific intention to update these statements.

Actual events and results may differ materially from those expressed or forecasted in forward-looking statements due to a number of factors. The principal risk factors that could cause our actual performance and future events and actions to differ materially from such forward-looking statements include, but are not limited to, changes in market prices of our investments in fixed maturity and equity securities, losses realized from derivative contracts, the occurrence of one or more catastrophic events, such as an earthquake, hurricane, act of terrorism or cyber attack that causes losses insured by our insurance subsidiaries and/or losses to our business operations, the frequency and severity of epidemics, pandemics or other outbreaks, including COVID-19, that negatively affect our operating results and restrict our access to borrowed funds through the capital markets at reasonable rates, changes in laws or regulations affecting our insurance, railroad, utilities and energy and finance subsidiaries, changes in federal income tax laws, and changes in general economic and market factors that affect the prices of securities or the industries in which we do business.

Part II

Item 5. Market for Registrant's Common Equity, Related Security Holder Matters and Issuer Purchases of Equity Securities

Market Information

Berkshire's Class A and Class B common stock are listed for trading on the New York Stock Exchange, trading symbols: BRK.A and BRK.B, respectively.

Shareholders

Berkshire had approximately 1,600 record holders of its Class A common stock and 18,900 record holders of its Class B common stock at February 16, 2021. Record owners included nominees holding at least 351,000 shares of Class A common stock and 1,332,000,000 shares of Class B common stock on behalf of beneficial-but-not-of-record owners.

Dividends

Berkshire has not declared a cash dividend since 1967.

Common Stock Repurchase Program

Berkshire's common stock repurchase program permits Berkshire to repurchase its Class A and Class B shares at any time that Warren Buffett, Berkshire's Chairman of the Board and Chief Executive Officer, and Charles Munger, Vice Chairman of the Board, believe that the repurchase price is below Berkshire's intrinsic value, conservatively determined. Repurchases may be in the open market or through privately negotiated transactions. Information with respect to Berkshire's Class A and Class B common stock repurchased during the fourth quarter of 2020 follows.

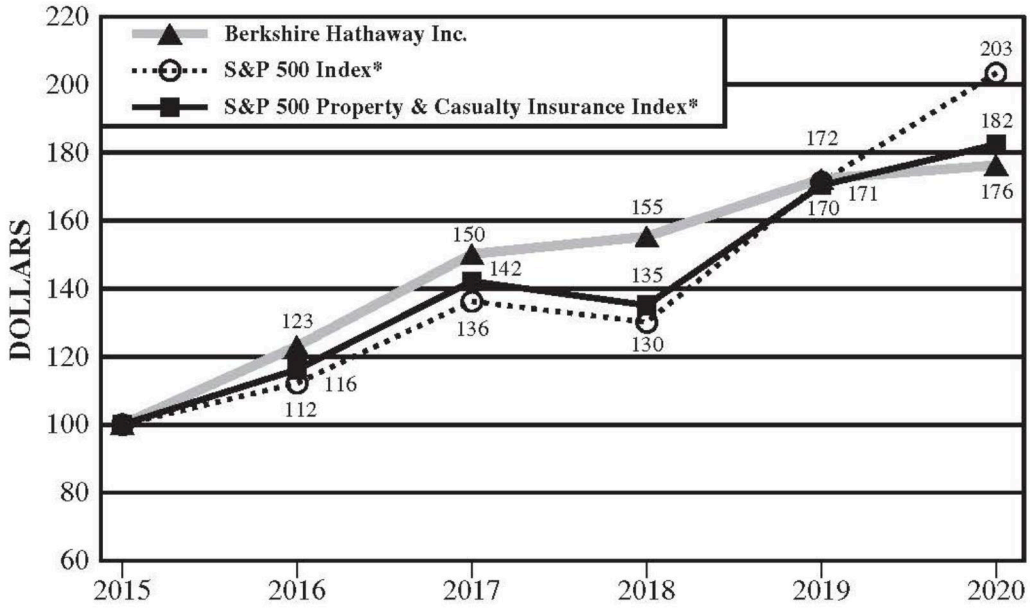
Period	Total number of shares purchased	Average price paid per share	Total number of shares purchased as part of publicly announced program	Maximum number or value of shares that yet may be repurchased under the program
October				
Class A common stock	1,894	$ 316,292.44	1,894	*
Class B common stock	11,097,536	$ 209.92	11,097,536	*
November				
Class A common stock	2,244	$ 341,117.06	2,244	*
Class B common stock	7,423,729	$ 219.12	7,423,729	*
December				
Class A common stock	1,787	$ 342,577.29	1,787	*
Class B common stock	12,605,335	$ 225.73	12,605,335	*

* *The program does not specify a maximum number of shares to be repurchased or obligate Berkshire to repurchase any specific dollar amount or number of Class A or Class B shares and there is no expiration date to the repurchase program. Berkshire will not repurchase its common stock if the repurchases reduce the total value of Berkshire's consolidated cash, cash equivalents and U.S. Treasury Bills holdings to less than $20 billion.*

Market for Registrant's Common Equity, Related Security Holder Matters and Issuer Purchases of Equity Securities *(Continued)*

Stock Performance Graph

The following chart compares the subsequent value of $100 invested in Berkshire common stock on December 31, 2015 with a similar investment in the Standard & Poor's 500 Stock Index and in the Standard & Poor's Property – Casualty Insurance Index**.



* *Cumulative return for the Standard & Poor's indices based on reinvestment of dividends.*

** *It would be difficult to develop a peer group of companies similar to Berkshire. The Corporation owns subsidiaries engaged in a number of diverse business activities of which the most important is the property and casualty insurance business and, accordingly, management has used the Standard & Poor's Property—Casualty Insurance Index for comparative purposes.*

Item 6. Selected Financial Data

Selected Financial Data for the Past Five Years

(dollars in millions except per-share data)

	2020	2019	2018	2017	2016
Revenues:					
Insurance premiums earned	$ 63,401	$ 61,078	$ 57,418	$ 60,597	$ 45,881
Sales and service revenues	127,044	134,989	133,336	130,243	123,053
Leasing revenue	5,209	5,856	5,732	2,552	2,553
Railroad, utilities and energy revenues	41,764	43,453	43,673	40,005	37,447
Interest, dividend and other investment income	8,092	9,240	7,678	6,536	6,180
Total revenues	$ 245,510	$ 254,616	$ 247,837	$ 239,933	$ 215,114
Investment and derivative gains/losses	$ 40,746	$ 72,607	$ (22,455)	$ 2,128	$ 8,304
Earnings:					
Net earnings attributable to Berkshire Hathaway (1)	$ 42,521	$ 81,417	$ 4,021	$ 44,940	$ 24,074
Net earnings per share attributable to Berkshire Hathaway shareholders (2)	$ 26,668	$ 49,828	$ 2,446	$ 27,326	$ 14,645
Year-end data:					
Total assets	$ 873,729	$ 817,729	$ 707,794	$ 702,095	$ 620,854
Notes payable and other borrowings:					
Insurance and other	41,522	37,590	34,975	40,409	42,559
Railroad, utilities and energy	75,373	65,778	62,515	62,178	59,085
Berkshire Hathaway shareholders' equity	443,164	424,791	348,703	348,296	282,070
Class A equivalent common shares outstanding, in thousands	1,544	1,625	1,641	1,645	1,644
Berkshire Hathaway shareholders' equity per outstanding Class A equivalent common share	$ 287,031	$ 261,417	$ 212,503	$ 211,750	$ 171,542

(1) Includes after-tax investment and derivative gains/losses of $31.6 billion in 2020, $57.4 billion in 2019, $(17.7) billion in 2018, $1.4 billion in 2017 and $6.5 billion in 2016. Beginning in 2018, investment gains/losses include the changes in fair values of equity securities during the period. Previously, investment gains/losses of equity securities were recognized in earnings when securities were sold. Net earnings in 2017 includes a one-time net benefit of $29.1 billion attributable to the enactment of the Tax Cuts and Jobs Act of 2017.

(2) Represents net earnings per average equivalent Class A share outstanding. Net earnings per average equivalent Class B common share outstanding is equal to 1/1,500 of such amount.

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

Results of Operations

Net earnings attributable to Berkshire Hathaway shareholders for each of the past three years are disaggregated in the table that follows. Amounts are after deducting income taxes and exclude earnings attributable to noncontrolling interests (in millions).

	2020	2019	2018
Insurance – underwriting	$ 657	$ 325	$ 1,566
Insurance – investment income	5,039	5,530	4,554
Railroad	5,161	5,481	5,219
Utilities and energy	3,091	2,840	2,621
Manufacturing, service and retailing	8,300	9,372	9,364
Investment and derivative gains/losses	31,591	57,445	(17,737)
Other*	(11,318)	424	(1,566)
Net earnings attributable to Berkshire Hathaway shareholders	$ 42,521	$ 81,417	$ 4,021

* *Includes goodwill and indefinite-lived intangible asset impairment charges of $11.0 billion in 2020, $435 million in 2019 and $3.0 billion in 2018, which includes our share of charges recorded by Kraft Heinz.*

Through our subsidiaries, we engage in a number of diverse business activities. We manage our operating businesses on an unusually decentralized basis. There are few centralized or integrated business functions. Our senior corporate management team participates in and is ultimately responsible for significant capital allocation decisions, investment activities and the selection of the Chief Executive to head each of the operating businesses. The business segment data (Note 27 to the accompanying Consolidated Financial Statements) should be read in conjunction with this discussion.

As the COVID-19 pandemic accelerated beginning in the second half of March, most of our businesses were negatively affected, with the effects to date ranging from relatively minor to severe. Revenues and earnings of most of our manufacturing, service and retailing businesses declined considerably, and in certain instances severely, in the second quarter due to closures of facilities where crowds gather, such as retail stores, restaurants and entertainment venues as well as from public travel restrictions and from closures of certain of our businesses. In each of the third and fourth quarters of 2020, several of these businesses experienced significant increases in revenues and earnings as compared to the second quarter.

Our businesses that were deemed essential continued to operate through the pandemic, including our railroad, utilities and energy, insurance and certain of our manufacturing, wholesale distribution and service businesses. In response to the effects of the pandemic, our businesses implemented various business continuity plans to protect our employees and customers. Such plans include a variety of actions, such as temporarily closing certain retail stores, manufacturing facilities and service centers of businesses that were not subject to government mandated closure. Our businesses also implemented practices to protect employees while at work. Such practices included work-from-home, staggered or reduced work schedules, increased cleaning and sanitation of workspaces, providing employee health screenings, eliminating non-essential travel and face-to-face meetings and providing general health reminders intended to lower the risk of spreading COVID-19.

We also took actions in response to the economic losses from reductions in consumer demand for products and services we offer and our temporary inability to produce goods and provide services at certain of our businesses. These actions included employee furloughs, wage and salary reductions, capital spending reductions and other actions intended to help mitigate the economic losses and preserve capital and liquidity. Certain of our businesses undertook and may continue to undertake restructuring activities to resize their operations to better fit expected customer demand. We cannot reliably predict future economic effects of the pandemic or when business activities at all of our numerous and diverse operations will normalize. Nor can we predict how these events will alter the future consumption patterns of consumers and businesses we serve.

Our insurance businesses generated after-tax earnings from underwriting of $657 million in 2020, $325 million in 2019 and $1.6 billion in 2018. In each year, we generated underwriting earnings from primary insurance and underwriting losses from reinsurance. Insurance underwriting results included after-tax losses from significant catastrophe events of approximately $750 million in 2020, $800 million in 2019 and $1.3 billion in 2018. Underwriting results in 2020 also reflected the effects of the pandemic, arising from premium reductions from the GEICO Giveback program, reduced claims frequencies for private passenger automobile insurance and increased loss estimates for certain commercial insurance and property and casualty reinsurance business.

Results of Operations *(Continued)*

After-tax earnings from insurance investment income in 2020 declined $491 million (8.9%) versus 2019, reflecting lower interest income primarily attributable to declines in interest rates on our substantial holdings of cash and U.S. Treasury Bills. After-tax earnings from insurance investment income in 2019 increased 21.4% over 2018, attributable to increases in interest and dividend income.

After-tax earnings of our railroad business decreased 5.8% in 2020 as compared to 2019. Earnings in 2020 reflected lower railroad operating revenues from lower shipping volumes, attributable to the negative effects of the COVID-19 pandemic, partly offset by lower operating costs and the effects of productivity improvements. After-tax earnings of our utilities and energy business increased 8.8% as compared to 2019. The increase reflected increased tax benefits from renewable energy and increased earnings from the real estate brokerage business. Earnings in 2020 from our manufacturing, service and retailing businesses declined 11.4% versus 2019. The effects of the COVID-19 pandemic varied among our manufacturing businesses relative to significance and duration.

Other earnings included after-tax goodwill and indefinite-lived intangible asset impairment charges of $11.0 billion in 2020, $435 million in 2019 and $3.0 billion in 2018. Such amounts included our share of impairment charges recorded by Kraft Heinz. Approximately $9.8 billion of the charges in 2020 were attributable to impairments of goodwill and identifiable intangible assets recorded in connection with Berkshire's acquisition of Precision Castparts in 2016. Other earnings in 2020 also included after-tax foreign exchange rate losses of $764 million related to non-U.S. Dollar denominated debt issued by Berkshire and its U.S.-based finance subsidiary, Berkshire Hathaway Finance Corporation ("BHFC").

After-tax earnings of our railroad business increased 5.0% in 2019 compared to 2018. Earnings in 2019 benefitted from higher rates per car/unit, a curtailment gain related to an amendment to defined benefit retirement plans and ongoing operating cost control initiatives, partly offset by lower freight volumes and incremental costs associated with the persistent flooding conditions and severe winter weather in the first half of 2019. After-tax earnings of our utilities and energy business increased 8.4% in 2019 compared to 2018.

Earnings from our manufacturing, service and retailing businesses in 2019 were relatively unchanged from 2018, reflecting mixed operating results with several of these businesses experiencing lower earnings in 2019 from a variety of factors. Revenues and pre-tax earnings in 2019 of certain of these businesses were negatively affected by the unfavorable effects of foreign currency translation attributable to a stronger U.S. Dollar, international trade tensions and U.S. trade tariffs.

Investment and derivative gains/losses in each of the three years presented included significant gains and losses on our investments in equity securities, including unrealized gains and losses from market price changes on securities we continue to hold. We believe that investment and derivative gains/losses, whether realized from dispositions or unrealized from changes in market prices of equity securities, are generally meaningless in understanding our reported results or evaluating the economic performance of our businesses. These gains and losses have caused and will continue to cause significant volatility in our periodic earnings.

Insurance—Underwriting

Our management views our insurance businesses as possessing two distinct activities – underwriting and investing. Underwriting decisions are the responsibility of the unit managers, while investing decisions are the responsibility of Berkshire's Chairman and CEO, Warren E. Buffett and Berkshire's corporate investment managers. Accordingly, we evaluate performance of underwriting operations without any allocation of investment income or investment gains/losses. We consider investment income as an integral component of our aggregate insurance operating results. However, we consider investment gains and losses, whether realized or unrealized as non-operating, based on our long-held strategy of acquiring securities and holding those securities for long periods. We believe that such gains and losses are not meaningful in understanding the operating results of our insurance businesses.

The timing and amount of catastrophe losses can produce significant volatility in our periodic underwriting results, particularly with respect to our reinsurance businesses. Generally, we consider pre-tax losses in excess of $100 million from a current year catastrophic event to be significant.

Changes in estimates for unpaid losses and loss adjustment expenses, including amounts established for occurrences in prior years, can also significantly affect our periodic underwriting results. Unpaid loss estimates, including estimates under retroactive reinsurance contracts, were approximately $120.8 billion as of December 31, 2020. Our periodic underwriting results may also include significant foreign currency transaction gains and losses arising from the changes in the valuation of non-U.S. Dollar denominated liabilities of our U.S. based insurance subsidiaries due to foreign currency exchange rate fluctuations.

Insurance—Underwriting *(Continued)*

Underwriting results in 2020 of certain of our commercial insurance and reinsurance businesses were negatively affected by estimated losses and costs associated with the COVID-19 pandemic, including estimated provisions for claims and uncollectible premiums and incremental operating costs to maintain customer service levels. The effects of the pandemic in the future may be further affected by judicial rulings and regulatory and legislative actions pertaining to insurance coverage and claims and by its effects on general economic activity, which we cannot reasonably estimate at this time.

We provide primary insurance and reinsurance products covering property and casualty risks, as well as life and health risks. Our insurance and reinsurance businesses are GEICO, Berkshire Hathaway Primary Group and Berkshire Hathaway Reinsurance Group ("BHRG").

Underwriting results of our insurance businesses are summarized below (dollars in millions).

	2020	2019	2018
Pre-tax underwriting earnings (loss):			
GEICO	$ 3,428	$ 1,506	$ 2,449
Berkshire Hathaway Primary Group	110	383	670
Berkshire Hathaway Reinsurance Group	(2,700)	(1,472)	(1,109)
Pre-tax underwriting earnings	838	417	2,010
Income taxes and noncontrolling interests	181	92	444
Net underwriting earnings	$ 657	$ 325	$ 1,566
Effective income tax rate	21.5%	24.2%	21.4%

GEICO

GEICO writes private passenger automobile insurance, offering coverages to insureds in all 50 states and the District of Columbia. GEICO markets its policies mainly by direct response methods where most customers apply for coverage directly to the company via the Internet or over the telephone. A summary of GEICO's underwriting results follows (dollars in millions).

	2020		2019		2018	
	Amount	%	Amount	%	Amount	%
Premiums written	$ 34,928		$ 36,016		$ 34,123	
Premiums earned	$ 35,093	100.0	$ 35,572	100.0	$ 33,363	100.0
Losses and loss adjustment expenses	26,018	74.1	28,937	81.3	26,278	78.8
Underwriting expenses	5,647	16.1	5,129	14.5	4,636	13.9
Total losses and expenses	31,665	90.2	34,066	95.8	30,914	92.7
Pre-tax underwriting earnings	$ 3,428		$ 1,506		$ 2,449	

2020 versus 2019

GEICO's pre-tax underwriting earnings for 2020 reflected significant declines in losses and loss adjustment expenses attributable to lower claims frequencies from the effects of less driving by policyholders during the COVID-19 pandemic offset by the effects of the GEICO Giveback program (see following paragraph) on earned premiums.

Premiums written decreased 3.0% compared to 2019. The GEICO Giveback program provided for a 15% premium credit to all voluntary auto and motorcycle policies renewing between April 8, 2020 and October 7, 2020, as well as to any new policies written during the same period. The GEICO Giveback program reduced premiums written in 2020 by approximately $2.9 billion. Premiums earned decreased 1.3% in 2020 compared to 2019, which included reductions of approximately $2.5 billion attributable to the GEICO Giveback program.

Insurance—Underwriting *(Continued)*

GEICO (Continued)

Voluntary auto policies-in-force at the end of 2020 increased approximately 820,000 (4.6%) compared to the end of 2019. The increase reflected a 7.3% decrease in new business sales and a 2.5% decrease in non-renewals and policy cancellations.

Losses and loss adjustment expenses decreased $2.9 billion (10.1%) in 2020 compared to 2019. GEICO's ratio of losses and loss adjustment expenses to premiums earned (the "loss ratio") was 74.1%, a decrease of 7.2 percentage points compared to 2019. The decrease in the loss ratio reflected declines in claims frequencies, partly offset by increases in claims severities and the impact of lower premiums earned attributable to the GEICO Giveback program.

Claims frequencies in 2020 were lower for property damage, bodily injury and personal injury protection coverages (twenty-eight to thirty percent range) and collision coverage (twenty-three to twenty-four percent range) compared to 2019. Average claims severities in 2020 were higher for property damage and collision coverages (eight to ten percent range) and bodily injury coverage (twelve to thirteen percent range).

Losses and loss adjustment expenses included net reductions of $253 million in 2020 for decreases in the ultimate loss estimates for prior years' loss events compared to net increases of $42 million in 2019. Losses incurred included $81 million in 2020 from Hurricanes Laura and Sally and U.S. wildfires. There were no losses from significant catastrophe events in 2019.

Underwriting expenses in 2020 increased $518 million (10.1%) compared to 2019, reflecting higher employee-related, advertising and technology costs partly offset by lower premium taxes. GEICO's expense ratio in 2020 (underwriting expenses to premiums earned) was 16.1%, an increase of 1.6 percentage points compared to 2019. The expense ratio increase was primarily attributable to the decline in earned premiums from the GEICO Giveback program.

2019 versus 2018

Premiums written and earned in 2019 increased 5.5% and 6.6%, respectively, compared to 2018. These increases were primarily attributable to voluntary auto policies-in-force growth of 6.4%, partially offset by a decrease in average premiums per auto policy. The increase in voluntary auto policies-in-force primarily resulted from an increase in new business sales and a decrease in policies cancelled or not renewed. Voluntary auto policies-in-force increased approximately 1,068,000 during 2019.

Losses and loss adjustment expenses in 2019 increased 10.1% compared to 2018. The loss ratio in 2019 was 81.3%, an increase of 2.5 percentage points over 2018, primarily due to increases in average claims severities.

Average claims severities in 2019 were higher versus 2018 for property damage and collision coverages (four to six percent range) and bodily injury coverage (seven to nine percent range). Claims frequencies in 2019 declined compared to 2018 for property damage and collision coverages (two to four percent range) and personal injury protection coverage (one to two percent range) and were relatively unchanged for bodily injury coverage. Losses and loss adjustment expenses included net increases of $42 million in 2019 and net of decreases $222 million in 2018 for changes in the ultimate loss estimates for prior years' loss events.

Underwriting expenses in 2019 increased $493 million (10.6%) over 2018. GEICO's underwriting expense ratio in 2019 was 14.5%, an increase of 0.6 percentage points compared to 2018. The underwriting expense increase was primarily attributable to increases in advertising expenses and employee-related costs, which reflected wage and staffing increases.

Insurance—Underwriting *(Continued)*

Berkshire Hathaway Primary Group

The Berkshire Hathaway Primary Group ("BH Primary") provides a variety of commercial insurance solutions, including healthcare malpractice, workers' compensation, automobile, general liability, property and various specialty coverages for small, medium and large clients. The largest of these insurers are Berkshire Hathaway Specialty Insurance ("BH Specialty"), Berkshire Hathaway Homestate Companies ("BHHC"), MedPro Group, Berkshire Hathaway GUARD Insurance Companies ("GUARD") and National Indemnity Company ("NICO Primary"). Other BH Primary insurers include U.S. Liability Insurance Company, Central States Indemnity Company and MLMIC Insurance Company ("MLMIC"), acquired October 1, 2018. A summary of BH Primary underwriting results follows (dollars in millions).

	2020		2019		2018	
	Amount	%	Amount	%	Amount	%
Premiums written	$ 10,212		$ 9,843		$ 8,561	
Premiums earned	$ 9,615	100.0	$ 9,165	100.0	$ 8,111	100.0
Losses and loss adjustment expenses	7,129	74.1	6,336	69.1	5,261	64.9
Underwriting expenses	2,376	24.7	2,446	26.7	2,180	26.9
Total losses and expenses	9,505	98.8	8,782	95.8	7,441	91.8
Pre-tax underwriting earnings	$ 110		$ 383		$ 670	

Premiums written increased $369 million (3.7%) in 2020 compared to 2019, reflecting increased premiums written from BH Specialty (34%) and MedPro Group (9%), partially offset by a 13% decrease in premiums written by our other primary insurers. The increase at BH Specialty was driven by increased casualty business globally and the increase at MedPro Group reflected increases across several product categories. The decline in volume by our other primary insurers was primarily due to lower workers' compensation and commercial automobile volumes and the effect of the divestiture of Applied Underwriters in October 2019. The declines in workers' compensation and commercial auto business written reflected the effects of reduced exposures and premium refunds related to the COVID-19 pandemic and volume reductions attributable to increased price competition in the market.

Premiums written increased $1.3 billion (15.0%) in 2019 compared to 2018. The increase was attributable to higher volumes from BH Specialty, MedPro Group and GUARD, as well as from the effects of the MLMIC acquisition. These increases were partly offset by lower volume at BHHC and the effect of the Applied Underwriters divestiture.

BH Primary's combined loss ratios were 74.1% in 2020, 69.1% in 2019 and 64.9% in 2018, which reflected the effects of significant catastrophe events during the year and changes in estimated losses for prior years' loss events. Losses and loss adjustment expenses attributable to significant catastrophe events were $207 million in 2020 (Hurricanes Laura and Sally and U.S. wildfires) and $190 million in 2018 (Hurricanes Florence and Michael and the wildfires in California). We incurred no losses from significant catastrophe events in 2019. Losses in 2020 also included $167 million attributable to the pandemic. Finally, losses and loss adjustment expenses were reduced $265 million in 2020, $499 million in 2019 and $715 million in 2018 for net reductions in estimated ultimate liabilities for prior years' loss events.

BH Primary insurers write significant levels of commercial and professional liability and workers' compensation insurance and the related claim costs may be subject to high severity and long claim-tails. Accordingly, we could experience significant increases in claims liabilities in the future attributable to higher-than-expected claim settlements, adverse litigation outcomes or judicial rulings and other factors not currently anticipated.

Berkshire Hathaway Reinsurance Group

We offer excess-of-loss and quota-share reinsurance coverages on property and casualty risks and life and health reinsurance to insurers and reinsurers worldwide through several subsidiaries, led by National Indemnity Company ("NICO"), Berkshire Hathaway Life Insurance Company of Nebraska ("BHLN") and General Reinsurance Corporation, General Reinsurance AG and General Re Life Corporation (collectively, "General Re"). We also periodically assume property and casualty risks under retroactive reinsurance contracts written through NICO. In addition, we write periodic payment annuity contracts predominantly through BHLN.

Insurance—Underwriting *(Continued)*

Berkshire Hathaway Reinsurance Group (Continued)

Generally, we strive to generate underwriting profits. However, time-value-of-money concepts are important elements in establishing prices for retroactive reinsurance and periodic payment annuity businesses due to the expected long durations of the liabilities. We expect to incur pre-tax underwriting losses from such businesses, primarily through deferred charge amortization and discount accretion charges. We receive premiums at the inception of these contracts, which are then available for investment. A summary of BHRG's premiums and pre-tax underwriting results follows (dollars in millions).

	Premiums written			Premiums earned			Pre-tax underwriting earnings (loss)		
	2020	2019	2018	2020	2019	2018	2020	2019	2018
Property/casualty	$ 13,295	$ 10,428	$ 9,413	$ 12,214	$ 9,911	$ 8,928	$ (799)	$ 16	$ (207)
Life/health	5,848	4,963	5,430	5,861	4,869	5,327	(18)	159	182
Retroactive reinsurance	38	684	517	38	684	517	(1,248)	(1,265)	(778)
Periodic payment annuity	566	863	1,156	566	863	1,156	(617)	(549)	(340)
Variable annuity	14	14	16	14	14	16	(18)	167	34
	$ 19,761	$ 16,952	$ 16,532	$ 18,693	$ 16,341	$ 15,944	$ (2,700)	$ (1,472)	$ (1,109)

Property/casualty

A summary of property/casualty reinsurance underwriting results follows (dollars in millions).

	2020		2019		2018	
	Amount	%	Amount	%	Amount	%
Premiums written	$ 13,295		$ 10,428		$ 9,413	
Premiums earned	$ 12,214	100.0	$ 9,911	100.0	$ 8,928	100.0
Losses and loss adjustment expenses	9,898	81.0	7,313	73.8	6,929	77.6
Underwriting expenses	3,115	25.5	2,582	26.0	2,206	24.7
Total losses and expenses	13,013	106.5	9,895	99.8	9,135	102.3
Pre-tax underwriting earnings (loss)	$ (799)		$ 16		$ (207)	

Premiums written in 2020 increased $2.9 billion (27.5%) compared to 2019. The increase was primarily attributable to new business, including a small number of contracts with very large premiums, and increased participations on renewals. Premiums written in 2019 increased $1.0 billion (10.8%) compared to 2018. The increase was primarily attributable to new business, net of non-renewals, and increased participations on renewal business, partly offset by the unfavorable foreign currency translation effects of a stronger U.S. Dollar.

Underwriting earnings in 2020 were negatively affected by an increase in losses and loss adjustment expenses of $2.6 billion (35.3%). The loss ratio in 2020 was 81.0%, an increase of 7.2 percentage points over 2019. Losses and loss adjustment expenses in 2020 included estimated losses of $964 million attributable to the COVID-19 pandemic and estimated losses from significant catastrophe events of $667 million from Hurricanes Laura and Sally and U.S. wildfires. Losses and loss adjustment expenses also reflected net increases in estimated ultimate liabilities for prior years' loss events of $162 million in 2020 primarily attributable to legacy environmental, asbestos and other latent injury claims. Such amount as a percentage of the related net unpaid claim liabilities as of the beginning of 2020 was 0.5%.

BHRG's loss ratio was 73.8% in 2019 and 77.6% in 2018. Losses in 2019 included approximately $1.0 billion from Typhoons Faxia and Hagibis and various U.S. and non-U.S. wildfires, while losses in 2018 included approximately $1.3 billion from Hurricanes Florence and Michael, Typhoon Jebi and wildfires in California. Losses and loss adjustment expenses also included net decreases of $295 million in 2019 and $469 million in 2018 for prior years' loss events. Such amounts as percentages of the related net unpaid claim liabilities as of the beginning of the applicable year were 1.0% in 2019 and 1.7% in 2018.

Underwriting expenses are primarily commissions and brokerage costs. Underwriting expenses in 2020 increased $533 million (20.6%) over 2019, and underwriting expenses in 2019 increased $376 million (17.0%) over 2018. The increases reflected the increases in premium volumes and changes in business mix.

Insurance—Underwriting *(Continued)*

Life/health

A summary of our life/health reinsurance underwriting results follows (dollars in millions).

	2020		2019		2018	
	Amount	%	Amount	%	Amount	%
Premiums written	$ 5,848		$ 4,963		$ 5,430	
Premiums earned	$ 5,861	100.0	$ 4,869	100.0	$ 5,327	100.0
Life and health insurance benefits	4,883	83.3	3,800	78.0	4,240	79.6
Underwriting expenses	996	17.0	910	18.7	905	17.0
Total benefits and expenses	5,879	100.3	4,710	96.7	5,145	96.6
Pre-tax underwriting earnings (loss)	$ (18)		$ 159		$ 182	

Life/health premiums written increased $885 million (17.8%) in 2020 compared to 2019. Approximately $480 million of the increase was attributable to a reinsurance contract covering U.S. health insurance risks that incepted in the fourth quarter of 2019, which was not renewed for 2021. The remainder of the increase was primarily from volume growth in the Asian and European life reinsurance markets.

Underwriting earnings in 2020 were negatively affected by increased life benefits from COVID-19-related claims (approximately $275 million) and continuing losses from increased liabilities from changes in underlying assumptions with respect to disability benefit liabilities in Australia, which were mostly offset by lower other life claims and reduced losses from U.S. long-term care business that is in run-off. The ratio of life and health insurance benefits to premiums earned was 83.3% in 2020 and 81.5% in 2019, which is before the effects of the BHLN contract amendment referred to below.

Life/health premiums written in 2019 decreased $467 million (8.6%) compared to 2018. In the first quarter of 2019, BHLN amended a yearly-renewable-term life reinsurance contract with a major reinsurer. BHLN recorded a reduction in earned premiums on this contract in 2019 of $49 million, while premiums earned in 2018 related to this contract were $954 million. In 2019, premiums earned also included $228 million from a new health reinsurance contract and reflected volume growth in life markets, partially offset by the unfavorable effects of foreign currency translation attributable to a stronger U.S. Dollar.

Underwriting earnings in 2019 included a one-time gain of $163 million attributable to the BHLN yearly-renewable-term life reinsurance contract amendment. Pre-tax underwriting earnings in 2019 also included losses from increased disability benefit liabilities in Australia, attributable to higher claims experience and changes to various underlying assumptions increased U.S. long-term care liabilities due to discount rate reductions and changes in other actuarial assumptions, and an increase in life claims in North America, partially offset by increased earnings from other international life business.

Retroactive reinsurance

There were no significant retroactive reinsurance contracts written in 2020. Premiums written were $684 million in 2019 and $517 million in 2018, attributable to a limited number of contracts in each year. Pre-tax underwriting losses in each year derived from deferred charge amortization and changes in the estimated timing and amounts of future claim payments. Underwriting results also include foreign currency exchange gains and losses from the effects of changes in foreign currency exchange rates on non-U.S. Dollar denominated liabilities of our U.S. subsidiaries. Underwriting results included pre-tax foreign currency losses of $139 million in 2020 and $76 million in 2019 and pre-tax gains of $169 million in 2018.

Pre-tax underwriting losses before foreign currency gains/losses were $1.1 billion in 2020, $1.2 billion in 2019 and $947 million in 2018. Overall, we decreased estimated ultimate liabilities $399 million in 2020 for prior years' contracts compared to an increase of $378 million in 2019. After adjustments to the related unamortized deferred charges from changes in the estimated timing and amount of the future claim payments, such changes produced pre-tax underwriting earnings of approximately $230 million in 2020 and pre-tax losses of $125 million in 2019.

Gross unpaid losses assumed under retroactive reinsurance contracts were $41.0 billion at December 31, 2020 and $42.4 billion at December 31, 2019. Unamortized deferred charge assets related to such reinsurance contracts were $12.4 billion at December 31, 2020 and $13.7 billion at December 31, 2019. Deferred charge assets will be charged to earnings over the expected remaining claims settlement periods through periodic amortization.

Insurance—Underwriting *(Continued)*

Periodic payment annuity

Periodic payment annuity premiums earned in 2020 decreased $297 million (34.4%) compared to 2019, which decreased $293 million (25.3%) from 2018. Periodic payment annuity business is price sensitive. The volumes written can change rapidly due to changes in prices, which are affected by prevailing interest rates, the perceived risks and durations associated with the expected annuity payments, as well as the level of competition.

Periodic payment annuity contracts normally produce pre-tax underwriting losses deriving from the recurring discount accretion of annuity liabilities. Underwriting results also include gains or losses from the effects of changes in mortality and interest rates and from foreign currency exchange rate changes on non-U.S. Dollar denominated liabilities of our U.S. subsidiaries. Pre-tax underwriting results included foreign currency losses of $67 million in 2020 and $40 million in 2019 compared to pre-tax gains of $93 million in 2018.

Excluding foreign currency gains/losses, pre-tax underwriting losses from periodic payment annuity contracts were $550 million in 2020, $509 million in 2019 and $433 million in 2018. These losses primarily derived from the recurring discount accretion of annuity liabilities, as well as from the impact of mortality and interest rate changes. Discounted annuity liabilities were $14.3 billion at December 31, 2020 and $13.5 billion at December 31, 2019. The weighted average discount rate was approximately 4.0%.

Variable annuity

Variable annuity guarantee reinsurance contracts produced pre-tax losses of $18 million in 2020 compared to pre-tax earnings of $167 million in 2019 and $34 million in 2018. The results of this business reflect changes in remaining liabilities for underlying guaranteed benefits reinsured, which are affected by changes in securities markets and interest rates and from the periodic amortization of expected profit margins. Underwriting results from these contracts can be volatile, reflecting the volatility of securities markets, interest rates and foreign currency exchange rates.

Insurance—Investment Income

A summary of net investment income attributable to our insurance operations follows (dollars in millions).

				Percentage change	
	2020	**2019**	**2018**	**2020 vs 2019**	**2019 vs 2018**
Interest and other investment income	$ 1,059	$ 2,075	$ 1,851	(49.0)%	12.1%
Dividend income	4,890	4,525	3,652	8.1	23.9
Pre-tax net investment income	5,949	6,600	5,503	(9.9)	19.9
Income taxes and noncontrolling interests	910	1,070	949		
Net investment income	$ 5,039	$ 5,530	$ 4,554		
Effective income tax rate	15.3%	16.1%	17.2%		

Interest and other investment income declined $1.0 billion (49.0%) in 2020 compared to 2019, primarily due to lower income from short-term investments. We continue to hold substantial balances of cash, cash equivalents and short-term U.S. Treasury Bills. Short-term interest rates declined over the second half of 2019 and the decline continued throughout 2020, which resulted in significantly lower interest income. We expect such rates, which are historically low, to remain low, negatively affecting our earnings from such investments in 2021. Nevertheless, we believe that maintaining ample liquidity is paramount and we insist on safety over yield with respect to short-term investments.

Insurance—Investment Income *(Continued)*

Dividend income increased $365 million (8.1%) in 2020 compared to 2019. The increase was primarily attributable to dividends from the investment in $10 billion liquidation value of 8% cumulative preferred stock of Occidental Petroleum Corporation ("Occidental") on August 8, 2019, partly offset by lower dividends from common stock investments.

Interest and other investment income increased $224 million (12.1%) in 2019 compared to 2018, primarily due to higher interest rates on short-term investments and interest from a term loan with Seritage Growth Properties, partially offset by lower income earned from fixed maturity securities and limited partnership investments. Dividend income increased $873 million (23.9%) in 2019 compared to 2018. The increase in dividend income was attributable to an overall increase in investment levels, including the investment in Occidental and increased dividends from common stock investments.

Invested assets of our insurance businesses derive from shareholder capital, including reinvested earnings, and from net liabilities under insurance and reinsurance contracts or "float." The major components of float are unpaid losses and loss adjustment expenses, including liabilities under retroactive reinsurance contracts, life, annuity and health insurance benefit liabilities, unearned premiums and other liabilities due to policyholders, which are reduced by insurance premiums and reinsurance receivables, deferred charges assumed under retroactive reinsurance contracts and deferred policy acquisition costs. Float approximated $138 billion at December 31, 2020, $129 billion at December 31, 2019 and $123 billion at December 31, 2018. Our combined insurance operations generated pre-tax underwriting earnings of approximately $838 million in 2020, $417 million in 2019 and $2.0 billion in 2018, and consequently, the average cost of float for each of those periods was negative.

A summary of cash and investments held in our insurance businesses as of December 31, 2020 and 2019 follows (in millions).

	December 31,	
	2020	**2019**
Cash, cash equivalents and U.S. Treasury Bills	$ 67,082	$ 64,908
Equity securities	269,498	240,126
Fixed maturity securities	20,317	18,537
Other	6,220	2,481
	$ 363,117	$ 326,052

Fixed maturity investments as of December 31, 2020 were as follows (in millions).

	Amortized cost	**Unrealized gains/losses**	**Carrying value**
U.S. Treasury, U.S. government corporations and agencies	$ 3,339	$ 55	$ 3,394
Foreign governments	11,232	105	11,337
Corporate bonds	4,678	462	5,140
Other	382	64	446
	$ 19,631	$ 686	$ 20,317

U.S. government obligations are rated AA+ or Aaa by the major rating agencies. Approximately 88% of all foreign government obligations were rated AA or higher by at least one of the major rating agencies. Foreign government securities include obligations issued or unconditionally guaranteed by national or provincial government entities.

Railroad ("Burlington Northern Santa Fe")

Burlington Northern Santa Fe, LLC ("BNSF") operates one of the largest railroad systems in North America, with approximately 32,500 route miles of track in 28 states. BNSF also operates in three Canadian provinces. BNSF classifies its major railroad business groups by type of product shipped which includes consumer products, industrial products, agricultural products and coal. A summary of BNSF's earnings follows (dollars in millions).

				Percentage change	
	2020	2019	2018	2020 vs 2019	2019 vs 2018
Railroad operating revenues	$ 20,181	$ 22,745	$ 22,999	(11.3)%	(1.1)%
Railroad operating expenses:					
Compensation and benefits	4,542	5,270	5,322	(13.8)	(1.0)
Fuel	1,789	2,944	3,346	(39.2)	(12.0)
Purchased services	1,954	2,049	2,131	(4.6)	(3.8)
Depreciation and amortization	2,460	2,389	2,306	3.0	3.6
Equipment rents, materials and other	1,684	2,028	2,110	(17.0)	(3.9)
Total	12,429	14,680	15,215	(15.3)	(3.5)
Railroad operating earnings	7,752	8,065	7,784	(3.9)	3.6
Other revenues (expenses):					
Other revenues	688	770	856	(10.6)	(10.0)
Other expenses, net	(611)	(515)	(736)	18.6	(30.0)
Interest expense	(1,037)	(1,070)	(1,041)	(3.1)	2.8
Pre-tax earnings	6,792	7,250	6,863	(6.3)	5.6
Income taxes	1,631	1,769	1,644	(7.8)	7.6
Net earnings	$ 5,161	$ 5,481	$ 5,219	(5.8)	5.0
Effective income tax rate	24.0%	24.4%	24.0%		

The following table summarizes BNSF's railroad freight volumes by business group (cars/units in thousands).

	Cars/Units			Percentage change	
	2020	2019	2018	2020 vs 2019	2019 vs 2018
Consumer products	5,266	5,342	5,597	(1.4)%	(4.6)%
Industrial products	1,622	1,931	1,991	(16.0)	(3.0)
Agricultural products	1,189	1,146	1,208	3.8	(5.1)
Coal	1,404	1,802	1,902	(22.1)	(5.3)
Total cars/units	9,481	10,221	10,698	(7.2)	(4.5)

2020 versus 2019

Railroad operating revenues declined 11.3% in 2020 versus 2019, reflecting a 7.2% decrease in volume and a 4.5% decrease in average revenue per car/unit. The decrease in revenue per car/unit was attributable to lower fuel surcharge revenue driven by lower fuel prices and business mix changes. The overall volume decrease was primarily due to the COVID-19 pandemic, which severely impacted volumes through the first half of the year. Volumes sequentially improved from earlier periods and recovered overall to pre-pandemic levels by the end of the year.

BNSF is an important component of the national and global supply chain and, as an essential business, has continued to operate throughout the duration of the COVID-19 pandemic. However, the pandemic caused significant economic disruptions that adversely affected the demand for transportation services. The pandemic continues to evolve, and the full extent to which it may impact BNSF's business, operating results, financial condition, or liquidity will depend on future developments. We believe BNSF's fundamental business remains strong and it has ample liquidity to continue business operations during this volatile period.

Railroad ("Burlington Northern Santa Fe") *(Continued)*

Pre-tax earnings were $6.8 billion in 2020, a decrease of 6.3% from 2019, principally due to the negative impacts of the pandemic on volumes. In addition, pre-tax earnings in 2019 included an operating revenue increase related to the favorable outcome of an arbitration hearing and a retirement plan curtailment gain that is included in other expenses, net in the preceding table. These effects were partially offset by significant improvements in 2020 in service, system velocity and cost performance compared to 2019, along with lower costs related to severe winter weather and flooding on parts of the network, which negatively affected expenses and service levels in 2019.

Operating revenues from consumer products of $7.3 billion in 2020 declined 7.6% compared to 2019, primarily due to a 6.3% decrease in average revenue per car/unit along with lower volumes. The volume decrease was primarily due to the impact of the pandemic. Lower international and automotive volumes were offset by higher domestic intermodal volumes. Increased retail sales, inventory replenishments by retailers and e-commerce activity produced recovery of intermodal volumes in the second half of 2020.

Operating revenues from industrial products were $5.0 billion in 2020, a decrease of 17.0% from 2019. The decrease was primarily attributable to the decline in volume and to a lesser extent lower average revenue per car/unit. Volumes decreased primarily due to lower U.S. industrial production driven by the pandemic, including reduced production and demand in the energy sector, which drove lower sand and petroleum products volume, along with reduced steel demand, which drove lower taconite volume.

Operating revenues from agricultural products increased 2.9% to $4.8 billion in 2020 compared to 2019. The increase was due to higher volumes, partially offset by slightly lower average revenue per car/unit. The volume increase was primarily due to higher grain and meal exports, partially offsetting adverse impacts of the pandemic, primarily for ethanol and sweeteners shipments.

Operating revenues from coal decreased 28.5% to $2.7 billion in 2020 compared to 2019. This decrease was primarily due to lower volumes, as well as lower revenues per car/unit. Volumes decreased primarily due to lower natural gas prices, lower electricity demand driven by the pandemic, utility coal plant retirements and mild temperatures.

Railroad operating expenses declined 15.3% to $12.4 billion in 2020 as compared to 2019. The ratio of railroad operating expenses to railroad operating revenues declined 2.9 percentage points to 61.6% in 2020 versus 2019. Railroad operating expenses in 2020 reflected lower volume-related costs, productivity improvements, the effects of cost control initiatives and improved weather conditions compared to 2019.

Compensation and benefits expenses decreased $728 million (13.8%) in 2020 compared to 2019, primarily due to lower employee counts associated with lower volume and due to improved workforce productivity. Fuel expenses decreased $1.2 billion (39.2%) compared to 2019, primarily due to lower average fuel prices, lower volumes and improved fuel efficiency. Purchased services expense declined $95 million (4.6%) compared to 2019. The decrease was primarily due to lower volume, improved productivity and higher insurance recoveries in 2020 related to network flooding in 2019. Equipment rents, materials and other expense decreased $344 million (17.0%) compared to 2019, primarily due to lower volume-related costs, the effects of cost controls and lower personal injury and derailment expenses.

2019 versus 2018

Railroad operating revenues were $22.7 billion in 2019, a decline of 1.1% versus 2018. During 2019, BNSF's revenues reflected a 3.6% comparative increase in average revenue per car/unit and a 4.5% decrease in volume. The increase in average revenue per car/unit was attributable to increased rates per car/unit and a favorable outcome of an arbitration hearing. Pre-tax earnings were approximately $7.3 billion in 2019, an increase of 5.6% over 2018. BNSF experienced severe winter weather and flooding on parts of the network, which negatively affected revenues, expenses and service levels. In addition to the impact of an increase in average revenue per car/unit, earnings in 2019 benefited from a reduction in total operating expenses.

Railroad ("Burlington Northern Santa Fe") *(Continued)*

Operating revenues from consumer products were $7.9 billion in 2019, a decrease of 0.5% compared to 2018, reflecting volume decreases and higher average revenue per car/unit. The volume decreases were driven by moderated demand and the availability of truck capacity, as well as lower west coast imports.

Operating revenues from industrial products were $6.1 billion in 2019, an increase of 1.7% from 2018. The increase was attributable to higher average revenue per car/unit, partially offset by a decrease in volume. Volumes decreased primarily due to overall softness in the industrial sector, lower sand volumes and reduced car loadings, due to the challenging weather conditions in 2019. Increased demand for petroleum products and liquefied petroleum gas, partially offset the other decreases in volumes.

Operating revenues from agricultural products decreased 0.3% in 2019 to $4.7 billion compared to 2018. The decrease was due to lower volumes and higher average revenue per car/unit. The volume decreases were attributable to export competition from non-U.S. sources, the impacts of international trade policies and the challenging weather conditions in 2019.

Operating revenues from coal decreased 7.4% in 2019 to $3.7 billion compared to 2018, reflecting lower average revenue per car/unit and lower volumes. Volumes were negatively impacted by adverse weather conditions, as well as from the effects of lower natural gas prices.

Railroad operating expenses were $14.7 billion in 2019, a decrease of $535 million compared to 2018. Our ratio of operating expenses to railroad operating revenues in 2019 of 64.5% decreased 1.7 percentage points versus 2018. Operating expenses in 2019 reflected lower volume-related costs, lower fuel prices and the effects of cost control initiatives, partially offset by the costs associated with the adverse weather conditions.

Fuel expenses decreased $402 million in 2019 compared to 2018, primarily due to lower average fuel prices, lower volumes and improved fuel efficiency. Purchased services expense decreased $82 million compared to 2018. The decrease was due to lower purchased transportation costs of our logistics services business, lower drayage, lower services expense and higher insurance recoveries. Equipment rents, materials and other expense decreased $82 million compared to 2018, due to lower locomotive and various other costs associated with lower volumes and cost controls. Other expenses, net decreased $221 million compared to 2018. In 2019, other expenses were net of a $120 million curtailment gain from an amendment to the company-sponsored defined benefit retirement plans.

Utilities and Energy ("Berkshire Hathaway Energy Company")

We currently own 91.1% of the outstanding common stock of Berkshire Hathaway Energy Company ("BHE"), which operates a global energy business. BHE's domestic regulated utility interests are comprised of PacifiCorp, MidAmerican Energy Company ("MEC") and NV Energy. In Great Britain, BHE subsidiaries operate two regulated electricity distribution businesses referred to as Northern Powergrid. BHE's natural gas pipelines consist of five domestic regulated interstate natural gas pipeline systems and a 25% interest in a liquefied natural gas export, import and storage facility in which BHE operates and consolidates for financial reporting purposes. Three of these systems were acquired on November 1, 2020 from Dominion Energy, Inc. ("BHE GT&S acquisition"). See Note 2 to accompanying Consolidated Financial Statements. Other energy businesses include a regulated electricity transmission-only business in Alberta, Canada ("AltaLink, L.P.") and a diversified portfolio of mostly renewable independent power projects. BHE also operates the largest residential real estate brokerage firm and one of the largest residential real estate brokerage franchise networks in the United States.

Utilities and Energy ("Berkshire Hathaway Energy Company") *(Continued)*

The rates our regulated businesses charge customers for energy and services are based in large part on the costs of business operations, including income taxes and a return on capital, and are subject to regulatory approval. To the extent such costs are not allowed in the approved rates, operating results will be adversely affected. A summary of BHE's net earnings follows (dollars in millions).

	2020	2019	2018
Revenues:			
Energy operating revenue	$ 15,556	$ 15,371	$ 15,573
Real estate operating revenue	5,396	4,473	4,214
Other income (loss)	79	270	200
Total revenue	21,031	20,114	19,987
Costs and expense:			
Energy cost of sales	4,187	4,586	4,769
Energy operating expense	7,539	6,824	6,969
Real estate operating costs and expense	4,885	4,251	4,000
Interest expense	1,941	1,835	1,777
Total costs and expense	18,552	17,496	17,515
Pre-tax earnings	2,479	2,618	2,472
Income tax expense (benefit)*	(1,010)	(526)	(452)
Net earnings after income taxes	3,489	3,144	2,924
Noncontrolling interests	71	18	23
Net earnings attributable to BHE	3,418	3,126	2,901
Noncontrolling interests and preferred stock dividends	327	286	280
Net earnings attributable to Berkshire Hathaway shareholders	$ 3,091	$ 2,840	$ 2,621
Effective income tax rate	(40.7)%	(20.1)%	(18.3)%

* *Includes significant production tax credits from wind-powered electricity generation.*

The discussion of BHE's operating results that follows is based on after-tax earnings, reflecting how the energy businesses are managed and evaluated. A summary of net earnings attributable to BHE follows (dollars in millions).

				Percentage change	
	2020	2019	2018	2020 vs 2019	2019 vs 2018
PacifiCorp	$ 741	$ 773	$ 739	(4.1)%	4.6%
MidAmerican Energy Company	818	781	669	4.7	16.7
NV Energy	410	365	317	12.3	15.1
Northern Powergrid	201	256	239	(21.5)	7.1
Natural gas pipelines	528	422	387	25.1	9.0
Other energy businesses	697	608	489	14.6	24.3
Real estate brokerage	375	160	145	134.4	10.3
Corporate interest and other	(352)	(239)	(84)	47.3	184.5
	$ 3,418	$ 3,126	$ 2,901	9.3	7.8

Utilities and Energy ("Berkshire Hathaway Energy Company") *(Continued)*

2020 versus 2019

PacifiCorp operates a regulated electric utility in portions of several Western states, including Utah, Oregon and Wyoming. PacifiCorp after-tax earnings decreased $32 million in 2020 compared to 2019. The decrease reflected higher operating expenses and net interest expense, partially offset by increased production tax credit benefits driven by repowered wind projects placed in-service, higher utility margin (operating revenue less cost of sales) and higher other income. The increase in operating expenses was largely due to costs associated with wildfires, a settlement agreement and pension benefits.

PacifiCorp utility margin was $3.3 billion in 2020, an increase of $47 million compared to 2019. The increase reflected higher operating revenue from favorable average retail prices and lower generation and purchased power costs, partially offset by lower operating revenue from a 1.4% decline in retail customer volumes. The decline in retail customer volumes was due to the impacts of the pandemic, partly offset by an increase in the average number of customers and the favorable impacts of weather.

MEC operates a regulated electric and natural gas utility primarily in Iowa and Illinois. After-tax earnings increased $37 million in 2020 compared to 2019. The increase reflected increased income tax benefits, primarily from production tax credits, driven by repowered and new wind projects placed in-service, and the effects of ratemaking. These effects were partially offset by higher depreciation expense from additional assets placed in-service, higher net interest expense, lower other income and lower electric and natural gas utility margins.

MEC electric utility margin decreased $10 million to $1.8 billion in 2020 compared to 2019. The electric utility margin decrease was attributable to lower operating revenue from unfavorable wholesale prices and price impacts from changes in retail sales mix. These effects were mostly offset by lower generation and purchased power costs and higher operating revenue from a 1.2% increase in retail customer volumes. The increase in electric retail customer volumes was primarily due to increased usage by certain industrial customers, partially offset by the impacts of the pandemic. Natural gas utility margin decreased $9 million in 2020 compared to 2019, due to the unfavorable impacts of weather.

NV Energy operates regulated electric and natural gas utilities in Nevada. After-tax earnings increased $45 million in 2020 compared to 2019. The increase reflected higher electric utility margin and lower income tax expense from the favorable impacts of ratemaking, partially offset by higher operating expenses. The increase in operating expenses was mainly due to higher earnings sharing accruals for customers at Nevada Power Company and higher depreciation expense from additional assets placed in-service.

NV Energy electric utility margin increased $100 million to $1.7 billion in 2020 compared to 2019. The increase was primarily due to higher operating revenue from a 1.5% increase in electric retail customer volumes, including distribution-only service customers and price impacts from changes in retail sales mix. The increase in electric retail customer volumes was primarily due to the favorable impacts of weather, partially offset by the impacts of the pandemic.

Northern Powergrid after-tax earnings decreased $55 million in 2020 as compared to 2019. The earnings decrease reflected write-offs of gas exploration costs and higher income tax expense, in large part from a change in the United Kingdom corporate income tax rate, partially offset by lower pension costs and interest expense.

Natural gas pipelines after-tax earnings increased $106 million in 2020 compared to 2019. The increase was primarily due to $73 million of earnings from the BHE GT&S acquisition, the favorable impact of a rate case settlement at Northern Natural Gas and higher transportation volume and rates, partially offset by higher depreciation, operating expenses and interest expenses.

Other energy business after-tax earnings in 2020 increased $89 million compared to 2019. The increase was primarily due to increased income tax benefits from renewable wind tax equity investments, largely from projects reaching commercial operation, partially offset by lower operating revenue and higher operating expenses from geothermal and natural gas units.

Utilities and Energy ("Berkshire Hathaway Energy Company") *(Continued)*

Real estate brokerage after-tax earnings increased $215 million in 2020 compared to 2019. The increase reflected higher earnings from mortgage and brokerage services. The increase in earnings from mortgage services was attributable to higher refinance activity from the favorable interest rate environment and the earnings increase from brokerage services was due to an increase of 13.1% in closed transaction dollar volume.

Corporate interest and other after-tax earnings decreased $113 million in 2020 compared to 2019. The decline was primarily due to higher interest expense and lower state income tax benefits.

2019 versus 2018

PacifiCorp after-tax earnings were $773 million in 2019, an increase of $34 million compared to 2018, reflecting slightly higher utility margin and higher other income, partly offset by higher depreciation expense from additional assets placed in-service. PacifiCorp utility margin was $3.3 billion in 2019, an increase of $4 million compared to 2018, as a 0.4% increase in retail customer volumes was largely offset by lower wholesale revenue mainly due to lower volumes.

MEC after-tax earnings of $781 million in 2019 increased $112 million as compared to 2018, primarily attributable to increases in electric utility margin, income tax benefits from higher production tax credits and the effects of ratemaking and other income. Electric utility margin in 2019 increased 2% to $1.8 billion, primarily due to higher wind generation and higher retail customer volumes of 1.4%, as a 4.0% increase in industrial volumes was largely offset by lower residential volumes from the unfavorable impacts of weather. These earnings increases were partially offset by increased depreciation expense from additional assets placed in-service (net of lower Iowa revenue sharing) and higher net interest expense.

NV Energy after-tax earnings were $365 million in 2019, an increase of $48 million compared to 2018, as lower operating expenses were partly offset by lower electric utility margin. Electric utility margin in 2019 was $1.6 billion, representing a decrease of $58 million (3%) versus 2018. The decrease was primarily due to a 1.4% decline in retail customer volumes, largely attributable to the impacts of weather, and rate reductions from the impact of the changes in U.S. income tax laws, partially offset by retail customer growth.

Northern Powergrid after-tax earnings increased in 2019 compared to 2018, reflecting higher distribution revenues and lower operating expenses, which were largely from lower pension settlement losses in 2019, partially offset by the unfavorable foreign currency translation effects of a strong average U.S. Dollar. Distribution revenues increased $18 million, attributable to higher tariff rates, partly offset by lower distributed units.

Natural gas pipelines after-tax earnings increased $35 million in 2019 compared to 2018, primarily due to higher transportation revenues from generally higher volumes and rates, favorable margins from system balancing activities and a decrease in operating expenses, partly offset by higher depreciation expense from increased spending on capital projects.

Other energy businesses after-tax earnings in 2019 increased $119 million compared to 2018. The increase was primarily due to improved earnings from renewable wind energy projects ($49 million from tax equity investments and $25 million from new and existing projects and activities), higher income from geothermal and natural gas units, largely due to higher generation and favorable margins and lower operating expenses, partly offset by lower earnings at a hydroelectric facility in the Philippines due to lower rainfall. The increase in earnings also reflected the effects of favorable regulatory decisions received in 2019 and the unfavorable impacts of a regulatory rate order received in 2018 at AltaLink L.P.

Real estate brokerage after-tax earnings increased in 2019 compared to 2018. The increase was primarily due to higher earnings at mortgage businesses due to increased refinance activity and earnings attributable to recent business acquisitions, partially offset by lower earnings at brokerage businesses, primarily from a decrease in closed units and lower margins.

Corporate interest and other after-tax earnings decreased $155 million in 2019 compared to 2018. The earnings decline was primarily due to income tax benefits recognized in 2018 related to the reduction of accrued repatriation taxes on undistributed foreign earnings in connection with the changes in U.S. income tax laws, higher interest expense and lower earnings from non-regulated energy services.

Manufacturing, Service and Retailing

A summary of revenues and earnings of our manufacturing, service and retailing businesses follows (dollars in millions).

	2020	2019	2018	Percentage change 2020 vs 2019	Percentage change 2019 vs 2018
Revenues					
Manufacturing	$ 59,079	$ 62,730	$ 61,883	(5.8)%	1.4%
Service and retailing	75,018	79,945	78,926	(6.2)	1.3
	$ 134,097	$ 142,675	$ 140,809	(6.0)	1.3
Pre-tax earnings *					
Manufacturing	$ 8,010	$ 9,522	$ 9,366	(15.9)%	1.7%
Service and retailing	2,879	2,843	2,942	1.3	(3.4)
	10,889	12,365	12,308	(11.9)	0.5
Income taxes and noncontrolling interests	2,589	2,993	2,944		
	$ 8,300	$ 9,372	$ 9,364		
Effective income tax rate	23.3%	23.7%	23.4%		
Pretax earnings as a percentage of revenues	8.1%	8.7%	8.7%		

* *Excludes certain acquisition accounting expenses, which primarily related to the amortization of identified intangible assets recorded in connection with our business acquisitions. The after-tax acquisition accounting expenses excluded from earnings above were $783 million in 2020, $788 million in 2019 and $932 million in 2018. In 2020, such expenses also exclude after-tax goodwill and indefinite-lived intangible asset impairment charges of $10.4 billion. These expenses are included in "Other" in the summary of earnings on page K-33 and in the "Other" earnings section on page K-56.*

Manufacturing

Our manufacturing group includes a variety of industrial, building and consumer products businesses. A summary of revenues and pre-tax earnings of our manufacturing operations follows (dollars in millions).

	2020	2019	2018	Percentage change 2020 vs 2019	Percentage change 2019 vs 2018
Revenues					
Industrial products	$ 25,667	$ 30,594	$ 30,679	(16.1)%	(0.3)%
Building products	21,244	20,327	18,677	4.5	8.8
Consumer products	12,168	11,809	12,527	3.0	(5.7)
	$ 59,079	$ 62,730	$ 61,883		
Pretax earnings					
Industrial products	$ 3,755	$ 5,635	$ 5,822	(33.4)%	(3.2)%
Building products	2,858	2,636	2,336	8.4	12.8
Consumer products	1,397	1,251	1,208	11.7	3.6
	$ 8,010	$ 9,522	$ 9,366		
Pre-tax earnings as a percentage of revenues					
Industrial products	14.6%	18.4%	19.0%		
Building products	13.5%	13.0%	12.5%		
Consumer products	11.5%	10.6%	9.6%		

Manufacturing, Service and Retailing *(Continued)*

Industrial products

The industrial products group includes specialty chemicals (The Lubrizol Corporation ("Lubrizol")), complex metal products for aerospace, power and general industrial markets (Precision Castparts Corp. ("PCC")), metal cutting tools/systems (IMC International Metalworking Companies ("IMC")), equipment and systems for the livestock and agricultural industries (CTB International ("CTB")), and a variety of industrial products for diverse markets (Marmon, Scott Fetzer and LiquidPower Specialty Products ("LSPI")). Marmon consists of more than 100 autonomous manufacturing and service businesses, including equipment leasing for the rail, intermodal tank container and mobile crane industries.

2020 versus 2019

Revenues of the industrial products group in 2020 declined $4.9 billion (16.1%) from 2019, while pre-tax earnings declined $1.9 billion (33.4%). Pre-tax earnings as a percentage of revenues for the group were 14.6% in 2020 compared to 18.4% in 2019.

PCC's revenues were $7.3 billion in 2020, a decrease of $3.0 billion (28.9%) compared to 2019. Historically, a significant portion of PCC's earnings have been dependent on sales related to the aerospace industry. The COVID-19 pandemic contributed to material declines in commercial air travel and aircraft production. Airlines responded to the pandemic by delaying delivery of aircraft orders or, in some cases, cancelling aircraft orders, resulting in significant reductions in build rates by aircraft manufacturers and significant inventory reduction initiatives by PCC's customers. Further, Boeing's 737 MAX aircraft production issues contributed to the declines in aerospace product sales across the industry in 2020. These factors resulted in significant declines in demand for PCC's aerospace products in 2020. In 2020, PCC's sales of products for power markets increased 2.2%, primarily driven by increases in industrial gas turbine products, offset by reductions in oil and gas products.

PCC's pre-tax earnings in 2020 were $650 million, a decrease of 64.5% compared to 2019, which reflected the decline in aerospace product sales as well as increased manufacturing inefficiencies attributable to lower volumes. In response to the effects of the pandemic, PCC has taken aggressive restructuring actions to resize operations in response to reduced expected volumes in aerospace markets. PCC's worldwide workforce was reduced by about 40% since the end of 2019. PCC recorded charges for restructuring and inventory and fixed asset charges of approximately $295 million in 2020. Although earnings as a percentage of revenues were negatively impacted in 2020 due to inefficiencies associated with aligning operations to reduced aircraft build rates, the restructuring actions taken contributed to improved margins in the fourth quarter compared to earlier in the year and further margin improvements are expected in the future. The level of aircraft production is currently expected to slowly increase beginning in the latter half of 2021. However, this is dependent of the timing and extent that COVID-19 infections are lowered on a sustained basis and the return to historical levels of air travel and subsequent demand for aerospace products.

Lubrizol's revenues were $5.95 billion in 2020, a decrease of 8.0% compared to 2019. The decline was primarily attributable to lower volumes from economic effects of the pandemic and a fire at an Additives manufacturing, blending and storage facility in Rouen, France at the end of the third quarter of 2019, which resulted in the temporary suspension of operations. Revenues in 2020 also reflected lower selling prices, partly offset by favorable changes in sales mix. Lubrizol's consolidated volume for the year declined 9% in 2020 compared to 2019, due to declines in the Additives and Engineered Materials product lines, partly offset by higher volumes in Life Science products. Overall, the effects of the pandemic on Lubrizol were more pronounced in the first half of the year, as volumes rebounded significantly in the second half.

Lubrizol's pre-tax earnings in 2020 were approximately $1.0 billion, essentially unchanged compared to 2019. The effects of lower sales volumes, including the effects from the Rouen fire and lower average selling prices were offset by lower average raw material costs, lower operating expenses and insurance recoveries in 2020 associated with the Rouen fire.

Marmon's revenues were $7.6 billion in 2020, a decrease of $681 million (8.2%) compared to 2019. Excluding the effects of business acquisitions, revenues decreased in essentially all sectors, primarily attributable to lower demand from the effects of the pandemic. The largest effects were experienced in the Transportation Products and Foodservice Technologies sectors. Additionally, revenues decreased due to lower metal prices in the Metal Services sector and the effect of business divestitures in 2019. Declines in oil prices in 2020 also adversely affected demand and revenues in the Rail & Leasing and Crane Services sectors.

Marmon's pre-tax earnings in 2020 decreased $312 million (24.3%) as compared to 2019. The decrease reflected the declines in revenues, increased restructuring charges and lower interest income. Restructuring initiatives were initiated in response to the lower product demand, particularly in the sectors most impacted by the pandemic.

Manufacturing, Service and Retailing *(Continued)*

Industrial products (Continued)

IMC's revenues declined 13.2% in 2020 compared to 2019, reflecting negative economic effects from the pandemic on demand for cutting tools in most geographic regions, partly offset by the effects of business acquisitions over the past year. IMC's pre-tax earnings declined 26.6% in 2020 versus 2019, attributable to declines in sales and margins due to lower volumes and to changes in sales mix.

2019 versus 2018

Revenues of the industrial products group were slightly lower in 2019 than in 2018 and pre-tax earnings declined 3.2% compared to 2018. Pre-tax earnings as a percentage of revenues for the group were 18.4% in 2019 compared to 19.0% in 2018.

PCC's revenues were $10.3 billion in 2019, an increase of $74 million (0.7%) compared to 2018. In 2019, PCC generated increased sales in aerospace markets, which was partially offset by lower sales in the power markets. The increase in aerospace sales was tempered due to significant efforts focused on the ramp-up requirements for certain new aerospace programs, such as LEAP, that created manufacturing inefficiencies and slowed production cycles contributing to delays in product deliveries and sales.

PCC's pre-tax earnings increased 5.1% in 2019 compared to 2018, reflecting increased sales of aerospace products and higher earnings from various non-recurring items in 2019, which were partially offset by lower earnings from the power markets due to the decrease in sales. Temporary unplanned shutdowns of certain metals facilities and metal press outages also negatively impacted earnings in 2018. PCC incurred incremental costs in 2019 to meet required deliveries to customers associated with the increased aerospace demand, which negatively affected margins and earnings. The production headwinds experienced were primarily attributable to shortages of qualified skilled labor and the rapid increase in requirements for newer, complex aerospace products.

Lubrizol's revenues were $6.5 billion in 2019, a decrease of 5.2% compared to 2018. The decline reflected lower volumes, including the effects from the Rouen fire, and unfavorable foreign currency translation effects, partly offset by higher average selling prices which were necessitated by raw material cost increases. Lubrizol's consolidated volume in 2019 declined 4% from 2018, primarily due to volume decline of 6% in the Additives product lines.

Lubrizol's pre-tax earnings in 2019 for the fourth quarter and year decreased 50.5% and 14.6%, respectively, compared to the same periods in 2018. Earnings in 2019 were significantly impacted by costs and lost business associated with the Rouen fire. Lubrizol's operating results in 2019 were also negatively affected by lower sales volumes, higher manufacturing expenses and unfavorable foreign currency translation effects, partly offset by improved material margins.

Marmon's revenues were $8.3 billion in 2019, an increase of $146 million (1.8%) compared to 2018. The revenue increase reflected the effects of business acquisitions, higher volumes in several business sectors, which were largely offset by lower distribution volumes in the Metals Services sector, unfavorable foreign currency translation and the impact of lower metal prices in the Electrical and Plumbing & Refrigeration sectors. Marmon's business acquisitions included the acquisition of the Colson Medical companies on October 31, 2019, resulting in a new Medical sector. Marmon's Rail & Leasing and Crane Services sectors benefitted from higher railcar equipment sales, railcar fleet utilization, railcar repair services, intermodal container leasing revenue and improved crane rental demand in the U.S. and Australia.

Marmon's pre-tax earnings increased $12 million in 2019 (1.0%) as compared to 2018. The earnings increase reflected the effects of business acquisitions, partly offset by lower gains from business divestitures. Earnings in 2019 also reflected increased earnings in sectors that experienced sales volume increases, which were substantially offset by lower earnings in the Metal Services and certain other sectors, the unfavorable impacts of foreign currency translation and increased interest and other expenses.

IMC's revenues declined 1.3% in 2019 as compared to 2018, reflecting unfavorable foreign currency translation effects of a stronger U.S. Dollar and lower sales in several regions, including Asia and Europe, mostly offset by increased revenues from recent business acquisitions. IMC's pre-tax earnings declined 12.8% in 2019 versus 2018, attributable to unfavorable foreign currency translation effects, changes in business mix to lower margin items and the effects of the U.S./China trade disputes.

Manufacturing, Service and Retailing *(Continued)*

Building products

The building products group includes manufactured and site-built home construction and related lending and financial services (Clayton Homes), flooring (Shaw), insulation, roofing and engineered products (Johns Manville), bricks and masonry products (Acme Building Brands), paint and coatings (Benjamin Moore), and residential and commercial construction and engineering products and systems (MiTek).

2020 versus 2019

Revenues of the building products group increased $917 million (4.5%) in 2020 compared to 2019 and pre-tax earnings increased $222 million (8.4%) over 2019. Pre-tax earnings as percentages of revenues were 13.5% in 2020 and 13.0% in 2019.

Clayton Homes' revenues were approximately $8.6 billion in 2020, an increase of $1.3 billion (17.1%) over 2019. The increase was primarily due to increases in home sales of $1.0 billion (18.4%), driven by increases in units sold and revenue per home sold and by changes in sales mix. Unit sales of site-built homes increased 28.6% in 2020 over 2019, while revenue per home increased slightly. Manufactured home unit sales increased 2.8% in 2020. Financial services revenues, which include mortgage services, insurance and interest income from lending activities increased 13.7% in 2020 compared to 2019, attributable to increased loan originations and average outstanding loan balances. Loan balances, net of allowances for credit losses, were approximately $17.1 billion at December 31, 2020 compared to $15.9 billion as of December 31, 2019.

Pre-tax earnings of Clayton Homes were approximately $1.25 billion in 2020, an increase of $152 million (13.9%) compared to 2019. The earnings increase reflected higher earnings from home sales, partly offset by higher materials costs, which lowered manufactured housing gross margin rates. Earnings in 2020 also benefitted from increased interest income, lower interest expense and higher earnings from mortgage services, partly offset by increased provisions for credit and insurance losses.

Aggregate revenues of our other building products businesses were approximately $12.6 billion in 2020, a decrease of 2.6% versus 2019. The revenue decrease reflected lower flooring volumes, partly attributable to the negative effects of the COVID-19 pandemic, partly offset by increased paint and coatings volumes, including volumes from a new agreement with Ace Hardware Stores, and increased volumes in residential markets.

Pre-tax earnings of the other building products businesses were approximately $1.6 billion in 2020, an increase of 4.6% over 2019. The earnings increase reflected the effects of lower average input costs, operating cost containment efforts and lower facilities closure costs.

2019 versus 2018

Revenues of the building products group in 2019 increased $1.65 billion (8.8%) compared to 2018, while pre-tax earnings increased 12.8% over 2018. Pre-tax earnings as percentages of revenues were 13.0% in 2019 and 12.5% in 2018.

Clayton Homes' revenues were approximately $7.3 billion in 2019, an increase of $1.3 billion (21.5%) over 2018. The comparative increase was primarily due to a 26% increase in home sales, reflecting a net increase in units sold and changes in sales mix. Unit sales of site-built homes increased 84% in 2019 over 2018, primarily due to business acquisitions, while average prices declined 5%. Manufactured home unit retail sales increased 5% and wholesale sales were 9% lower in 2019. Interest income from lending activities increased 6.7% in 2019 compared to 2018, attributable to increased originations and average outstanding loan balances. Aggregate loan balances outstanding were approximately $15.9 billion at December 31, 2019 compared to $14.7 billion as of December 31, 2018.

Clayton Homes' pre-tax earnings were $1.1 billion in 2019, an increase of $182 million (20.0%) compared to 2018. The increase was attributable to home building activities, which benefitted from the increases in home sales, and to financial services activities. Pre-tax earnings from lending and finance activities increased 12%, primarily due to an increase in interest income attributable to higher average loan balances, increased earnings from other financial services and lower credit losses, partially offset by higher interest expense, attributable to higher average borrowings and interest rates, and by higher other operating costs.

Aggregate revenues of our other building products businesses were $13.0 billion in 2019, an increase of 2.8% versus 2018. Revenues increased for paint and coatings, hard surface flooring and roofing products, attributable to a combination of increased volumes, product mix changes and increased average selling prices, while sales of brick products declined.

Manufacturing, Service and Retailing *(Continued)*

Building products (Continued)

Pre-tax earnings of the other building products businesses were $1.5 billion in 2019, an increase of 8.2% over 2018. Earnings in 2019 benefitted from a combination of increases in selling prices in certain product categories, declining raw material costs for certain commodities and operating cost control initiatives, which were partly offset by the effects of increased facilities closure costs.

Consumer products

The consumer products group includes leisure vehicles (Forest River), several apparel and footwear operations (including Fruit of the Loom, Garan, H.H. Brown Shoe Group and Brooks Sports) and a manufacturer of high-performance alkaline batteries (Duracell). This group also includes custom picture framing products (Larson Juhl) and jewelry products (Richline).

2020 versus 2019

Consumer products revenues increased of $359 million (3.0%) in 2020 versus 2019, while pre-tax earnings increased $146 million (11.7%). Pre-tax earnings as a percentage of revenues in 2020 increased 0.9 percentage points to 11.5%.

The comparative increase in revenues reflected revenue increases from Forest River and Duracell, partially offset by lower apparel and footwear revenues. Forest River revenues increased 11.7% in 2020 compared to 2019, primarily attributable to a significant increase in recreational vehicle unit sales over the last half of the year and changes in sales mix. Unit sales in the second half of 2020 increased 31% over the second half of 2019. Revenues from Duracell increased 10.0% in 2020 compared to 2019, reflecting the effects of changes in sales mix and increased volume. Apparel and footwear revenues declined 6.1% in 2020 compared to 2019.

Apparel and footwear sales volumes in the first half of 2020, particularly in the second quarter, reflected the negative effects of the pandemic, which included retail store closures, reduced or cancelled orders and pandemic-related disruptions at certain manufacturing facilities. Sales recovered somewhat in the second half of 2020, attributable to higher consumer demand and inventory restocking by retailers. Brooks Sports revenues were higher, partly attributable to the effect of the reduced sales in 2019 that were caused by shipping delays at a new distribution facility.

The comparative increase in pre-tax earnings was primarily attributable to Forest River and Duracell, partially offset by lower earnings from apparel and footwear. The increase reflected the effects of sales volumes changes and ongoing expense management efforts.

2019 versus 2018

Consumer products revenues declined $718 million (5.7%) in 2019 versus 2018, driven by a 12.9% revenue decline from Forest River, primarily due to lower unit sales. Revenues of Duracell increased 1.3% and apparel and footwear revenues declined 1.1% compared to 2018. Although revenues from Brooks Sports increased 3.5% in 2019, its operating results were negatively affected by lost sales associated with problems encountered at a distribution center that opened in the second quarter. In addition, our other apparel and other footwear businesses continue to experience lower sales volumes for certain products, reflecting the shift by major retailers towards private label products.

Consumer products pre-tax earnings increased $43 million (3.6%) in 2019 compared to 2018. The increase was primarily attributable to continuing cost containment efforts across several of the businesses and the effects of a new Duracell product launch, partially offset by the impact of lower recreational vehicle sales at Forest River.

Manufacturing, Service and Retailing *(Continued)*

Service and retailing

A summary of revenues and pre-tax earnings of our service and retailing businesses follows (dollars in millions).

				Percentage change	
	2020	2019	2018	2020 vs 2019	2019 vs 2018
Revenues					
Service	$ 12,346	$ 13,496	$ 13,333	(8.5)%	1.2%
Retailing	15,832	15,991	15,606	(1.0)	2.5
McLane Company	46,840	50,458	49,987	(7.2)	0.9
	$ 75,018	$ 79,945	$ 78,926		
Pre-tax earnings					
Service	$ 1,600	$ 1,681	$ 1,836	(4.8)%	(8.4)%
Retailing	1,028	874	860	17.6	1.6
McLane Company	251	288	246	(12.8)	17.1
	$ 2,879	$ 2,843	$ 2,942		
Pre-tax earnings as a percentage of revenues					
Service	13.0%	12.5%	13.8%		
Retailing	6.5%	5.5%	5.5%		
McLane Company	0.5%	0.6%	0.5%		

Service

Our service business group offers shared ownership programs for general aviation aircraft (NetJets) and high technology training products and services to operators of aircraft (FlightSafety). We also distribute electronic components (TTI), franchise and service a network of quick service restaurants (Dairy Queen) and offer third party logistics services that primarily serve the petroleum and chemical industries (Charter Brokerage). Other service businesses include transportation equipment leasing (XTRA) and furniture leasing (CORT), electronic news distribution, multimedia and regulatory filings (Business Wire) and the operation of a television station in Miami, Florida (WPLG).

2020 versus 2019

Service group revenues declined $1.15 billion (8.5%) in 2020 compared to 2019 and pre-tax earnings decreased $81 million (4.8%). Pre-tax earnings of the group as a percentage of revenues were 13.0% in 2020 compared to 12.5% in 2019.

The aggregate revenues of NetJets and FlightSafety in 2020 declined $816 million (13.5%) compared to 2019, reflecting lower demand for air travel and aviation services attributable to the COVID-19 pandemic. NetJets experienced a decline in flight hours of 27% and FlightSafety's commercial and corporate simulator training hours declined 30% from 2019. The comparative service group revenue decline was also attributable to the effects of the disposition of the newspaper operations in March of 2020 and lower revenues from CORT, which was driven by lower demand attributable to the effects of the pandemic. Partially offsetting these declines were revenue increases at TTI and at WPLG.

The decline in earnings reflected lower earnings from NetJets, TTI and CORT and from the effects of the divestiture of the newspaper operations, partly offset by higher earnings from XTRA, Business Wire, WPLG and FlightSafety. TTI's earnings decline reflected lower average gross margin rates, attributable to product mix changes and sales price pressures deriving from ample inventory availability. The decline at NetJets was primarily attributable to increased asset impairment charges and restructuring costs, partly offset by lower general and administrative expenses and a slight net increase in margins. The decline at CORT was driven by lower revenues, partly offset by the effects of cost control initiatives. The increase at FlightSafety was attributable to the effects of contract losses recorded in 2019 with respect to an existing government contract and cost control efforts in 2020, which more than offset significantly lower earnings from commercial and corporate training services.

Manufacturing, Service and Retailing *(Continued)*

Service (Continued)

2019 versus 2018

Service group revenues increased $163 million (1.2%) in 2019 compared to 2018, primarily attributable to increased sales at TTI and higher aviation-related services revenues (NetJets and FlightSafety), partially offset by decreases from the media businesses and Charter Brokerage. TTI's sales increased 2% in 2019 compared to the exceptionally high sales levels in 2018. TTI's sales slowed throughout 2019, attributable to softening customer demand, lower average selling prices and the effects of U.S. trade tariffs. The increase in NetJets' revenues in 2019 reflected increased lease revenue, primarily attributable to an increase in aircraft on lease, and increased flight hours, partly offset by lower revenue from prepaid flight cards. The revenue decline at Charter Brokerage was attributable to the divesture of a high revenue, low margin business in mid-2019.

Pre-tax earnings of the service group decreased $155 million (8.4%) compared to 2018. The comparative earnings decline was primarily due to lower earnings from TTI and FlightSafety, partly offset by higher earnings from NetJets. TTI's earnings decline was primarily attributable to lower gross margins, unfavorable foreign currency translation effects and higher operating expenses. The earnings decline at FlightSafety was attributable to pre-tax losses of approximately $165 million recorded in the fourth quarter of 2019 in connection with an existing government contract, partly offset by lower training equipment impairment charges. Earnings from NetJets increased in 2019, primarily attributable to increased revenues and improved fleet and operating efficiencies, which improved operating margins.

Retailing

Our largest retailing business is Berkshire Hathaway Automotive ("BHA"), which consists of over 80 auto dealerships that sell new and pre-owned automobiles and offer repair services and related products and represented 62.6% of our combined retailing revenue in 2020. BHA also operates two insurance businesses, two auto auctions and an automotive fluid maintenance products distributor. Our retailing businesses also include four home furnishings retailing businesses (Nebraska Furniture Mart, R.C. Willey, Star Furniture and Jordan's), which sell furniture, appliances, flooring and electronics and represented 20.6% of the combined retailing revenues in 2020.

Other retailing businesses include three jewelry retailing businesses (Borsheims, Helzberg and Ben Bridge), See's Candies (confectionary products), Pampered Chef (high quality kitchen tools), Oriental Trading Company (party supplies, school supplies and toys and novelties) and Detlev Louis Motorrad ("Louis"), a retailer of motorcycle accessories based in Germany.

2020 versus 2019

Retailing group revenues in 2020 declined $159 million (1.0%) compared to 2019. The spread of COVID-19 throughout the U.S. resulted in the temporary closures or restricted operations at several of our retailing businesses and effected consumer spending patterns during 2020. The severity and duration of the effects from the pandemic varied widely at our retail operations.

BHA's revenues decreased 2.9% in 2020 compared to 2019. BHA's revenues in 2020 reflected decreases in new and pre-owned vehicle sales of 2.6% as well as lower vehicle service and repair revenues. Home furnishings revenues were essentially unchanged in 2020 compared to 2019. The group experienced lower revenues in the first half of 2020, attributable to restricted store hours, which were substantially offset by increased revenues over the second half of the year. However, supply chain disruptions had a negative effect on obtaining product at certain times, which negatively affected sales levels.

The effects of the pandemic contributed to significantly lower sales in 2020 for our jewelry stores, See's Candy and Oriental Trading Company, which were more than offset by significant revenue increases from Pampered Chef and Louis. Sales volumes generally increased and operating results improved beginning in the latter part of the second quarter as our operations slowly reopened.

Manufacturing, Service and Retailing *(Continued)*

Retailing (Continued)

Retail group pre-tax earnings increased $154 million (17.6%) in 2020 from 2019. BHA's pre-tax earnings increased 37.7%, primarily due to lower selling, general and administrative expenses, lower floorplan interest expense and higher average gross sales margin rates. Aggregate pre-tax earnings for the remainder of our retailing group increased 1.1% in 2020 compared to 2019, reflecting higher earnings from the home furnishings businesses and from Pampered Chef, which were substantially offset by lower earnings from our other retailing operations.

Home furnishings group pre-tax earnings increased $79 million (36%) in 2020 versus 2019, reflecting generally higher average gross margin rates, sales mix changes and fewer sales promotions, and from lower advertising and other operating expenses. Certain of our other operations, including Pampered Chef and Louis experienced significant earnings increases in 2020, while others, including See's Candy and Oriental Trading Company, experienced significant declines driven by the negative effects of the pandemic.

2019 versus 2018

Retailing group revenues increased $385 million (2.5%) in 2019 compared to 2018. BHA's revenues increased 4.1% in 2019 over 2018, primarily attributable to an 11.5% increase in pre-owned vehicle sales, vehicle pricing increases, improvement in vehicle finance and service contract activities and vehicle repair work as compared to 2018. New vehicle sales in 2019 were relatively unchanged from 2018. Home furnishings group revenues declined 1.3% in 2019 compared to 2018, as sales were relatively unchanged or lower in each of our home furnishings operations.

Retail group pre-tax earnings increased $14 million (1.6%) in 2019 over 2018. BHA's pre-tax earnings increased 22.7%, primarily due to the increases in earnings from finance and service contract activities, partly offset by higher floorplan interest expense. Home furnishings group pre-tax earnings declined 14.7% versus 2018, reflecting the decline in revenues and generally higher operating expenses.

McLane Company

McLane operates a wholesale distribution business that provides grocery and non-food consumer products to retailers and convenience stores ("grocery") and to restaurants ("foodservice"). McLane also operates businesses that are wholesale distributors of distilled spirits, wine and beer ("beverage"). The grocery and foodservice businesses generate high sales and very low profit margins. These businesses have several significant customers, including Walmart, 7-Eleven, Yum! Brands and others. Grocery sales comprised about two-thirds of McLane's consolidated sales in 2020 with food service comprising most of the remainder. A curtailment of purchasing by any of its significant customers could have an adverse impact on periodic revenues and earnings.

2020 versus 2019

Revenues declined $3.6 billion (7.2%) in 2020 compared to 2019. The decline was attributable to COVID-19 related restaurant closures (particularly in the casual dining category) in the foodservice business and lower sales in certain product categories within the grocery business. McLane operates on a 52/53-week fiscal year and 2020 included 52 weeks compared to 53 weeks in 2019. Otherwise, revenues declined 5.2% in the grocery business and 7.7% in the foodservice business in 2020 as compared to 2019.

Pre-tax earnings decreased $37 million (12.8%) in 2020 as compared to 2019. The earnings decrease included the effects of increased LIFO inventory reserves of $22 million, credit and inventory losses of $12 million in the foodservice operations and the impact of lower sales. McLane continues to operate in an intensely competitive business environment, which is negatively affecting its current operating results. We expect that these operating conditions will continue.

2019 versus 2018

Revenues increased $471 million (0.9%) in 2019 compared to 2018. McLane's results in 2019 included 53 weeks compared to 52 weeks in 2018. Otherwise, revenues decreased roughly 3% in the grocery business and increased 3% in the foodservice business in 2019 as compared to 2018. Pre-tax earnings increased $42 million (17.1%) as compared to 2018. The earnings increase in 2019 reflected an increase in average gross margin rates and changes in business mix, partly offset by increased operating expenses, the largest portion of which was employee costs.

Investment and Derivative Gains (Losses)

A summary of investment and derivative gains and losses follows (dollars in millions).

	2020	2019	2018
Investment gains (losses)	$ 40,905	$ 71,123	$ (22,155)
Derivative gains (losses)	(159)	1,484	(300)
Gains (losses) before income taxes and noncontrolling interests	40,746	72,607	(22,455)
Income taxes and noncontrolling interests	9,155	15,162	(4,718)
Net gains (losses)	$ 31,591	$ 57,445	$ (17,737)
Effective income tax rate	21.7%	20.9%	20.8%

Investment gains (losses)

We are required to include the unrealized gains and losses arising from changes in market prices of investments in equity securities in earnings, which significantly increases the volatility of our periodic net earnings due to the magnitude of our equity securities portfolio and the inherent volatility of equity securities prices. Pre-tax investment gains included net unrealized gains of approximately $55.0 billion in 2020 attributable to changes in market prices of equity securities we held at December 31, 2020 and net losses of approximately $14.0 billion from changes in market prices during 2020 on securities sold during 2020. We recorded pre-tax unrealized investment gains of approximately $69.6 billion in 2019 attributable to changes in market prices in 2019 on equity securities we held at December 31, 2019. Pre-tax unrealized investment losses of approximately $22.7 billion were recorded in 2018 attributable to market price changes in 2018 on equity securities we held at December 31, 2018. Taxable investment gains on equity securities sold, which is the difference between sales proceeds and the original cost basis of the securities sold, were $6.2 billion in 2020, $3.2 billion in 2019 and $3.3 billion in 2018.

We believe that investment gains/losses, whether realized from sales or unrealized from changes in market prices, are often meaningless in terms of understanding our reported consolidated earnings or evaluating our periodic economic performance. We continue to believe the investment gains/losses recorded in earnings, including the changes in market prices for equity securities, in any given period has little analytical or predictive value.

Derivative gains (losses)

Derivative contract gains/losses include the changes in fair value of our equity index put option contract liabilities, which relate to contracts that were originated prior to March 2008. Substantially all remaining contracts will expire by February 2023. The periodic changes in the fair values of these liabilities are recorded in earnings and can be significant, primarily due to the volatility of underlying equity markets. As of December 31, 2020, the intrinsic value of our equity index put option contracts was $727 million and our recorded liability at fair value was approximately $1.1 billion. Our ultimate payment obligations, if any, under our contracts will be determined as of the contract expiration dates based on the intrinsic value as defined under the contracts.

Equity index put option contracts produced pre-tax losses of $159 million in 2020, pre-tax gains of $1.5 billion in 2019 and pre-tax losses of $300 million in 2018. These gains and losses reflected changes in the equity index values and shorter remaining contract durations. Settlement payments to counterparties were relatively insignificant in each of the three years.

Other

A summary of after-tax other earnings/losses follows (in millions).

	2020	2019	2018
Equity method earnings (losses)	$ 665	$ 1,023	$ (1,419)
Acquisition accounting expenses	(783)	(788)	(831)
Goodwill and intangible asset impairments	(10,381)	(96)	(280)
Corporate interest expense, before foreign currency effects	(334)	(280)	(311)
Foreign currency exchange rate gains (losses) on Berkshire and BHFC non-U.S. Dollar senior notes	(764)	58	289
Income tax expense adjustments	(60)	(377)	—
Other, principally corporate investment income	339	884	986
	$ (11,318)	$ 424	$ (1,566)

Other *(Continued)*

After-tax equity method earnings (losses) include our proportionate share of earnings attributable to our investments in Kraft Heinz, Pilot, Berkadia and Electric Transmission of Texas. Our after-tax earnings from Kraft Heinz were $170 million in 2020 and $488 million in 2019 and our after-tax losses were $1,859 million in 2018. Our earnings from Kraft Heinz included our after-tax share of goodwill and other intangible asset impairment charges recorded by Kraft Heinz in each year. Our after-tax share of such charges was $611 million in 2020, $339 million in 2019 and approximately $2.7 billion in 2018.

After-tax acquisition accounting expenses include charges arising from the application of the acquisition method in connection with certain of Berkshire's past business acquisitions. Such charges arise primarily from the amortization or impairment of intangible assets recorded in connection with those business acquisitions. Goodwill and intangible asset impairments in 2020 included after-tax charges of $9.8 billion attributable to impairments of goodwill and certain identifiable intangible assets that were recorded in connection with our acquisition of PCC in 2016. See Critical Accounting Policies on page K-63 for additional details.

Foreign currency exchange rate gains and losses pertain to Berkshire's outstanding Euro denominated debt (€6.85 billion par) and Japanese Yen denominated debt (¥625.5 billion par), and BHFC's Great Britain Pound denominated debt (£1.75 billion par). Changes in foreign currency exchange rates produced gains and losses from the periodic revaluation of these liabilities into U.S. Dollars. The gains and losses recorded in any given period can be significant due to the magnitude of the borrowings and the inherent volatility in foreign currency exchange rates.

The income tax expense adjustments relate to investments that were made between 2015 and 2018 in certain tax equity investment funds. Our investments in these funds aggregated approximately $340 million. In December 2018, we first learned of allegations by federal authorities of fraudulent conduct by the sponsor of these funds. In January 2020, the principals involved in creating the investment funds plead guilty to criminal charges related to the sale of the investments. In the first quarter of 2019, we concluded that it is more likely than not that the previously recognized income tax benefits were not valid.

Financial Condition

Our consolidated balance sheet continues to reflect very significant liquidity and a very strong capital base. Consolidated shareholders' equity at December 31, 2020 was $443.2 billion, an increase of $18.4 billion since December 31, 2019, which was net of common stock repurchases of $24.7 billion. Net earnings attributable to Berkshire shareholders was $42.5 billion and included after-tax gains on our investments of approximately $31.7 billion. During each of the last three years, changes in the market prices of our investments in equity securities produced exceptional volatility in our earnings. Our results in 2020 also included after-tax goodwill and other intangible asset impairments charges of $11.0 billion.

At December 31, 2020, our insurance and other businesses held cash, cash equivalents and U.S. Treasury Bills of $135.0 billion, which included $112.8 billion in U.S. Treasury Bills. Investments in equity and fixed maturity securities (excluding our investment in Kraft Heinz) were $301.6 billion.

Berkshire parent company debt outstanding at December 31, 2020 was $22.7 billion, an increase of $2.8 billion since December 31, 2019. In 2020, Berkshire repaid maturing senior notes of €1.0 billion and issued €1.0 billion of 0.0% senior notes due in 2025. Berkshire also issued ¥195.5 billion of senior notes (approximately $1.8 billion) with a weighted average interest rate of 1.07% and maturity dates ranging from 2023 to 2060. In the first quarter of 2021, senior notes of $1.7 billion will mature, including $665 million (€550 million) that matured in January. In January 2021, Berkshire issued €600 million of 0.5% senior notes due in 2041.

Berkshire's insurance and other subsidiary outstanding borrowings were approximately $18.9 billion at December 31, 2020, which included senior note borrowings of BHFC, a wholly-owned financing subsidiary, of approximately $13.1 billion. BHFC's borrowings are used to fund a portion of loans originated and acquired by Clayton Homes and equipment held for lease by our railcar leasing business. In 2020, BHFC repaid $900 million of maturing senior notes and issued $3.0 billion of senior notes with maturity dates ranging from 2030 to 2050 and a weighted average interest rate of 2.3%. Berkshire guarantees the full and timely payment of principal and interest with respect to BHFC's senior notes. In January 2021, $750 million of BHFC debt matured and BHFC issued $750 million of 2.5% senior notes due in 2051.

Our railroad, utilities and energy businesses (conducted by BNSF and BHE) maintain very large investments in capital assets (property, plant and equipment) and will regularly make significant capital expenditures in the normal course of business. Capital expenditures of these two operations were $9.8 billion in 2020 and we forecast a similar amount of capital expenditures in 2021.

Financial Condition *(Continued)*

BNSF's outstanding debt was $23.2 billion as of December 31, 2020. In 2020, BNSF issued $575 million of 3.05% senior unsecured debentures due in 2051. Outstanding borrowings of BHE and its subsidiaries were $52.2 billion at December 31, 2020, an increase of $9.6 billion since December 31, 2019. In 2020, BHE and its subsidiaries issued new term debt of approximately $7.6 billion with maturity dates ranging from 2025 to 2062 and repaid approximately $3.2 billion of debt. BHE also assumed $5.6 billion in debt in connection with the business acquired from Dominion Energy in November 2020. Berkshire does not guarantee the repayment of debt issued by BNSF, BHE or any of their subsidiaries and is not committed to provide capital to support BNSF, BHE or any of their subsidiaries.

Berkshire's common stock repurchase program as amended permits Berkshire to repurchase its Class A and Class B shares at prices below Berkshire's intrinsic value, as conservatively determined by Warren Buffett, Berkshire's Chairman of the Board and Chief Executive Officer, and Charlie Munger, Vice Chairman of the Board. The program allows share repurchases in the open market or through privately negotiated transactions and does not specify a maximum number of shares to be repurchased. The program is expected to continue indefinitely. We will not repurchase our stock if it reduces the total amount of Berkshire's consolidated cash, cash equivalents and U.S. Treasury Bill holdings below $20 billion. Financial strength and redundant liquidity will always be of paramount importance at Berkshire. In 2020, Berkshire paid $24.7 billion to repurchase shares of its Class A and B common stock.

Contractual Obligations

We are party to contracts associated with ongoing business and financing activities, which will result in cash payments to counterparties in future periods. Certain obligations are included in our Consolidated Balance Sheets, such as notes payable, which require future payments on contractually specified dates and in fixed and determinable amounts. Other obligations pertaining to the acquisition of goods or services in the future, such as certain purchase obligations, are not currently reflected in the financial statements and will be recognized in future periods as the goods are delivered or services are provided. The timing and amount of the payments under insurance and reinsurance contracts are contingent upon the outcome of future events. Actual payments will likely vary, perhaps materially, from the estimated liabilities currently recorded in our Consolidated Balance Sheet.

A summary of our contractual obligations as of December 31, 2020 follows (in millions). Actual payments will likely vary, perhaps significantly, from estimates reflected in the table.

	Estimated payments due by period				
	Total	**2021**	**2022-2023**	**2024-2025**	**After 2025**
Notes payable and other borrowings, including interest	$ 182,004	$ 13,456	$ 23,393	$ 19,596	$ 125,559
Operating leases	6,318	1,342	2,016	1,269	1,691
Purchase obligations *(1)*	48,413	14,552	7,947	5,939	19,975
Unpaid losses and loss adjustment expenses *(2)*	120,820	27,617	28,623	16,144	48,436
Life, annuity and health insurance benefits *(3)*	36,920	2,623	269	540	33,488
Other	26,524	3,136	7,762	1,684	13,942
Total	$ 420,999	$ 62,726	$ 70,010	$ 45,172	$ 243,091

(1) Primarily related to fuel, capacity, transmission and maintenance contracts and capital expenditure commitments of BHE and BNSF and aircraft purchase commitments of NetJets.

(2) Includes unpaid losses and loss adjustment expenses under retroactive reinsurance contracts.

(3) Amounts represent estimated undiscounted benefits, net of estimated future premiums, as applicable.

Critical Accounting Policies

Certain accounting policies require us to make estimates and judgments in determining the amounts reflected in the Consolidated Financial Statements. Such estimates and judgments necessarily involve varying, and possibly significant, degrees of uncertainty. Accordingly, certain amounts currently recorded in the financial statements will likely be adjusted in the future based on new available information and changes in other facts and circumstances. A discussion of our principal accounting policies that required the application of significant judgments as of December 31, 2020 follows.

Property and casualty insurance unpaid losses

We record liabilities for unpaid losses and loss adjustment expenses (also referred to as "gross unpaid losses" or "claim liabilities") based upon estimates of the ultimate amounts payable for losses occurring on or before the balance sheet date. The timing and amount of ultimate loss payments are contingent upon, among other things, the timing of claim reporting from insureds and ceding companies and the final determination of the loss amount through the loss adjustment process. We use a variety of techniques in establishing claim liabilities and all techniques require significant judgments and assumptions.

As of the balance sheet date, recorded claim liabilities include liabilities for reported claims and for claims not yet reported. The period between the loss occurrence date and loss settlement date is the "claim-tail." Property claims usually have relatively short claim-tails, absent litigation. Casualty claims usually have longer claim-tails, occasionally extending for decades. Casualty claims may be more susceptible to litigation and the impact of changing contract interpretations. The legal environment and judicial process further contribute to extending claim-tails.

Our consolidated claim liabilities as of December 31, 2020 were approximately $120.8 billion (including liabilities from retroactive reinsurance), of which 83% related to GEICO and the Berkshire Hathaway Reinsurance Group. Additional information regarding significant uncertainties inherent in the processes and techniques for estimating unpaid losses of these businesses follows.

GEICO

GEICO predominantly writes private passenger auto insurance. As of December 31, 2020, GEICO's gross unpaid losses were $22.9 billion and claim liabilities, net of reinsurance recoverable, were $21.8 billion.

GEICO's claim reserving methodologies produce liability estimates based upon the individual claims. The key assumptions affecting our liability estimates include projections of ultimate claim counts ("frequency") and average loss per claim ("severity"). We utilize a combination of several actuarial estimation methods, including Bornhuetter-Ferguson and chain-ladder methodologies.

Claim liability estimates for automobile liability coverages (such as bodily injury ("BI"), uninsured motorists, and personal injury protection) are more uncertain due to the longer claim-tails, so we establish additional case development estimates. As of December 31, 2020, case development liabilities averaged approximately 33% of the case reserves. We select case development factors through analysis of the overall adequacy of historical case liabilities.

Incurred-but-not-reported ("IBNR") claim liabilities are based on projections of the ultimate number of claims expected (reported and unreported) for each significant coverage. We use historical claim count data to develop age-to-age projections of the ultimate counts by quarterly accident period, from which we deduct reported claims to produce the number of unreported claims. We estimate the average costs per unreported claim and apply such estimates to the unreported claim counts, producing an IBNR liability estimate. We may record additional IBNR estimates when actuarial techniques are difficult to apply.

We test the adequacy of the aggregate claim liabilities using one or more actuarial projections based on claim closure models and paid and incurred loss triangles. Each type of projection analyzes loss occurrence data for claims occurring in a given period and projects the ultimate cost.

Our claim liability estimates recorded at the end of 2019 were reduced by $253 million during 2020, which produced a corresponding increase to pre-tax earnings. The assumptions used to estimate liabilities at December 31, 2020 reflect the most recent frequency and severity results. Future development of recorded liabilities will depend on whether actual frequency and severity of claims are more or less than anticipated.

Property and casualty losses *(Continued)*

GEICO (Continued)

With respect to liabilities for BI claims, we believe it is reasonably possible that average severities will change by at least one percentage point from the severities used in establishing the recorded liabilities at December 31, 2020. We estimate that a one percentage point increase or decrease in BI severities would produce a $300 million increase or decrease in recorded liabilities, with a corresponding decrease or increase in pre-tax earnings. Many of the economic forces that would likely cause BI severity to differ from expectations would likely also cause severities for other injury coverages to differ in the same direction.

Berkshire Hathaway Reinsurance Group

BHRG's liabilities for unpaid losses and loss adjustment expenses derive primarily from reinsurance contracts issued through NICO and General Re. A summary of BHRG's property and casualty unpaid losses and loss adjustment expenses, other than retroactive reinsurance losses and loss adjustment expenses, as of December 31, 2020 follows (in millions).

	Property	Casualty	Total
Reported case liabilities	$ 5,714	$ 9,497	$ 15,211
IBNR liabilities	5,821	14,615	20,436
Gross unpaid losses and loss adjustment expenses	11,535	24,112	35,647
Reinsurance recoverable	181	864	1,045
Net unpaid losses and loss adjustment expenses	$ 11,354	$ 23,248	$ 34,602

Gross unpaid losses and loss adjustment expenses consist primarily of traditional property and casualty coverages written primarily under excess-of-loss and quota-share treaties. Under certain contracts, coverage can apply to multiple lines of business written and the ceding company may not report loss data by such lines consistently, if at all. In those instances, we allocate losses to property and casualty coverages based on internal estimates.

In connection with reinsurance contracts, the nature, extent, timing and perceived reliability of loss information received from ceding companies varies widely depending on the type of coverage and the contractual reporting terms. Contract terms, conditions and coverages also tend to lack standardization and may evolve more rapidly than primary insurance policies.

The nature and extent of loss information provided under many facultative (individual risk) or per occurrence excess contracts may be comparable to the information received under a primary insurance contract. However, loss information is often less detailed with respect to aggregate excess-of-loss and quota-share contracts and is often in a summary format rather than on an individual claim basis. Loss data includes recoverable paid losses, as well as case loss estimates. Ceding companies infrequently provide reliable IBNR estimates to reinsurers.

Loss reporting to reinsurers is typically slower in comparison to primary insurers. In the U.S., such reporting is generally required at quarterly intervals ranging from 30 to 90 days after the end of the quarterly period, while outside of the U.S., reinsurance reporting practices may vary further. In certain countries, clients report annually from 90 to 180 days after the end of the annual period. Reinsurers may assume and cede underlying risks from other reinsurers, which may further delay the reporting of claims. The relative impact of reporting delays on the reinsurer may vary depending on the type of coverage, contractual reporting terms, the magnitude of the claim relative to the attachment point of the reinsurance coverage, and for other reasons.

As reinsurers, the premium and loss data we receive is at least one level removed from the underlying claimant, so there is a risk that the loss data reported is incomplete, inaccurate or the claim is outside the coverage terms. We maintain certain internal procedures in order to determine that the information is complete and in compliance with the contract terms. Generally, our reinsurance contracts permit us to access the ceding company's records with respect to the subject business, thus providing the ability to audit the reported information. In the normal course of business, disputes occasionally arise concerning whether claims are covered by our reinsurance policies. We resolve most coverage disputes through negotiation with the client. If disputes cannot be resolved, our contracts generally provide arbitration or alternative dispute resolution processes. There are no coverage disputes at this time for which an adverse resolution would likely have a material impact on our consolidated results of operations or financial condition.

Property and casualty losses *(Continued)*

Berkshire Hathaway Reinsurance Group (Continued)

Establishing claim liability estimates for reinsurance requires evaluation of loss information received from our clients. We generally rely on the ceding companies reported case loss estimates. We independently evaluate certain reported case losses and if appropriate, we use our own case liability estimate. For instance, as of December 31, 2020, our case loss estimates exceeded ceding company estimates by approximately $800 million for certain legacy workers' compensation claims occurring over 10 years ago. We also periodically conduct detailed reviews of individual client claims, which may cause us to adjust our case estimates.

Although liabilities for losses are initially determined based on pricing and underwriting analysis, BHRG uses a variety of actuarial methodologies that place reliance on the extrapolation of actual historical data, loss development patterns, industry data and other benchmarks, as appropriate. The estimate of the required IBNR liabilities also requires judgment by actuaries and management to reflect the impact of additional factors like change in business mix, volume, claim reporting and handling practices, inflation, social and legal environment and the terms and conditions of the contracts. The methodologies generally fall into one of the following categories or are hybrids of one or more of the following categories:

Paid and incurred loss development methods – these methods consider expected case loss emergence and development patterns, together with expected loss ratios by year. Factors affecting our loss development analysis include, but are not limited to, changes in the following: client claims reporting and settlement practices; the frequency of client company claim reviews; policy terms and coverage (such as loss retention levels and occurrence and aggregate policy limits); loss trends; and legal trends that result in unanticipated losses. Collectively, these factors influence our selections of expected case loss emergence patterns.

Incurred and paid loss Bornhuetter-Ferguson methods – these methods consider actual paid and incurred losses and expected patterns of paid and incurred losses, taking the initial expected ultimate losses into account to determine an estimate of the expected unpaid or unreported losses.

Frequency and severity methods – these methods commonly focus on a review of the number of anticipated claims and the anticipated claims severity and may also rely on development patterns to derive such estimates. However, our processes and techniques for estimating liabilities in such analyses generally rely more on a per-policy assessment of the ultimate cost associated with the individual loss rather than with an analysis of historical development patterns of past losses.

Additional Analysis – in some cases we have established reinsurance claim liabilities on a contract-by-contract basis, determined from case loss estimates reported by the ceding company and IBNR liabilities that are primarily a function of an anticipated loss ratio for the contract and the reported case loss estimate. Liabilities are adjusted upward or downward over time to reflect case losses reported versus expected case losses, which we use to form revised judgement on the adequacy of the expected loss ratio and the level of IBNR liabilities required for unreported claims. Anticipated loss ratios are also revised to include estimates of known major catastrophe events.

Our claim liability estimation process for short-tail lines, primarily property exposures, utilizes a combination of the paid and incurred loss development methods and the incurred and paid loss Bornhuetter-Ferguson methods. Certain catastrophe, individual risk and aviation excess-of-loss contracts tend to generate low frequency/high severity losses. Our processes and techniques for estimating liabilities under such contracts generally rely more on a per contract assessment of the ultimate cost associated with the individual loss event rather than with an analysis of the historical development patterns of past losses.

For our long-tail lines, primarily casualty exposures, we may rely on different methods depending on the maturity of the business, with estimates for the most recent years being based on priced loss expectations and more mature years reflecting the paid or incurred development pattern indications.

In 2020, certain workers' compensation claims reported losses were less than expected. As a result, we reduced estimated ultimate losses for prior years' loss events by $160 million. We estimate that increases of ten percent in the tail of the expected loss emergence pattern and in the expected loss ratios would produce a net increase of approximately $1.1 billion in IBNR liabilities, producing a corresponding decrease in pre-tax earnings. We believe it is reasonably possible for these assumptions to increase at these rates.

Property and casualty losses *(Continued)*

Berkshire Hathaway Reinsurance Group (Continued)

For other casualty losses, excluding asbestos, environmental, and other latent injury claims, the overall change in estimates for prior years' events was not significant in 2020. However, the potential for significant changes in future periods remains. For certain significant casualty and general liability portfolios, we estimate that increases of five percent in the claim-tails of the expected loss emergence patterns and in the expected loss ratios would produce a net increase in our nominal IBNR liabilities and a corresponding reduction in pre-tax earnings of approximately $900 million, although outcomes of greater than or less than $900 million are possible given the diversification in worldwide business.

Estimated ultimate liabilities for asbestos, environmental and other latent injury claims, excluding amounts assumed under retroactive reinsurance contracts increased $468 million in 2020, which produced a corresponding reduction in pre-tax earnings. Net liabilities for such claims were approximately $2.1 billion at December 31, 2020. Loss estimations for these exposures are difficult to determine due to the changing legal environment and increases may be required in the future if new exposures or claimants are identified, new claims are reported or new theories of liability emerge.

Retroactive reinsurance

Our retroactive reinsurance contracts cover loss events occurring before the contract inception dates. Claim liabilities relating to our retroactive reinsurance contracts are predominately related to casualty or liability exposures. We expect the claim-tails to be very long. As of December 31, 2020, gross unpaid losses were $41.0 billion and deferred charge assets were $12.4 billion.

Our contracts are generally subject to maximum limits of indemnifications and, as such, we currently expect that maximum remaining gross losses payable under our retroactive policies will not exceed $56 billion. Absent significant judicial or legislative changes affecting asbestos, environmental or latent injury exposures, we also currently believe it unlikely that losses will develop upward to the maximum losses payable or downward by more than 15% of our estimated gross liability.

We establish liability estimates by individual contract, considering exposure and development trends. In establishing our liability estimates, we often analyze historical aggregate loss payment patterns and project expected ultimate losses under various scenarios. We assign judgmental probability factors to these scenarios and an expected outcome is determined. We then monitor subsequent loss payment activity and review ceding company reports and other available information concerning the underlying losses. We re-estimate the expected ultimate losses when significant events or significant deviations from expected results are revealed.

Certain of our retroactive reinsurance contracts include asbestos, environmental and other latent injury claims. Our estimated liabilities for such claims were approximately $12.5 billion at December 31, 2020. We do not consistently receive reliable detailed data regarding asbestos, environmental and latent injury claims from all ceding companies, particularly with respect to multi-line or aggregate excess-of-loss policies. When possible, we conduct a detailed analysis of the underlying loss data to make an estimate of ultimate reinsured losses. When detailed loss information is unavailable, we develop estimates by applying recent industry trends and projections to aggregate client data. Judgments in these areas necessarily consider the stability of the legal and regulatory environment under which we expect claims will be adjudicated. Legal reform and legislation could also have a significant impact on our ultimate liabilities.

We reduced estimated ultimate liabilities for prior years' retroactive reinsurance contracts by $399 million in 2020, which after the changes in related deferred charge assets, resulted in pre-tax earnings of $230 million. In 2020, we paid losses and loss adjustment expenses of $1.1 billion with respect to these contracts.

Property and casualty losses *(Continued)*

Retroactive reinsurance (Continued)

In connection with our retroactive reinsurance contracts, we also record deferred charge assets, which at contract inception represents the excess, if any, of the estimated ultimate liability for unpaid losses over premiums received. We amortize deferred charge assets, which produces charges to pre-tax earnings in future periods based on the expected timing and amount of loss payments. We also adjust deferred charge balances due to changes in the expected timing and ultimate amount of claim payments. Significant changes in such estimates may have a significant effect on unamortized deferred charge balances and the amount of periodic amortization. Based on the contracts in effect as of December 31, 2020, we currently estimate that amortization expense in 2021 will approximate $1.1 billion.

Other Critical Accounting Policies

Our Consolidated Balance Sheet at December 31, 2020 includes goodwill of acquired businesses of $73.7 billion and other indefinite-lived intangible assets of $18.3 billion. We evaluate these assets for impairment annually in the fourth quarter and on an interim basis if the facts and circumstances lead us to believe that more-likely-not there has been an impairment.

Goodwill and indefinite-lived intangible asset impairment reviews include determining the estimated fair values of our reporting units and intangible assets. The key assumptions and inputs used in such determinations may include forecasting revenues and expenses, cash flows and capital expenditures, as well as an appropriate discount rate and other inputs. Significant judgment by management is required in estimating the fair value of a reporting unit and in performing impairment reviews. Due to the inherent subjectivity and uncertainty in forecasting future cash flows and earnings over long periods of time, actual results may vary materially from the forecasts. If the carrying value of the indefinite-lived intangible asset exceeds fair value, the excess is charged to earnings as an impairment loss. If the carrying value of a reporting unit exceeds the estimated fair value of the reporting unit, then the excess, limited to the carrying amount of goodwill, will be charged to earnings as an impairment loss.

In response to the adverse effects of the COVID-19 pandemic, we considered whether goodwill needed to be reevaluated for impairment during the second quarter of 2020. We determined it was necessary to quantitively reevaluate goodwill for impairment for certain reporting units, and most significantly for PCC. As a result of our reviews, we recorded pre-tax goodwill impairment charges of $10.0 billion and indefinite-lived intangible asset impairment charges of $638 million of which approximately $10 billion related to PCC.

Prior to the reevaluation, the carrying value of goodwill related to PCC was approximately $17 billion. Additionally, the carrying value of PCC's indefinite-lived intangible assets was approximately $14 billion. Substantially all of these amounts were recorded in connection with Berkshire's acquisition of PCC in 2016. The effects of the COVID-19 pandemic on commercial airlines and aircraft manufacturers is particularly severe. We considered a number of factors in our reevaluation, including but not limited to the announcements by airlines concerning potential future demand, employment levels and aircraft orders, announcements by manufacturers on reduced aircraft production, and the actions we are taking or may take to restructure our operations to fit lower expected demand. In our judgment, the timing and extent of the recovery in the commercial airline and aerospace industries may be dependent on the development and wide-scale distribution of medicines and vaccines that effectively treat the virus. Consequently, we deemed it prudent under the prevailing circumstances to increase discount rates and reduce prior long-term forecasts of future cash flows for purposes of reviewing for impairments.

As of December 31, 2020, we concluded it is more likely than not that goodwill recorded in our Consolidated Balance Sheet was not impaired. Making estimates of the fair value of reporting units at this time is and will likely be significantly affected by assumptions on the severity, duration or long-term effects of the pandemic on the reporting unit's business, which we cannot reliably predict. Consequently, any fair value estimates in such instances can be subject to wide variations. The effects of the COVID-19 pandemic could prove to be worse than we currently estimate and could lead us to record additional goodwill or indefinite-lived intangible asset impairment charges in 2021.

We primarily use discounted projected future earnings or cash flow methods in determining fair values. The key assumptions and inputs used in such methods may include forecasting revenues and expenses, cash flows and capital expenditures, as well as an appropriate discount rate and other inputs. A significant amount of judgment is required in estimating the fair value of a reporting unit and in performing goodwill impairment tests.

Market Risk Disclosures

Our Consolidated Balance Sheets include substantial amounts of assets and liabilities whose fair values are subject to market risks. Our significant market risks are primarily associated with equity prices, interest rates, foreign currency exchange rates and commodity prices. The fair values of our investment portfolios and equity index put option contracts remain subject to considerable volatility. The following sections address the significant market risks associated with our business activities.

Equity Price Risk

Equity securities represent a significant portion of our investment portfolio. Strategically, we strive to invest in businesses that possess excellent economics and able and honest management, and we prefer to invest a meaningful amount in each investee. Historically, equity investments have been concentrated in relatively few issuers. At December 31, 2020, approximately 68% of the total fair value of equity securities was concentrated in four issuers.

We often hold our equity investments for long periods and short-term price volatility has occurred in the past and will occur in the future. We also strive to maintain significant levels of shareholder capital and ample liquidity to provide a margin of safety against short-term price volatility.

We are also subject to equity price risk with respect to our equity index put option contracts. Our ultimate liability with respect to these contracts is determined from the movement of the underlying stock index between the contract inception date and expiration date. The fair values of our liabilities arising from these contracts are also affected by changes in other factors such as interest rates and the remaining duration of the contracts.

The following table summarizes our equity securities and derivative contract liabilities with significant equity price risk as of December 31, 2020 and 2019 and the estimated effects of a hypothetical 30% increase and a 30% decrease in market prices as of those dates. The selected 30% hypothetical increase and decrease does not reflect the best or worst case scenario. Indeed, results from declines could be far worse due both to the nature of equity markets and the aforementioned concentrations existing in our equity investment portfolio. Dollar amounts are in millions.

	Fair Value	Hypothetical Price Change	Estimated Fair Value after Hypothetical Change in Prices	Estimated Increase (Decrease) in Net Earnings *(1)*
December 31, 2020				
Investments in equity securities	$ 281,170	30% increase	$ 362,830	$ 63,321
		30% decrease	199,547	(63,293)
Equity index put option contract liabilities	1,065	30% increase	257	638
		30% decrease	2,702	(1,293)
December 31, 2019				
Investments in equity securities	$ 248,027	30% increase	$ 319,445	$ 56,493
		30% decrease	176,749	(56,382)
Equity index put option contract liabilities	968	30% increase	267	554
		30% decrease	2,776	(1,428)

(1) The estimated increase (decrease) is after income taxes at the statutory rate in effect as of the balance sheet date.

Market Risk Disclosures *(Continued)*

Interest Rate Risk

We may also invest in bonds, loans or other interest rate sensitive instruments. Our strategy is to acquire or originate such instruments at prices considered appropriate relative to the perceived credit risk. We also issue debt in the ordinary course of business to fund business operations, business acquisitions and for other general purposes. We attempt to maintain high credit ratings, in order to minimize the cost of our debt. We infrequently utilize derivative products, such as interest rate swaps, to manage interest rate risks.

The fair values of our fixed maturity investments, loans and finance receivables, and notes payable and other borrowings will fluctuate in response to changes in market interest rates. In addition, changes in interest rate assumptions used in our equity index put option contract models cause changes in the reported liabilities. Increases and decreases in interest rates generally translate into decreases and increases in fair values of these instruments. Additionally, fair values of interest rate sensitive instruments may be affected by the creditworthiness of the issuer, prepayment options, relative values of alternative investments, the liquidity of the instrument and other general market conditions.

The following table summarizes the estimated effects of hypothetical changes in interest rates on our significant assets and liabilities that are subject to significant interest rate risk at December 31, 2020 and 2019. We assumed that the interest rate changes occur immediately and uniformly to each category of instrument and that there were no significant changes to other factors used to determine the value of the instrument. The hypothetical changes in interest rates do not reflect the best or worst case scenarios. Actual results may differ from those reflected in the table. Dollars are in millions.

		Estimated Fair Value after Hypothetical Change in Interest Rates (bp=basis points)			
	Fair Value	100 bp decrease	100 bp increase	200 bp increase	300 bp increase
December 31, 2020					
Assets:					
Investments in fixed maturity securities	$ 20,410	$ 20,622	$ 20,139	$ 19,879	$ 19,628
Investments in equity securities*	8,891	9,408	8,413	7,970	7,559
Loans and finance receivables	20,554	21,472	19,916	19,219	18,570
Liabilities:					
Notes payable and other borrowings:					
Insurance and other	46,677	50,754	42,785	39,514	36,739
Railroad, utilities and energy	92,593	102,926	83,070	75,484	69,093
Equity index put option contracts	1,065	1,125	1,008	953	900
December 31, 2019					
Assets:					
Investments in fixed maturity securities	$ 18,685	$ 19,008	$ 18,375	$ 18,075	$ 17,787
Investments in equity securities*	10,314	11,016	9,671	9,081	8,539
Loans and finance receivables	17,861	18,527	17,240	16,660	16,116
Liabilities:					
Notes payable and other borrowings:					
Insurance and other	40,589	44,334	37,454	34,799	32,534
Railroad, utilities and energy	76,237	84,758	69,160	63,218	58,193
Equity index put option contracts	968	1,065	877	792	713

* *Occidental Petroleum Cumulative Perpetual Preferred Stock*

Foreign Currency Risk

Certain of our subsidiaries operate in foreign jurisdictions and we transact business in foreign currencies. In addition, we hold investments in common stocks of major multinational companies, who have significant foreign business and foreign currency risk of their own. We generally do not attempt to match assets and liabilities by currency and do not use derivative contracts to manage foreign currency risks in a meaningful way.

Our net assets subject to financial statement translation into U.S. Dollars are primarily in our insurance, utilities and energy and certain manufacturing and service subsidiaries. A portion of our financial statement translation-related impact from changes in foreign currency rates is recorded in other comprehensive income. In addition, we include gains or losses in net earnings related to certain liabilities of Berkshire and U.S. insurance subsidiaries that are denominated in foreign currencies, due to changes in exchange rates. A summary of these gains (losses), after-tax, for each of the years ending December 31, 2020 and 2019 follows (in millions).

	2020	2019
Non-U.S. denominated debt included in net earnings	$ (764)	$ 58
Net liabilities under certain reinsurance contracts included in net earnings	(163)	(92)
Foreign currency translation included in other comprehensive income	1,264	257

Commodity Price Risk

Our subsidiaries use commodities in various ways in manufacturing and providing services. As such, we are subject to price risks related to various commodities. In most instances, we attempt to manage these risks through the pricing of our products and services to customers. To the extent that we are unable to sustain price increases in response to commodity price increases, our operating results will likely be adversely affected. We do not utilize derivative contracts to manage commodity price risks to any significant degree.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

See "Market Risk Disclosures" contained in Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations."

Management's Report on Internal Control Over Financial Reporting

Management of Berkshire Hathaway Inc. is responsible for establishing and maintaining adequate internal control over financial reporting, as such term is defined in the Securities Exchange Act of 1934 Rule 13a-15(f). Under the supervision and with the participation of our management, including our principal executive officer and principal financial officer, we conducted an evaluation of the effectiveness of the Company's internal control over financial reporting as of December 31, 2020 as required by the Securities Exchange Act of 1934 Rule 13a-15(c). In making this assessment, we used the criteria set forth in the framework in *Internal Control—Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission. Based on our evaluation under the framework in *Internal Control—Integrated Framework* (2013), our management concluded that our internal control over financial reporting was effective as of December 31, 2020.

The effectiveness of our internal control over financial reporting as of December 31, 2020 has been audited by Deloitte & Touche LLP, an independent registered public accounting firm, as stated in their report which appears on page K-67.

Berkshire Hathaway Inc.
February 27, 2021

Item 8. Financial Statements and Supplementary Data

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Shareholders and the Board of Directors of
Berkshire Hathaway Inc.
Omaha, Nebraska

Opinions on the Financial Statements and Internal Control over Financial Reporting
We have audited the accompanying consolidated balance sheets of Berkshire Hathaway Inc. and subsidiaries (the "Company") as of December 31, 2020 and 2019, the related consolidated statements of earnings, comprehensive income, changes in shareholders' equity, and cash flows, for each of the three years in the period ended December 31, 2020, and the related notes (collectively referred to as the "financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2020, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2020 and 2019, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2020, in conformity with accounting principles generally accepted in the United States of America. Also, in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2020, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by COSO.

Basis for Opinions
The Company's management is responsible for these financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on these financial statements and an opinion on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the US federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the financial statements included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures to respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting
A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current-period audit of the financial statements that were communicated or required to be communicated to the audit committee and that (1) relate to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Unpaid Losses and Loss Adjustment Expenses— Refer to Notes 1 and 15 to the financial statements

Critical Audit Matter Description

The Company's unpaid losses and loss adjustment expenses ("claim liabilities") under short duration property and casualty insurance and reinsurance contracts are $79,854 million as of December 31, 2020. The key assumptions affecting certain claim liabilities include expected loss and expense ("loss") ratios, expected claim count emergence patterns, expected loss payment emergence patterns and expected loss reporting emergence patterns.

Given the subjectivity of estimating these key assumptions, performing audit procedures to evaluate whether claim liabilities were appropriately recorded as of December 31, 2020, required a high degree of auditor judgment and an increased extent of effort, including the need to involve our actuarial specialists.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to the key assumptions affecting certain claim liabilities included the following, among others:

- We tested the operating effectiveness of controls over claim liabilities, including those over the key assumptions.
- We evaluated the methods and assumptions used by management to estimate the claim liabilities by:
  - Testing the underlying data that served as the basis for the actuarial analysis, such as historical claims and earned premium, to test that the inputs to the actuarial estimate were reasonable.
  - Comparing management's prior-year claim liabilities to actual development during the current year to identify potential bias in the determination of the claim liabilities.
- With the assistance of our actuarial specialists:
  - We developed independent estimates of the claim liabilities, including loss data and industry claim development factors as needed, and compared our estimates to management's estimates.
  - We compared management's change in ultimate loss and loss adjustment expense to prior year estimates to test the reasonableness of the prior year estimates and assessed unexpected development.

Unpaid Losses and Loss Adjustment Expenses Under Retroactive Reinsurance Contracts — Refer to Notes 1 and 16 to the financial statements

Critical Audit Matter Description

The Company's unpaid losses and loss adjustment expenses ("claim liabilities") for property and casualty retroactive reinsurance contracts are $40,966 million as of December 31, 2020. The key assumptions affecting certain claim liabilities and related deferred charge reinsurance assumed assets ("related assets") include expected loss and expense ("loss") ratios, expected loss payment emergence patterns and expected loss reporting emergence.

Given the subjectivity of estimating these key assumptions, performing audit procedures to evaluate whether claim liabilities were appropriately recorded as of December 31, 2020, required a high degree of auditor judgment and an increased extent of effort, including the need to involve our actuarial specialists.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to the key assumptions affecting claim liabilities and related assets included the following, among others:

- We tested the operating effectiveness of controls over claim liabilities and related assets, including those over the key assumptions.

- We evaluated the methods and assumptions used by management to estimate the claim liabilities and related assets by:
  - Testing the underlying data that served as the basis for the actuarial analysis, including historical claims, to test that the inputs to the actuarial estimate were reasonable.
  - Comparing management's prior-year claim liabilities to actual development during the current year to identify potential bias in the determination of the claim liabilities and related assets.
- With the assistance of our actuarial specialists:
  - We developed independent claim liability estimates for certain retroactive reinsurance contracts and compared our estimates to management's estimates. For other retroactive reinsurance contracts and related assets, we evaluated the process used by management to develop the estimated claim liabilities and related assets.
  - We compared management's change in ultimate loss and loss adjustment expense to prior year estimates, assessed unexpected development and assessed internal rates of return.

Goodwill and Indefinite-Lived Intangible Assets — Refer to Notes 1 and 13 to the financial statements

Critical Audit Matter Description

The Company's evaluation of goodwill and indefinite-lived intangible assets for impairment involves the comparison of the fair value of each reporting unit or asset to its carrying value. The Company evaluates goodwill and indefinite-lived intangible assets for impairment at least annually. When evaluating goodwill and indefinite-lived intangible assets for impairment, the fair value of each reporting unit or asset is estimated. Significant judgment is required in estimating fair values and performing impairment tests. The Company primarily uses discounted projected future net earnings or net cash flows and multiples of earnings to estimate fair value, which requires management to make significant estimates and assumptions related to forecasts of future revenue, earnings before interest and taxes ("EBIT"), and discount rates. Changes in these assumptions could have a significant impact on the fair value of reporting units and indefinite-lived intangible assets.

The Precision Castparts Corp. ("PCC") reporting unit reported approximately $31 billion of goodwill and indefinite-lived intangible assets as of December 31, 2019. During the second quarter of 2020, the Company performed an interim reevaluation of the goodwill and indefinite-lived intangible assets at the PCC reporting unit. This determination was made due to disruptions arising from the COVID-19 pandemic that had an adverse impact on the industries in which PCC operates. As a result of the reevaluation, the Company recognized goodwill and indefinite-lived intangible asset impairment charges in the amount of approximately $10 billion, as the fair values of the PCC reporting unit and indefinite-lived intangible assets were less than their respective carrying values. As a result, PCC reported goodwill and indefinite-lived intangible assets of approximately $21 billion as of December 31, 2020.

Given the significant judgments made by management to estimate the fair value of the PCC reporting unit and certain customer relationships with indefinite lives along with the difference between their fair values and carrying values, performing audit procedures to evaluate the reasonableness of management's estimates and assumptions related to forecasts of future revenue and EBIT and the selection of the discount rate required a high degree of auditor judgment and an increased extent of effort, including the need to involve our fair value specialists.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to forecasts of future revenue and EBIT and the selection of the discount rate for the PCC reporting unit and certain customer relationships included the following, among others:

- We tested the effectiveness of controls over goodwill and indefinite-lived intangible assets, including those over the forecasts of future revenue and EBIT and the selection of the discount rate.
- We evaluated management's ability to accurately forecast future revenue and EBIT by comparing prior year forecasts to actual results in the respective years.
- We evaluated the reasonableness of management's current revenue and EBIT forecasts by comparing the forecasts to historical results and forecasted information included in analyst and industry reports and certain peer companies' disclosures.
- With the assistance of our fair value specialists, we evaluated the valuation methodologies, the long-term growth rates and discount rate, including testing the underlying source information and the mathematical accuracy of the calculations, and developed a range of independent estimates and compared those to the long-term growth rates and discount rate selected by management.

/s/ Deloitte & Touche LLP
Omaha, Nebraska
February 27, 2021

We have served as the Company's auditor since 1985.

BERKSHIRE HATHAWAY INC.
and Subsidiaries
CONSOLIDATED BALANCE SHEETS
(dollars in millions)

	December 31,	
	2020	**2019**
ASSETS		
Insurance and Other:		
Cash and cash equivalents*	$ 44,714	$ 61,151
Short-term investments in U.S. Treasury Bills	90,300	63,822
Investments in fixed maturity securities	20,410	18,685
Investments in equity securities	281,170	248,027
Equity method investments	17,303	17,505
Loans and finance receivables	19,201	17,527
Other receivables	32,310	32,418
Inventories	19,208	19,852
Property, plant and equipment	21,200	21,438
Equipment held for lease	14,601	15,065
Goodwill	47,121	57,052
Other intangible assets	29,462	31,051
Deferred charges under retroactive reinsurance contracts	12,441	13,747
Other	14,580	13,232
	664,021	630,572
Railroad, Utilities and Energy:		
Cash and cash equivalents*	3,276	3,024
Receivables	3,542	3,417
Property, plant and equipment	151,216	137,838
Goodwill	26,613	24,830
Regulatory assets	3,440	2,881
Other	21,621	15,167
	209,708	187,157
	$ 873,729	$ 817,729

* *Includes U.S. Treasury Bills with maturities of three months or less when purchased of $23.2 billion at December 31, 2020 and $37.1 billion at December 31, 2019.*

See accompanying Notes to Consolidated Financial Statements

BERKSHIRE HATHAWAY INC.
and Subsidiaries
CONSOLIDATED BALANCE SHEETS
(dollars in millions)

	December 31,	
	2020	**2019**
LIABILITIES AND SHAREHOLDERS' EQUITY		
Insurance and Other:		
Unpaid losses and loss adjustment expenses	$ 79,854	$ 73,019
Unpaid losses and loss adjustment expenses under retroactive reinsurance contracts	40,966	42,441
Unearned premiums	21,395	19,782
Life, annuity and health insurance benefits	21,616	20,155
Other policyholder liabilities	8,670	7,723
Accounts payable, accruals and other liabilities	29,279	27,611
Derivative contract liabilities	1,065	968
Aircraft repurchase liabilities and unearned lease revenues	5,856	5,281
Notes payable and other borrowings	41,522	37,590
	250,223	234,570
Railroad, Utilities and Energy:		
Accounts payable, accruals and other liabilities	15,224	14,708
Regulatory liabilities	7,475	7,311
Notes payable and other borrowings	75,373	65,778
	98,072	87,797
Income taxes, principally deferred	74,098	66,799
Total liabilities	422,393	389,166
Shareholders' equity:		
Common stock	8	8
Capital in excess of par value	35,626	35,658
Accumulated other comprehensive income	(4,243)	(5,243)
Retained earnings	444,626	402,493
Treasury stock, at cost	(32,853)	(8,125)
Berkshire Hathaway shareholders' equity	443,164	424,791
Noncontrolling interests	8,172	3,772
Total shareholders' equity	451,336	428,563
	$ 873,729	$ 817,729

See accompanying Notes to Consolidated Financial Statements

BERKSHIRE HATHAWAY INC.
and Subsidiaries
CONSOLIDATED STATEMENTS OF EARNINGS
(dollars in millions except per share amounts)

	Year Ended December 31,		
	2020	**2019**	**2018**
Revenues:			
Insurance and Other:			
Insurance premiums earned	$ 63,401	$ 61,078	$ 57,418
Sales and service revenues	127,044	134,989	133,336
Leasing revenues	5,209	5,856	5,732
Interest, dividend and other investment income	8,092	9,240	7,678
	203,746	211,163	204,164
Railroad, Utilities and Energy:			
Freight rail transportation revenues	20,750	23,357	23,703
Energy operating revenues	15,540	15,353	15,555
Service revenues and other income	5,474	4,743	4,415
	41,764	43,453	43,673
Total revenues	245,510	254,616	247,837
Investment and derivative contract gains/losses:	40,746	72,607	(22,455)
Costs and expenses:			
Insurance and Other:			
Insurance losses and loss adjustment expenses	43,951	44,456	39,906
Life, annuity and health insurance benefits	5,812	4,986	5,699
Insurance underwriting expenses	12,798	11,200	9,793
Cost of sales and services	101,091	107,041	106,083
Cost of leasing	3,520	4,003	4,061
Selling, general and administrative expenses	19,809	19,226	17,856
Goodwill and intangible asset impairments	10,671	96	382
Interest expense	1,105	1,056	1,035
	198,757	192,064	184,815
Railroad, Utilities and Energy:			
Freight rail transportation expenses	13,120	15,436	16,045
Utilities and energy cost of sales and other expenses	11,638	11,296	11,641
Other expenses	4,796	4,002	3,895
Interest expense	2,978	2,905	2,818
	32,532	33,639	34,399
Total costs and expenses	231,289	225,703	219,214
Earnings before income taxes and equity method earnings (losses)	54,967	101,520	6,168
Equity method earnings (losses)	726	1,176	(2,167)
Earnings before income taxes	55,693	102,696	4,001
Income tax expense (benefit)	12,440	20,904	(321)
Net earnings	43,253	81,792	4,322
Earnings attributable to noncontrolling interests	732	375	301
Net earnings attributable to Berkshire Hathaway shareholders	$ 42,521	$ 81,417	$ 4,021
Net earnings per average equivalent Class A share	$ 26,668	$ 49,828	$ 2,446
Net earnings per average equivalent Class B share*	$ 17.78	$ 33.22	$ 1.63
Average equivalent Class A shares outstanding	1,594,469	1,633,946	1,643,795
Average equivalent Class B shares outstanding	2,391,703,454	2,450,919,020	2,465,692,368

* *Class B shares are economically equivalent to one-fifteen-hundredth of a Class A share. Accordingly, net earnings per average equivalent Class B share outstanding is equal to one-fifteen-hundredth of the equivalent Class A amount. See Note 22.*

See accompanying Notes to Consolidated Financial Statements

BERKSHIRE HATHAWAY INC.
and Subsidiaries
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME
(dollars in millions)

	Year Ended December 31,		
	2020	**2019**	**2018**
Net earnings	$ 43,253	$ 81,792	$ 4,322
Other comprehensive income:			
Unrealized appreciation of investments	74	142	(438)
Applicable income taxes	(19)	(31)	84
Foreign currency translation	1,284	323	(1,531)
Applicable income taxes	3	(28)	62
Defined benefit pension plans	(355)	(711)	(571)
Applicable income taxes	74	155	143
Other, net	(42)	(48)	(12)
Other comprehensive income, net	1,019	(198)	(2,263)
Comprehensive income	44,272	81,594	2,059
Comprehensive income attributable to noncontrolling interests	751	405	249
Comprehensive income attributable to Berkshire Hathaway shareholders	$ 43,521	$ 81,189	$ 1,810

BERKSHIRE HATHAWAY INC.
and Subsidiaries
CONSOLIDATED STATEMENTS OF CHANGES IN SHAREHOLDERS' EQUITY
(dollars in millions)

	Berkshire Hathaway shareholders' equity					
	Common stock and capital in excess of par value	**Accumulated other comprehensive income**	**Retained earnings**	**Treasury stock**	**Non-controlling interests**	**Total**
Balance December 31, 2017	$ 35,702	$ 58,571	$255,786	$ (1,763)	$ 3,658	$351,954
Adoption of new accounting pronouncements	—	(61,375)	61,305	—	—	(70)
Net earnings	—	—	4,021	—	301	4,322
Other comprehensive income, net	—	(2,211)	—	—	(52)	(2,263)
Issuance (acquisition) of common stock	59	—	—	(1,346)	—	(1,287)
Transactions with noncontrolling interests	(46)	—	—	—	(110)	(156)
Balance December 31, 2018	35,715	(5,015)	321,112	(3,109)	3,797	352,500
Net earnings	—	—	81,417	—	375	81,792
Other comprehensive income, net	—	(228)	—	—	30	(198)
Issuance (acquisition) of common stock	21	—	—	(5,016)	—	(4,995)
Transactions with noncontrolling interests	(70)	—	(36)	—	(430)	(536)
Balance December 31, 2019	35,666	(5,243)	402,493	(8,125)	3,772	428,563
Net earnings	—	—	42,521	—	732	43,253
Adoption of new accounting pronouncement	—	—	(388)	—	—	(388)
Other comprehensive income, net	—	1,000	—	—	19	1,019
Issuance (acquisition) of common stock	—	—	—	(24,728)	—	(24,728)
Transactions with noncontrolling interests	(32)	—	—	—	3,649	3,617
Balance December 31, 2020	$ 35,634	$ (4,243)	$444,626	$ (32,853)	$ 8,172	$451,336

See accompanying Notes to Consolidated Financial Statements

BERKSHIRE HATHAWAY INC.
and Subsidiaries
CONSOLIDATED STATEMENTS OF CASH FLOWS
(dollars in millions)

	Year Ended December 31,		
	2020	**2019**	**2018**
Cash flows from operating activities:			
Net earnings	$ 43,253	$ 81,792	$ 4,322
Adjustments to reconcile net earnings to operating cash flows:			
Investment (gains) losses	(40,905)	(71,123)	22,155
Depreciation and amortization	10,596	10,064	9,779
Other, including asset impairment charges	11,263	(1,254)	2,957
Changes in operating assets and liabilities:			
Losses and loss adjustment expenses	4,819	6,087	3,449
Deferred charges reinsurance assumed	1,307	357	1,174
Unearned premiums	1,587	1,707	1,794
Receivables and originated loans	(1,609)	(2,303)	(3,443)
Other assets	(1,109)	(2,011)	(1,832)
Other liabilities	3,376	190	2,002
Income taxes	7,195	15,181	(4,957)
Net cash flows from operating activities	39,773	38,687	37,400
Cash flows from investing activities:			
Purchases of equity securities	(30,161)	(18,642)	(43,210)
Sales of equity securities	38,756	14,336	18,783
Purchases of U.S. Treasury Bills and fixed maturity securities	(208,429)	(136,123)	(141,844)
Sales of U.S. Treasury Bills and fixed maturity securities	31,873	15,929	39,693
Redemptions and maturities of U.S. Treasury Bills and fixed maturity securities	149,709	137,767	113,045
Purchases of loans and finance receivables	(772)	(75)	(1,771)
Collections of loans and finance receivables	393	345	342
Acquisitions of businesses, net of cash acquired	(2,532)	(1,683)	(3,279)
Purchases of property, plant and equipment and equipment held for lease	(13,012)	(15,979)	(14,537)
Other	(3,582)	(1,496)	(71)
Net cash flows from investing activities	(37,757)	(5,621)	(32,849)
Cash flows from financing activities:			
Proceeds from borrowings of insurance and other businesses	5,925	8,144	2,409
Repayments of borrowings of insurance and other businesses	(2,700)	(5,095)	(7,395)
Proceeds from borrowings of railroad, utilities and energy businesses	8,445	5,400	7,019
Repayments of borrowings of railroad, utilities and energy businesses	(3,761)	(2,638)	(4,213)
Changes in short term borrowings, net	(1,118)	266	(1,943)
Acquisition of treasury stock	(24,706)	(4,850)	(1,346)
Other	(429)	(497)	(343)
Net cash flows from financing activities	(18,344)	730	(5,812)
Effects of foreign currency exchange rate changes	92	25	(140)
Increase (decrease) in cash and cash equivalents and restricted cash	(16,236)	33,821	(1,401)
Cash and cash equivalents and restricted cash at beginning of year	64,632	30,811	32,212
Cash and cash equivalents and restricted cash at end of year *	$ 48,396	$ 64,632	$ 30,811
** Cash and cash equivalents and restricted cash at end of year are comprised of the following:*			
Insurance and Other	*$ 44,714*	*$ 61,151*	*$ 27,749*
Railroad, Utilities and Energy	*3,276*	*3,024*	*2,612*
Restricted cash, included in other assets	*406*	*457*	*450*
	$ 48,396	*$ 64,632*	*$ 30,811*

See accompanying Notes to Consolidated Financial Statements

(1) Significant accounting policies and practices

(a) Nature of operations and basis of consolidation

Berkshire Hathaway Inc. ("Berkshire") is a holding company owning subsidiaries engaged in a number of diverse business activities, including insurance and reinsurance, freight rail transportation, utilities and energy, manufacturing, service and retailing. In these notes the terms "us," "we," or "our" refer to Berkshire and its consolidated subsidiaries. Further information regarding our reportable business segments is contained in Note 27. Information concerning business acquisitions completed over the past three years appears in Note 2. We believe that reporting the Railroad, Utilities and Energy subsidiaries separately is appropriate given the relative significance of their long-lived assets, capital expenditures and debt, which is not guaranteed by Berkshire.

The accompanying Consolidated Financial Statements include the accounts of Berkshire consolidated with the accounts of all subsidiaries and affiliates in which we hold a controlling financial interest as of the financial statement date. Normally a controlling financial interest reflects ownership of a majority of the voting interests. We consolidate variable interest entities ("VIE") when we possess both the power to direct the activities of the VIE that most significantly affect its economic performance, and we (a) are obligated to absorb the losses that could be significant to the VIE or (b) hold the right to receive benefits from the VIE that could be significant to the VIE. Intercompany accounts and transactions have been eliminated.

(b) Use of estimates in preparation of financial statements

We prepare our Consolidated Financial Statements in conformity with accounting principles generally accepted in the United States ("GAAP") which requires us to make estimates and assumptions that affect the reported amounts of assets and liabilities at the balance sheet date and the reported amounts of revenues and expenses during the period. Our estimates of unpaid losses and loss adjustment expenses are subject to considerable estimation error due to the inherent uncertainty in projecting ultimate claim costs. In addition, estimates and assumptions associated with the amortization of deferred charges on retroactive reinsurance contracts, determinations of fair values of certain financial instruments and evaluations of goodwill and identifiable intangible assets for impairment require considerable judgment. Actual results may differ from the estimates used in preparing our Consolidated Financial Statements.

The novel coronavirus ("COVID-19") spread rapidly across the world in 2020 and was declared a pandemic by the World Health Organization. The government and private sector responses to contain its spread began to significantly affect our operating businesses in March. COVID-19 has since adversely affected nearly all of our operations, although the effects are varying significantly. The duration and extent of the effects over longer terms cannot be reasonably estimated at this time. The risks and uncertainties resulting from the pandemic that may affect our future earnings, cash flows and financial condition include the time necessary to distribute safe and effective vaccines and to vaccinate a significant number of people in the U.S. and throughout the world as well as the long-term effect from the pandemic on the demand for certain of our products and services. Accordingly, significant estimates used in the preparation of our financial statements including those associated with evaluations of certain long-lived assets, goodwill and other intangible assets for impairment, expected credit losses on amounts owed to us and the estimations of certain losses assumed under insurance and reinsurance contracts may be subject to significant adjustments in future periods.

(c) Cash and cash equivalents and short-term investments in U.S. Treasury Bills

Cash equivalents consist of demand deposit and money market accounts and investments (including U.S. Treasury Bills) with maturities of three months or less when purchased. Short-term investments in U.S. Treasury Bills consist of U.S. Treasury Bills with maturities exceeding three months at the time of purchase and are stated at amortized cost, which approximates fair value.

(1) Significant accounting policies and practices *(Continued)*

(d) Investments in fixed maturity securities

We classify investments in fixed maturity securities on the acquisition date and at each balance sheet date. Securities classified as held-to-maturity are carried at amortized cost, reflecting the ability and intent to hold the securities to maturity. Securities classified as trading are acquired with the intent to sell in the near term and are carried at fair value with changes in fair value reported in earnings. All other securities are classified as available-for-sale and are carried at fair value. Substantially all of these investments are classified as available-for-sale. We amortize the difference between the original cost and maturity value of a fixed maturity security to earnings using the interest method.

We record investment gains and losses on available-for-sale fixed maturity securities when the securities are sold, as determined on a specific identification basis. For securities in an unrealized loss position, we recognize a loss in earnings for the excess of amortized cost over fair value if we intend to sell before the price recovers. Otherwise, we evaluate as of the balance sheet date whether the unrealized losses are attributable to credit losses or other factors. We consider the severity of the decline in value, creditworthiness of the issuer and other relevant factors. We record an allowance for credit losses, limited to the excess of amortized cost over fair value, along with a corresponding charge to earnings if the present value of estimated cash flows is less than the present value of contractual cash flows. The allowance may be subsequently increased or decreased based on the prevailing facts and circumstances. The portion of the unrealized loss that we believe is not related to a credit loss is recognized in other comprehensive income.

(e) Investments in equity securities

We carry substantially all investments in equity securities at fair value and record the subsequent changes in fair values in the Consolidated Statements of Earnings as a component of investment gains/losses.

(f) Investments under the equity method

We utilize the equity method to account for investments when we possess the ability to exercise significant influence, but not control, over the operating and financial policies of the investee. The ability to exercise significant influence is presumed when the investor possesses more than 20% of the voting interests of the investee. This presumption may be overcome based on specific facts and circumstances that demonstrate that the ability to exercise significant influence is restricted. We apply the equity method to investments in common stock and to other investments when such other investments possess substantially identical subordinated interests to common stock.

In applying the equity method, we record the investment at cost and subsequently increase or decrease the carrying amount of the investment by our proportionate share of the net earnings or losses and other comprehensive income of the investee. We record dividends or other equity distributions as reductions in the carrying value of the investment. In the event that net losses of the investee reduce the carrying amount to zero, additional net losses may be recorded if other investments in the investee are at-risk, even if we have not committed to provide financial support to the investee. Such additional equity method losses, if any, are based upon the change in our claim on the investee's book value.

(g) Receivables

Receivables primarily consist of balances due from customers, insurance premiums receivable and reinsurance losses recoverable. Trade receivables, insurance premium receivables and other receivables are primarily short-term in nature with stated collection terms of less than one year from the date of origination. Reinsurance recoverables are comprised of amounts ceded under reinsurance contracts or pursuant to mandatory government-sponsored insurance programs. Reinsurance recoverables relate to claims for unpaid losses and loss adjustment expenses arising from property and casualty contracts and claim benefits under life and health insurance contracts. Receivables are stated net of estimated allowances for uncollectible balances. Prior to 2020, we recorded provisions for uncollectible balances when it was probable counterparties or customers would be unable to pay all amounts due based on the contractual terms and historical loss history.

As of January 1, 2020, we adopted a new accounting pronouncement that affects the measurement of allowances for credit losses. See Note 1(w). In measuring credit loss allowances, we primarily utilize credit loss history, with adjustments to reflect current or expected future economic conditions when reasonable and supportable forecasts of losses deviate from historical experience. In evaluating expected credit losses of reinsurance recoverable on unpaid losses, we review the credit quality of the counterparty and consider right-of-offset provisions within reinsurance contracts and other forms of credit enhancement including, collateral, guarantees and other available information. We charge-off receivables against the allowances after all reasonable collection efforts are exhausted.

(1) Significant accounting policies and practices *(Continued)*

(h) Loans and finance receivables

Loans and finance receivables are primarily manufactured home loans, and to lesser extent, commercial loans and site-built home loans. We carry substantially all of these loans at amortized cost, net of allowances for expected credit losses, based on our ability and intent to hold such loans to maturity. Acquisition costs and loan origination and commitment costs paid or fees received along with acquisition premiums or discounts are amortized as yield adjustments over the lives of the loans.

Prior to 2020, credit losses were measured when non-collection was considered probable based on the prevailing facts and circumstances. Beginning in 2020, measurements of expected credit losses include provisions for non-collection, whether the risk is probable or remote. Expected credit losses on manufactured housing installment loans are based on the net present value of future principal payments less estimated expenses related to the charge-off and foreclosure of expected uncollectible loans and include provisions for loans that are not in foreclosure. Our principal credit quality indicator is whether the loans are performing. Expected credit loss estimates consider historical default rates, collateral recovery rates, historical runoff rates, interest rates, reductions of future cash flows for modified loans and the historical time elapsed from last payment until foreclosure, among other factors. In addition, our estimates consider current conditions and reasonable and supportable forecasts.

Loans are considered delinquent when payments are more than 30 days past due. We place loans over 90 days past due on nonaccrual status and accrued but uncollected interest is reversed. Subsequent collections on the loans are first applied to the principal and interest owed for the most delinquent amount. We resume interest income accrual once a loan is less than 90 days delinquent.

Loans are considered non-performing when the foreclosure process has started. Once a loan is in the process of foreclosure, interest income is not recognized unless the foreclosure is cured or the loan is modified. Once a modification is complete, interest income is recognized based on the terms of the new loan. Foreclosed loans are charged off when the collateral is sold. Loans not in foreclosure are evaluated for charge-off based on individual circumstances concerning the future collectability of the loan and the condition of the collateral securing the loan.

(i) Derivatives

We carry derivative contracts in our Consolidated Balance Sheets at fair value, net of reductions permitted under master netting agreements with counterparties. We record the changes in fair value of derivative contracts that do not qualify as hedging instruments for financial reporting purposes in earnings or, if such contracts involve our regulated utilities subsidiaries, as regulatory assets or liabilities when inclusion in regulated rates is probable.

(j) Fair value measurements

As defined under GAAP, fair value is the price that would be received to sell an asset or paid to transfer a liability between market participants in the principal market or in the most advantageous market when no principal market exists. Adjustments to transaction prices or quoted market prices may be required in illiquid or disorderly markets in estimating fair value. Alternative valuation techniques may be appropriate under the circumstances to determine the value that would be received to sell an asset or paid to transfer a liability in an orderly transaction. Market participants are assumed to be independent, knowledgeable, and able and willing to transact an exchange and not acting under duress. Our nonperformance or credit risk is considered in determining the fair value of liabilities. Considerable judgment may be required in interpreting market data used to develop the estimates of fair value. Accordingly, estimates of fair value presented herein are not necessarily indicative of the amounts that could be realized in a current or future market exchange.

(k) Inventories

Inventories consist of manufactured goods, goods or products acquired for resale and homes constructed for sale. Manufactured inventory costs include materials, direct and indirect labor and factory overhead. At December 31, 2020, we used the last-in-first-out ("LIFO") method to value approximately 35% of consolidated inventories with the remainder primarily determined under first-in-first-out and average cost methods. Non-LIFO inventories are stated at the lower of cost or net realizable value. The excess of current or replacement costs over costs determined under LIFO was approximately $1.1 billion as of December 31, 2020 and $950 million as of December 31, 2019.

(1) Significant accounting policies and practices *(Continued)*

(l) Property, plant and equipment

We record additions to property, plant and equipment used in operations at cost, which includes asset additions, improvements and betterments. With respect to constructed assets, all materials, direct labor and contract services as well as certain indirect costs are capitalized. Indirect costs include interest over the construction period. With respect to constructed assets of our utility and energy subsidiaries that are subject to authoritative guidance for regulated operations, capitalized costs also include an allowance for funds used during construction, which represents the cost of equity funds used to finance the construction of the regulated facilities. Normal repairs and maintenance and other costs that do not improve the property, extend useful lives or otherwise do not meet capitalization criteria are charged to expense as incurred.

Depreciation of assets of our regulated utilities and railroad is generally determined using group depreciation methods where rates are based on periodic depreciation studies approved by the applicable regulator. Under group depreciation, a composite rate is applied to the gross investment in a particular class of property, despite differences in the service life or salvage value of individual property units within the same class. When such assets are retired or sold, no gain or loss is recognized. Gains or losses on disposals of all other assets are recorded through earnings.

We depreciate property, plant and equipment used by our other businesses to estimated salvage value primarily using the straight-line method over estimated useful lives. Ranges of estimated useful lives of depreciable assets used in our other businesses are as follows: buildings and improvements – 5 to 50 years, machinery and equipment – 3 to 25 years and furniture, fixtures and other – 3 to 15 years. Ranges of estimated useful lives of depreciable assets unique to our railroad business are as follows: track structure and other roadway – 10 to 100 years and locomotives, freight cars and other equipment – 6 to 43 years. Ranges of estimated useful lives of assets unique to our regulated utilities and energy businesses are as follows: utility generation, transmission and distribution systems – 5 to 80 years, interstate natural gas pipeline assets – 3 to 80 years and independent power plants and other assets – 3 to 40 years.

We evaluate property, plant and equipment for impairment when events or changes in circumstances indicate that the carrying value of such assets may not be recoverable or when the assets are held for sale. Upon the occurrence of a triggering event, we assess whether the estimated undiscounted cash flows expected from the use of the asset and the residual value from the ultimate disposal of the asset exceeds the carrying value. If the carrying value exceeds the estimated recoverable amounts, we reduce the carrying value to fair value and record an impairment loss in earnings, except with respect to impairment of assets of our regulated utility and energy subsidiaries where the impacts of regulation are considered in evaluating the carrying value.

(m) Leases

We are party to contracts where we lease property to others (“lessor” contracts) and where we lease property from others (“lessee” contracts). We record acquisitions of and additions to equipment that we lease to others at cost. We depreciate equipment held for lease to estimated salvage value primarily using the straight-line method over estimated useful lives ranging from 3 to 35 years. We use declining balance deprecation methods for assets when the revenue-earning power of the asset is relatively greater during the earlier years of its life and maintenance and repair costs increase during the later years. We also evaluate equipment held for lease for impairment consistent with policies for property, plant and equipment.

When we lease assets from others, we record right-of-use assets and lease liabilities. Right-of-use assets represent our right to use an underlying asset for the lease term and lease liabilities represent our obligation to make lease payments arising from the lease. In this regard, lease payments include fixed payments and variable payments that depend on an index or rate. The lease term is generally the non-cancellable lease period. Certain lease contracts contain renewal options or other terms that provide for variable payments based on performance or usage. Options are not included in determining right-of-use assets or lease liabilities unless it is reasonably certain that options will be exercised. Generally, incremental borrowing rates are used in measuring lease liabilities. Right-of-use assets are subject to review for impairment.

(1) Significant accounting policies and practices *(Continued)*

(n) Goodwill and other intangible assets

Goodwill represents the excess of the acquisition price of a business over the fair value of identified net assets of that business. We evaluate goodwill for impairment at least annually. When evaluating goodwill for impairment, we estimate the fair value of the reporting unit. Several methods may be used to estimate a reporting unit's fair value, including market quotations, asset and liability fair values and other valuation techniques, including, but not limited to, discounted projected future net earnings or net cash flows and multiples of earnings. If the carrying amount of a reporting unit, including goodwill, exceeds the estimated fair value, then the excess is charged to earnings as an impairment loss.

Intangible assets with indefinite lives are also tested for impairment at least annually and when events or changes in circumstances indicate that, more-likely-than-not, the asset is impaired. Significant judgment is required in estimating fair values and performing goodwill and indefinite-life intangible asset impairment tests. We amortize intangible assets with finite lives in a pattern that reflects the expected consumption of related economic benefits or on a straight-line basis over the estimated economic useful lives. Intangible assets with finite lives are reviewed for impairment when events or changes in circumstances indicate that the carrying amount may not be recoverable.

(o) Revenue recognition

We earn insurance premiums on prospective property/casualty insurance and reinsurance contracts over the loss exposure or coverage period in proportion to the level of protection provided. In most cases, such premiums are earned ratably over the term of the contract with unearned premiums computed on a monthly or daily pro-rata basis. Premiums on retroactive property/casualty reinsurance contracts are earned at the inception of the contracts, as all underlying loss events covered by the policies occurred prior to contract inception. Premiums for life reinsurance and annuity contracts are earned when due. Premiums earned are stated net of amounts ceded to reinsurers. Premiums earned on contracts with experience-rating provisions reflect estimated loss experience under such contracts.

Sales and service revenues are recognized when goods or services are transferred to a customer. A good or service is transferred when (or as) the customer obtains control of that good or service. Revenues are based on the consideration we expect to receive in connection with our promises to deliver goods and services to our customers.

We manufacture and/or distribute a wide variety of industrial, building and consumer products. Our sales contracts provide customers with these products through wholesale and retail channels in exchange for consideration specified under the contracts. Contracts generally represent customer orders for individual products at stated prices. Sales contracts may contain either single or multiple performance obligations. In instances where contracts contain multiple performance obligations, we allocate the revenue to each obligation based on the relative stand-alone selling prices of each product or service.

Sales revenue reflects reductions for returns, allowances, volume discounts and other incentives, some of which may be contingent on future events. In certain customer contracts, sales revenue includes certain state and local excise taxes billed to customers on specified products when those taxes are levied directly upon us by the taxing authorities. Sales revenue excludes sales taxes and value-added taxes collected on behalf of taxing authorities. Sales revenue includes consideration for shipping and other fulfillment activities performed prior to the customer obtaining control of the goods. We also elect to treat consideration for such services performed after control has passed to the customer as sales revenue.

Our product sales revenues are generally recognized at a point in time when control of the product transfers to the customer, which coincides with customer pickup or product delivery or acceptance, depending on terms of the arrangement. We recognize sales revenues and related costs with respect to certain contracts over time, primarily from certain castings, forgings and aerostructures contracts. Control of the product units under these contracts transfers continuously to the customer as the product is manufactured. These products generally have no alternative use and the contract requires the customer to provide reasonable compensation if terminated for reasons other than breach of contract.

(1) Significant accounting policies and practices *(Continued)*

(o) Revenue recognition (Continued)

Our energy revenue derives primarily from tariff-based sales arrangements approved by various regulatory commissions. These tariff-based revenues are mainly comprised of energy, transmission, distribution and natural gas and have performance obligations to deliver energy products and services to customers which are satisfied over time as energy is delivered or services are provided. Our nonregulated energy revenue primarily relates to our renewable energy business. Energy revenues are equivalent to the amounts we have the right to invoice and correspond directly with the value to the customer of the performance to date and include billed and unbilled amounts. Payments from customers are generally due within 30 days of billing. Rates charged for energy products and services are established by regulators or contractual arrangements that establish the transaction price, as well as the allocation of price among the separate performance obligations. When preliminary regulated rates are permitted to be billed prior to final approval by the applicable regulator, certain revenue collected may be subject to refund and a liability for estimated refunds is accrued.

The primary performance obligation under our freight rail transportation service contracts is to move freight from a point of origin to a point of destination. The performance obligations are represented by bills of lading which create a series of distinct services that have a similar pattern of transfer to the customer. The revenues for each performance obligation are based on various factors including the product being shipped, the origin and destination pair and contract incentives, which are outlined in various private rate agreements, common carrier public tariffs, interline foreign road agreements and pricing quotes. The transaction price is generally a per car/unit amount to transport railcars from a specified origin to a specified destination. Freight revenues are recognized over time as the service is performed because the customer simultaneously receives and consumes the benefits of the service. Revenues recognized represent the proportion of the service completed as of the balance sheet date. Invoices for freight transportation services are generally issued to customers and paid within 30 days or less. Customer incentives, which are primarily provided for shipping a specified cumulative volume or shipping to/from specific locations, are recorded as a reduction to revenue on a pro-rata basis based on actual or projected future customer shipments.

Other service revenues derive from contracts with customers in which performance obligations are satisfied over time, where customers receive and consume benefits as we perform the services, or at a point in time when the services are provided. Other service revenues primarily derive from real estate brokerage, automotive repair, aircraft management, aviation training and franchising and news distribution services.

Leasing revenue is generally recognized ratably over the term of the lease or based on usage, if applicable under the terms of the contract. A substantial portion of our leases are classified as operating leases.

(p) Losses and loss adjustment expenses

We record liabilities for unpaid losses and loss adjustment expenses under property/casualty insurance and reinsurance contracts for loss events that have occurred on or before the balance sheet date. Such liabilities represent the estimated ultimate payment amounts without discounting for time value.

We base liability estimates on (1) loss reports from policyholders and cedents, (2) individual case estimates and (3) estimates of incurred but not reported losses. Losses and loss adjustment expenses in the Consolidated Statements of Earnings include paid claims, claim settlement costs and changes in estimated claim liabilities. Losses and loss adjustment expenses charged to earnings are net of amounts recovered and estimates of amounts recoverable under ceded reinsurance contracts. Reinsurance contracts do not relieve the ceding company of its obligations to indemnify policyholders with respect to the underlying insurance and reinsurance contracts.

(q) Retroactive reinsurance contracts

We record liabilities for unpaid losses and loss adjustment expenses under short duration retroactive reinsurance contracts consistent with other short duration property/casualty insurance and reinsurance contracts described in Note 1(p). With respect to retroactive reinsurance contracts, we also record deferred charge assets at the inception of the contracts, representing the excess, if any, of the estimated ultimate claim liabilities over the premiums earned. We subsequently amortize the deferred charge assets over the expected claim settlement periods using the interest method. Changes to the estimated timing or amount of future loss payments also produce changes in deferred charge balances. We apply changes in such estimates retrospectively and the resulting changes in deferred charge balances, together with periodic amortization, are included in insurance losses and loss adjustment expenses in the Consolidated Statements of Earnings.

(1) Significant accounting policies and practices *(Continued)*

(r) Insurance policy acquisition costs

We capitalize the incremental costs that directly relate to the successful sale of insurance contracts, subject to ultimate recoverability, and we subsequently amortize such costs to underwriting expenses as the related premiums are earned. Direct incremental acquisition costs include commissions, premium taxes and certain other costs associated with successful efforts. We expense all other underwriting costs as incurred. The recoverability of capitalized insurance policy acquisition costs generally reflects anticipation of investment income. The unamortized balances are included in other assets and were approximately $3.25 billion and $2.95 billion at December 31, 2020 and 2019, respectively.

(s) Life and annuity insurance benefits

We compute liabilities for insurance benefits under life contracts based upon estimated future investment yields, expected mortality, morbidity and lapse or withdrawal rates, as well as estimates of premiums we expect to receive and expenses we expect to incur in the future. These assumptions, as applicable, also include a margin for adverse deviation and may vary with the characteristics of the contract's date of issuance, policy duration and country of risk. The interest rate assumptions used may vary by contract or jurisdiction. We discount periodic payment annuity liabilities based on the implicit rate as of the inception of the contracts such that the present value of the liabilities equals the premiums. Discount rates for most contracts range from 3% to 7%.

(t) Regulated utilities and energy businesses

Certain energy subsidiaries prepare their financial statements in accordance with authoritative guidance for regulated operations, reflecting the economic effects of regulation from the ability to recover certain costs from customers and the requirement to return revenues to customers in the future through the regulated rate-setting process. Accordingly, certain costs are deferred as regulatory assets and certain income is accrued as regulatory liabilities. Regulatory assets and liabilities will be amortized into operating expenses and revenues over various future periods.

Regulatory assets and liabilities are continually assessed for probable future inclusion in regulatory rates by considering factors such as applicable regulatory or legislative changes and recent rate orders received by other regulated entities. If future inclusion in regulatory rates ceases to be probable, the amount no longer probable of inclusion in regulatory rates is charged or credited to earnings (or other comprehensive income, if applicable) or returned to customers.

(u) Foreign currency

The accounts of our non-U.S. based subsidiaries are measured, in most instances, using functional currencies other than the U.S. Dollar. Revenues and expenses in the financial statements of these subsidiaries are translated into U.S. Dollars at the average exchange rate for the period and assets and liabilities are translated at the exchange rate as of the end of the reporting period. The net effects of translating the financial statements of these subsidiaries are included in shareholders' equity as a component of accumulated other comprehensive income. Gains and losses arising from transactions denominated in a currency other than the functional currency of the reporting entity, including gains and losses from the remeasurement of assets and liabilities due to changes in currency exchange rates, are included in earnings.

(v) Income taxes

Berkshire files a consolidated federal income tax return in the United States, which includes eligible subsidiaries. In addition, we file income tax returns in state, local and foreign jurisdictions as applicable. Provisions for current income tax liabilities are calculated and accrued on income and expense amounts expected to be included in the income tax returns for the current year. Income taxes reported in earnings also include deferred income tax provisions.

Deferred income tax assets and liabilities are computed on differences between the financial statement bases and tax bases of assets and liabilities at the enacted tax rates. Changes in deferred income tax assets and liabilities associated with components of other comprehensive income are charged or credited directly to other comprehensive income. Otherwise, changes in deferred income tax assets and liabilities are included as a component of income tax expense. The effect on deferred income tax assets and liabilities attributable to changes in enacted tax rates are charged or credited to income tax expense in the period of enactment. Valuation allowances are established for certain deferred tax assets when realization is not likely.

(1) Significant accounting policies and practices *(Continued)*

(v) Income taxes (Continued)

Assets and liabilities are established for uncertain tax positions taken or positions expected to be taken in income tax returns when such positions, in our judgment, do not meet a more-likely-than-not threshold based on the technical merits of the positions. Estimated interest and penalties related to uncertain tax positions are included as a component of income tax expense.

(w) New accounting pronouncements adopted in 2020

We adopted Accounting Standards Codification ("ASC") 326 "Financial Instruments-Credit Losses" on January 1, 2020. ASC 326 provides for the measurement of expected credit losses on financial assets that are carried at amortized cost based on the net amounts expected to be collected. Measurements of expected credit losses therefore include provisions for non-collection, whether the risk is probable or remote. Prior to the adoption of ASC 326, credit losses were measured when non-collection was considered probable based on the prevailing facts and circumstances. We do not measure an allowance for expected credit losses on accrued interest and instead, as permitted, we elected to reverse uncollectible accrued interest through interest income on a timely basis. Upon adoption of ASC 326, we recorded a charge to retained earnings of $388 million representing the cumulative after-tax increase in our allowances for credit losses, which was primarily related to our manufactured housing loans.

(x) New accounting pronouncements adopted in 2019

Berkshire adopted ASC 842 "Leases" on January 1, 2019. Most significantly, ASC 842 requires a lessee to recognize a liability to make operating lease payments and an asset with respect to its right to use the underlying asset for the lease term. In adopting and applying ASC 842, we elected to use practical expedients, including but not limited to, not reassessing past lease and easement accounting, not separating lease components from non-lease components by class of asset and not recording assets or liabilities for leases with terms of one year or less. We adopted ASC 842 as of January 1, 2019 with respect to contracts in effect as of that date and elected to not restate prior period financial statements.

Upon the adoption of ASC 842, we recognized operating lease right-of-use assets of approximately $6.2 billion and lease liabilities of $5.9 billion. We also reduced other assets by approximately $300 million. Consequently, our consolidated assets and liabilities increased by approximately $5.9 billion. ASC 842 did not have a material effect on our accounting for our lessor contracts or for lessee contracts classified as financing leases.

(y) New accounting pronouncements adopted in 2018

On January 1, 2018, we adopted Accounting Standards Update ("ASU") 2016-01 "Financial Instruments—Recognition and Measurement of Financial Assets and Financial Liabilities," ASU 2018-02 "Reclassification of Certain Tax Effects from Accumulated Other Comprehensive Income" and ASC 606 "Revenues from Contracts with Customers." Prior year financial statements were not restated. A summary of the effects of the initial adoption of ASU 2016-01, ASU 2018-02 and ASC 606 on our shareholders' equity follows (in millions).

	ASU 2016-01	ASU 2018-02	ASC 606	Total
Increase (decrease):				
Accumulated other comprehensive income	$ (61,459)	$ 84	$ —	$ (61,375)
Retained earnings	61,459	(84)	(70)	61,305
Shareholders' equity	—	—	(70)	(70)

In adopting ASU 2016-01, as of January 1, 2018, we reclassified the net after-tax unrealized gains on equity securities from accumulated other comprehensive income to retained earnings. Thereafter, the unrealized gains and losses from the changes during the period in the fair values of our equity securities are included within investment gains/losses in the Consolidated Statements of Earnings. In adopting ASU 2018-02, we reclassified certain deferred income tax effects as of January 1, 2018 attributable to the reduction in the U.S. statutory income tax rate under the Tax Cuts and Jobs Act of 2017 from accumulated other comprehensive income to retained earnings. In adopting ASC 606, we recorded increases to certain assets and other liabilities, with the cumulative net effect recorded to retained earnings.

(1) Significant accounting policies and practices *(Continued)*

(z) New accounting pronouncements to be adopted subsequent to December 31, 2020

In August 2018, the FASB issued ASU 2018-12 "Targeted Improvements to the Accounting for Long-Duration Contracts." ASU 2018-12 requires periodic reassessment of actuarial and discount rate assumptions used to value policyholder liabilities and deferred acquisition costs of long-duration insurance and reinsurance contracts, with the effects of changes in cash flow assumptions reflected in earnings and the effects of changes in discount rate assumptions reflected in other comprehensive income. Under current GAAP, the actuarial and discount rate assumptions are set at the contract inception date and not subsequently changed, except under limited circumstances. ASU 2018-12 requires new disclosures and is effective for fiscal years beginning after December 15, 2022, with early adoption permitted. We are evaluating the effect this standard will have on our Consolidated Financial Statements.

(2) Business acquisitions

Our long-held acquisition strategy is to acquire businesses that have consistent earning power, good returns on equity and able and honest management. Financial results attributable to business acquisitions are included in our Consolidated Financial Statements beginning on their respective acquisition dates.

In July 2020, Berkshire Hathaway Energy ("BHE") reached a definitive agreement with Dominion Energy, Inc. ("Dominion") to acquire substantially all of Dominion's natural gas transmission and storage business. On October 5, 2020, BHE and Dominion also agreed, as permitted under the acquisition agreement, to provide for the acquisition of all originally agreed upon businesses, except for certain pipeline assets (the "Excluded Assets") and entered into a second acquisition agreement with respect to the Excluded Assets. The acquisition of the Dominion businesses, other than the Excluded Assets, was completed on November 1, 2020 and included more than 5,400 miles of natural gas transmission, gathering and storage pipelines, about 420 billion cubic feet of operated natural gas storage capacity and partial ownership of a liquefied natural gas export, import and storage facility ("Cove Point"). Under the terms of the second acquisition agreement, BHE agreed to acquire the Excluded Assets for approximately $1.3 billion in cash. The closing of this second acquisition is subject to receiving necessary regulatory approvals and other customary closing conditions and is expected to occur during the first half of 2021.

The cost of the acquisition completed on November 1, 2020, was approximately $2.5 billion after post-closing adjustments as provided in the agreement. The preliminary fair values of identified assets acquired and liabilities assumed and residual goodwill are summarized as follows (in millions).

Property, plant and equipment	$	9,254
Goodwill		1,732
Other		2,376
Assets acquired	$	13,362
Notes payable and other borrowings	$	5,615
Other		1,317
Liabilities assumed		6,932
Noncontrolling interests		3,916
Net assets	$	2,514

As part of this acquisition, BHE acquired an indirect 25% economic interest in Cove Point, consisting of 100% of the general partnership interest and 25% of the limited partnership interests. We concluded that Cove Point is a VIE and that we have the power to direct the activities that most significantly impact its economic performance as well as the obligation to absorb losses and receive benefits which could be significant to Cove Point. Therefore, we treat Cove Point as a consolidated subsidiary. The noncontrolling interests in the preceding table is attributable to the limited partner interests held by third parties.

On October 1, 2018, we acquired MLMIC Insurance Company ("MLMIC"), a writer of medical professional liability insurance domiciled in New York. The acquisition price was approximately $2.5 billion. As of the acquisition date, the fair value of MLMIC's assets was approximately $6.1 billion, primarily investments ($5.2 billion), and the fair value of its liabilities was approximately $3.6 billion, primarily unpaid losses and loss adjustment expenses ($3.2 billion).

(2) Business acquisitions *(Continued)*

In each of the past three years, we also completed several smaller-sized business acquisitions, which we consider as "bolt-ons" to several of our existing business operations. Aggregate consideration paid for bolt-on acquisitions, net of cash acquired was approximately $130 million in 2020, $1.7 billion in 2019 and $1.0 billion in 2018. We do not believe that these acquisitions are material, individually or in the aggregate to our Consolidated Financial Statements.

(3) Investments in fixed maturity securities

Investments in fixed maturity securities as of December 31, 2020 and 2019 are summarized by type below (in millions).

	Amortized Cost	Unrealized Gains	Unrealized Losses	Fair Value
December 31, 2020				
U.S. Treasury, U.S. government corporations and agencies	$ 3,348	$ 55	$ —	$ 3,403
Foreign governments	11,233	110	(5)	11,338
Corporate bonds	4,729	464	(2)	5,191
Other	414	66	(2)	478
	$ 19,724	$ 695	$ (9)	$ 20,410
December 31, 2019				
U.S. Treasury, U.S. government corporations and agencies	$ 3,054	$ 37	$ (1)	$ 3,090
Foreign governments	8,584	63	(9)	8,638
Corporate bonds	5,896	459	(3)	6,352
Other	539	67	(1)	605
	$ 18,073	$ 626	$ (14)	$ 18,685

Investments in foreign governments include securities issued by national and provincial government entities as well as instruments that are unconditionally guaranteed by such entities. As of December 31, 2020, approximately 88% of our foreign government holdings were rated AA or higher by at least one of the major rating agencies.

The amortized cost and estimated fair value of fixed maturity securities at December 31, 2020 are summarized below by contractual maturity dates. Amounts are in millions. Actual maturities may differ from contractual maturities due to early call or prepayment rights held by issuers.

	Due in one year or less	Due after one year through five years	Due after five years through ten years	Due after ten years	Mortgage-backed securities	Total
Amortized cost	$ 10,379	$ 8,323	$ 373	$ 337	$ 312	$ 19,724
Fair value	10,448	8,456	496	640	370	20,410

(4) Investments in equity securities

Investments in equity securities as of December 31, 2020 and 2019 are summarized based on the primary industry of the investee in the table below (in millions).

	Cost Basis	Net Unrealized Gains	Fair Value
December 31, 2020 *			
Banks, insurance and finance	$ 26,312	$ 40,167	$ 66,479
Consumer products	34,747	111,583	146,330
Commercial, industrial and other	47,561	20,800	68,361
	$ 108,620	$ 172,550	$ 281,170

* *Approximately 68% of the aggregate fair value was concentrated in four companies (American Express Company – $18.3 billion; Apple Inc. – $120.4 billion; Bank of America Corporation – $31.3 billion and The Coca-Cola Company – $21.9 billion).*

	Cost Basis	Net Unrealized Gains	Fair Value
December 31, 2019 *			
Banks, insurance and finance	$ 40,419	$ 61,976	$ 102,395
Consumer products	38,887	60,747	99,634
Commercial, industrial and other	31,034	14,964	45,998
	$ 110,340	$ 137,687	$ 248,027

* *Approximately 60% of the aggregate fair value was concentrated in four companies (American Express Company – $18.9 billion; Apple Inc. – $73.7 billion; Bank of America Corporation – $33.4 billion and The Coca-Cola Company – $22.1 billion).*

On August 8, 2019, Berkshire invested a total of $10 billion in Occidental Corporation ("Occidental") newly issued Occidental Cumulative Perpetual Preferred Stock with an aggregate liquidation value of $10 billion and warrants to purchase up to 80 million shares of Occidental common stock at an exercise price of $62.50 per share. In accordance with the terms of the warrants, on August 3, 2020, the number of shares of common stock that can be purchased was increased to 83.86 million shares and the exercise price was reduced to $59.62 per share. The preferred stock accrues dividends at 8% per annum and is redeemable at the option of Occidental commencing in 2029 at a redemption price equal to 105% of the liquidation preference plus any accumulated and unpaid dividends, or is mandatorily redeemable under certain specified capital return events. Dividends on the preferred stock may be paid in cash or, at Occidental's option, in shares of Occidental common stock. The warrants are exercisable in whole or in part until one year after the redemption of the preferred stock. Our investments in Occidental are included in the commercial, industrial and other category in the preceding tables.

(5) Equity method investments

Berkshire and its subsidiaries hold investments in certain businesses that are accounted for pursuant to the equity method. Currently, the most significant of these is our investment in the common stock of The Kraft Heinz Company ("Kraft Heinz"). Kraft Heinz is one of the world's largest manufacturers and marketers of food and beverage products, including condiments and sauces, cheese and dairy, meals, meats, refreshment beverages, coffee and other grocery products. Berkshire currently owns 325,442,152 shares of Kraft Heinz common stock representing 26.6% of the outstanding shares.

(5) Equity method investments *(Continued)*

We recorded equity method earnings from our investment in Kraft Heinz of $95 million in 2020, $493 million in 2019 and losses of approximately $2.7 billion in 2018. Equity method earnings (losses) included the effects of goodwill and identifiable intangible asset impairment charges recorded by Kraft Heinz. Our share of such charges was approximately $850 million in 2020, $450 million in 2019 and $3.7 billion in 2018. We received dividends from Kraft Heinz of $521 million in each of 2020 and 2019 and $814 million in 2018, which we recorded as reductions in our carrying value.

Shares of Kraft Heinz common stock are publicly-traded and the fair value of our investment was approximately $11.3 billion at December 31, 2020 and $10.5 billion at December 31, 2019. The carrying value of our investment was approximately $13.3 billion at December 31, 2020 and $13.8 billion at December 31, 2019. As of December 31, 2020, the carrying value of our investment exceeded the fair value based on the quoted market price by $2.0 billion (15% of carrying value). In light of this fact, we evaluated our investment in Kraft Heinz for impairment. We utilize no bright-line tests in such evaluations. Based on the available facts and information regarding the operating results of Kraft Heinz, our ability and intent to hold the investment until recovery, the relative amount of the decline and the length of time that fair value was less than carrying value, we concluded that recognition of an impairment loss in earnings was not required. However, we will continue to monitor this investment and it is possible that an impairment loss will be recorded in earnings in a future period based on changes in facts and circumstances or intentions.

Summarized financial information of Kraft Heinz follows (in millions).

	December 26, 2020	December 28, 2019
Assets	$ 99,830	$ 101,450
Liabilities	49,587	49,701

	Year ending December 26, 2020	Year ending December 28, 2019	Year ending December 29, 2018
Sales	$ 26,185	$ 24,977	$ 26,268
Net earnings (losses) attributable to Kraft Heinz common shareholders	$ 356	$ 1,935	$ (10,192)

Other investments accounted for pursuant to the equity method include our investments in Berkadia Commercial Mortgage LLC (“Berkadia”), Pilot Travel Centers LLC (“Pilot”) and Electric Transmission Texas, LLC (“ETT”). The carrying value of our investments in these entities was approximately $4.0 billion as of December 31, 2020 and $3.7 billion as of December 31, 2019. Our equity method earnings in these entities were $631 million in 2020, $683 million in 2019 and $563 million in 2018. Additional information concerning these investments follows.

We own a 50% interest in Berkadia, with Jefferies Financial Group Inc. (“Jefferies”) owning the other 50% interest. Berkadia is a servicer of commercial real estate loans in the U.S., performing primary, master and special servicing functions for U.S. government agency programs, commercial mortgage-backed securities transactions, banks, insurance companies and other financial institutions. A source of funding for Berkadia’s operations is through its issuance of commercial paper, which is currently limited to $1.5 billion. On December 31, 2020, Berkadia’s commercial paper outstanding was $1.47 billion. The commercial paper is supported by a surety policy issued by a Berkshire insurance subsidiary. Jefferies is obligated to indemnify us for one-half of any losses incurred under the policy.

A Berkshire Hathaway Energy Company subsidiary owns a 50% interest in ETT, an owner and operator of electric transmission assets in the Electric Reliability Council of Texas footprint. American Electric Power owns the other 50% interest.

On October 3, 2017, we entered into an investment agreement and an equity purchase agreement whereby we acquired a 38.6% interest in Pilot, headquartered in Knoxville, Tennessee. Pilot is the largest operator of travel centers in North America, supplying more than 11 billion gallons of fuel per year via more than 950 retail locations across 44 U.S. states and six Canadian provinces and through wholesale distribution. The Haslam family currently owns a 50.1% interest in Pilot and a third party owns the remaining 11.3% interest. We also entered into an agreement to acquire in 2023 an additional 41.4% interest in Pilot with the Haslam family retaining a 20% interest. As a result, Berkshire will become the majority owner of Pilot in 2023.

(6) Investment gains/losses

Investment gains/losses for each of the three years ending December 31, 2020 are summarized below (in millions).

	2020	2019	2018
Equity securities:			
Change in unrealized investment gains/losses during the year on securities held at the end of the period	$ 54,951	$ 69,581	$ (22,729)
Investment gains/losses during the year on securities sold	(14,036)	1,585	291
	40,915	71,166	(22,438)
Fixed maturity securities:			
Gross realized gains	56	87	480
Gross realized losses	(27)	(25)	(227)
Other	(39)	(105)	30
	$ 40,905	$ 71,123	$ (22,155)

Equity securities gains and losses include unrealized gains and losses from changes in fair values during the period on equity securities we still own, as well as gains and losses on securities we sold during the period. As reflected in the Consolidated Statements of Cash Flows, we received proceeds of approximately $38.8 billion in 2020, $14.3 billion in 2019 and $18.8 billion in 2018 from sales of equity securities. In the preceding table, investment gains/losses on equity securities sold reflect the difference between proceeds from sales and the fair value of the equity security sold at the beginning of the period or the purchase date, if later. Our taxable gains on equity securities sold during the year, which are generally the difference between the proceeds from sales and our original cost, were $6.2 billion in 2020, $3.2 billion in 2019 and $3.3 billion in 2018.

(7) Loans and finance receivables

Loans and finance receivables are summarized as follows (in millions).

	December 31,	
	2020	2019
Loans and finance receivables before allowances and discounts	$ 20,436	$ 18,199
Allowances for uncollectible loans	(712)	(167)
Unamortized acquisition discounts and points	(523)	(505)
	$ 19,201	$ 17,527

Loans and finance receivables are principally manufactured home loans, and to a lesser extent, commercial loans and site-built home loans. Reconciliations of the allowance for credit losses on loans and finance receivables for 2020 and 2019 follow (in millions).

	2020	2019
Balance at beginning of year	$ 167	$ 177
Adoption of ASC 326	486	—
Provision for credit losses	177	125
Charge-offs, net of recoveries	(118)	(135)
Balance at December 31	$ 712	$ 167

At December 31, 2020, approximately 99% of home loan balances were evaluated collectively for impairment. At December 31, 2020, we considered approximately 97% of the loan balances to be current as to payment status. A summary of performing and non-performing home loans before discounts and allowances by year of loan origination as of December 31, 2020 follows (in millions).

	Loans and Financing Receivables by Origination Year						
	2020	2019	2018	2017	2016	Prior	Total
Performing	$ 4,430	$ 2,537	$ 1,928	$ 1,424	$ 1,276	$ 6,645	$ 18,240
Non-performing	3	5	7	7	7	43	72
Total	$ 4,433	$ 2,542	$ 1,935	$ 1,431	$ 1,283	$ 6,688	$ 18,312

(7) Loans and finance receivables *(Continued)*

We are party to an agreement with Seritage Growth Properties to provide a $2.0 billion term loan facility, which expires on July 31, 2023. The outstanding loan under the facility was approximately $1.6 billion at December 31, 2020 and 2019, and is secured by mortgages on real estate properties. In 2020, we provided a loan to Lee Enterprises, Inc. in connection with its acquisition of our newspaper operations and the repayment by Lee of its then outstanding credit facilities. The loan balance as of December 31, 2020 was $524 million. We are the sole lender to each of these entities and each of these loans is current as to payment status.

(8) Other receivables

Other receivables of insurance and other businesses are comprised of the following (in millions).

	December 31,	
	2020	2019
Insurance premiums receivable	$ 14,025	$ 13,379
Reinsurance recoverables	4,805	4,470
Trade receivables	11,521	12,275
Other	2,637	2,712
Allowances for uncollectible accounts	(678)	(418)
	$ 32,310	$ 32,418

Receivables of our railroad and utilities and energy businesses are comprised of the following (in millions).

	December 31,	
	2020	2019
Trade receivables	$ 3,235	$ 3,120
Other	438	388
Allowances for uncollectible accounts	(131)	(91)
	$ 3,542	$ 3,417

Provisions for credit losses on receivables in the preceding tables were $564 million in 2020 and $363 million in 2019. Net charge-offs were $401 million in 2020 and $350 million in 2019.

(9) Inventories

Inventories are comprised of the following (in millions).

	December 31,	
	2020	2019
Raw materials	$ 4,821	$ 4,492
Work in process and other	2,541	2,700
Finished manufactured goods	4,412	4,821
Goods acquired for resale	7,434	7,839
	$ 19,208	$ 19,852

(10) Property, plant and equipment

A summary of property, plant and equipment of our insurance and other businesses follows (in millions).

	December 31,	
	2020	2019
Land, buildings and improvements	$ 13,799	$ 13,259
Machinery and equipment	25,488	24,285
Furniture, fixtures and other	4,530	4,666

	43,817	42,210
Accumulated depreciation	(22,617)	(20,772)
	$ 21,200	$ 21,438

(10) Property, plant and equipment *(Continued)*

A summary of property, plant and equipment of railroad and utilities and energy businesses follows (in millions). The utility generation, transmission and distribution systems and interstate natural gas pipeline assets are owned by regulated public utility and natural gas pipeline subsidiaries.

	December 31,	
	2020	2019
Railroad:		
Land, track structure and other roadway	$ 63,824	$ 62,404
Locomotives, freight cars and other equipment	13,523	13,482
Construction in progress	916	748
	78,263	76,634
Accumulated depreciation	(13,175)	(12,101)
	65,088	64,533
Utilities and energy:		
Utility generation, transmission and distribution systems	86,730	81,127
Interstate natural gas pipeline assets	16,667	8,165
Independent power plants and other assets	12,671	8,817
Construction in progress	3,308	3,732
	119,376	101,841
Accumulated depreciation	(33,248)	(28,536)
	86,128	73,305
	$ 151,216	$ 137,838

Depreciation expense for each of the three years ending December 31, 2020 is summarized below (in millions).

	2020	2019	2018
Insurance and other	$ 2,320	$ 2,269	$ 2,186
Railroad, utilities and energy	5,799	5,297	5,098
	$ 8,119	$ 7,566	$ 7,284

(11) Equipment held for lease

Equipment held for lease includes railcars, aircraft, over-the-road trailers, intermodal tank containers, cranes, storage units and furniture. Equipment held for lease is summarized below (in millions).

	December 31,	
	2020	2019
Railcars	$ 9,402	$ 9,260
Aircraft	8,204	8,093
Other	4,868	4,862
	22,474	22,215
Accumulated depreciation	(7,873)	(7,150)
	$ 14,601	$ 15,065

Depreciation expense for equipment held for lease was $1,200 million in 2020, $1,181 million in 2019 and $1,102 million in 2018. Fixed and variable operating lease revenues for each of the two years ending December 31, 2020 are summarized below (in millions).

	2020	2019
Fixed lease revenue	$ 4,262	$ 4,415
Variable lease revenue	947	1,441
	$ 5,209	$ 5,856

(11) Equipment held for lease *(Continued)*

A summary of future operating lease receipts as of December 31, 2020 follows (in millions).

2021	2022	2023	2024	2025	Thereafter	Total
$ 2,618	$ 1,962	$ 1,429	$ 905	$ 443	$ 387	$ 7,744

(12) Leases

We are party to contracts where we lease property from others. As a lessee, we primarily lease office and operating facilities, locomotives, freight cars, energy generation facilities and transmission assets. Operating lease right-of-use assets were $5,579 million and lease liabilities were $5,469 million at December 31, 2020. Operating lease right-of-use assets were $5,941 million and lease liabilities were $5,882 million at December 31, 2019. Such amounts were included in other assets and accounts payable, accruals and other liabilities in our Consolidated Balance Sheet. The weighted average term of these leases was approximately 7.3 years at December 31, 2020 and 7.7 years at December 31, 2019. The weighted average discount rate used to measure lease liabilities was approximately 3.6% at December 31, 2020 and 3.8% at December 31, 2019. A summary of our remaining operating lease payments as of December 31, 2020 and December 31, 2019 follows (in millions).

	Year 1	Year 2	Year 3	Year 4	Year 5	Thereafter	Total lease payments	Amount representing interest	Lease liabilities
December 31:									
2020	$ 1,342	$ 1,111	$ 905	$ 725	$ 544	$ 1,691	$ 6,318	$ (849)	$ 5,469
2019	1,374	1,183	950	764	620	1,988	6,879	(997)	5,882

Components of operating lease costs for the years ending December 31, 2020 and 2019, by type, are summarized in the following table (in millions). Operating lease expense was $1,649 million in 2018.

	2020	2019
Operating lease cost	$ 1,413	$ 1,459
Short-term lease cost	145	178
Variable lease cost	228	276
Sublease income	(10)	(24)
Total lease cost	$ 1,776	$ 1,889

(13) Goodwill and other intangible assets

Reconciliations of the changes in the carrying value of goodwill during 2020 and 2019 follows (in millions).

	December 31,	
	2020	2019
Balance at beginning of year	$ 81,882	$ 81,025
Acquisitions of businesses	1,758	890
Impairment charges	(10,033)	(90)
Other, including foreign currency translation	127	57
Balance at end of year*	$ 73,734	$ 81,882

* *Net of accumulated goodwill impairments of $11.0 billion as of December 31, 2020 and $1.1 billion as of December 31, 2019.*

(13) Goodwill and other intangible assets *(Continued)*

The gross carrying amounts and related accumulated amortization of other intangible assets are summarized as follows (in millions).

	December 31, 2020		December 31, 2019	
	Gross carrying amount	Accumulated amortization	Gross carrying amount	Accumulated amortization
Insurance and other:				
Customer relationships	$ 27,374	$ 5,756	$ 27,943	$ 5,025
Trademarks and trade names	5,206	779	5,286	759
Patents and technology	4,766	3,313	4,560	3,032
Other	3,339	1,375	3,364	1,286
	$ 40,685	$ 11,223	$ 41,153	$ 10,102
Railroad, utilities and energy:				
Customer relationships	$ 678	$ 361	$ 678	$ 324
Trademarks, trade names and other	1,003	98	325	84
	$ 1,681	$ 459	$ 1,003	$ 408

Intangible asset amortization expense was $1,277 million in 2020, $1,317 million in 2019 and $1,393 million in 2018. Estimated amortization expense over the next five years is as follows (in millions): 2021 – $1,262; 2022 – $1,190; 2023 – $1,108; 2024 – $986 and 2025 – $906. Intangible assets with indefinite lives were $18.3 billion as of December 31, 2020 and $19.0 billion as of December 31, 2019 and primarily related to certain customer relationships and trademarks and trade names.

During 2020, we concluded it was necessary to reevaluate goodwill and indefinite-lived intangible assets of certain of our reporting units for impairment due to the disruptions arising from the COVID-19 pandemic. We believed that the most significant of these disruptions related to the air travel and commercial aerospace and supporting industries. We recorded pre-tax goodwill impairment charges of approximately $10 billion and pre-tax indefinite-lived intangible asset impairment charges of $638 million in the second quarter of 2020. Approximately $10 billion of these charges related to Precision Castparts Corp. (“PCC”), the largest business within Berkshire's manufacturing segment. The carrying value of PCC-related goodwill and indefinite-lived intangible assets prior to the impairment charges was approximately $31 billion.

The impairment charges were determined based on discounted cash flow methods and reflected our assessments of the risks and uncertainties associated with the aerospace industry. Significant judgment is required in estimating the fair value of a reporting unit and in performing impairment tests. Due to the inherent uncertainty in forecasting cash flows and earnings, actual results in the future may vary significantly from the forecasts.

(14) Derivative contracts

We are party to derivative contracts through certain of our subsidiaries. The most significant derivative contracts consist of equity index put option contracts. Information related to these contracts follows (dollars in millions).

	December 31,	
	2020	2019
Balance sheet liabilities - at fair value	$ 1,065	$ 968
Notional value	10,991	14,385
Intrinsic value	727	397
Weighted average remaining life (in years)	1.2	1.8

The equity index put option contracts are European style options written prior to March 2008 on four major equity indexes. Notional value in the preceding table represents the aggregate undiscounted amounts payable assuming that the value of each index is zero at each contract’s expiration date. Intrinsic value is the undiscounted liability assuming the contracts are settled based on the index values and foreign currency exchange rates as of the balance sheet date. Substantially all open contracts as of December 31, 2020 will expire by February 2023.

(14) Derivative contracts *(Continued)*

Future payments, if any, under any given contract will be required if the prevailing index value is below the contract strike price at the expiration date. We received aggregate premiums of $1.9 billion on the contract inception dates with respect to unexpired contracts as of December 31, 2020 and we have no counterparty credit risk.

We recorded derivative contract losses of $159 million in 2020, gains of $1,484 million in 2019 and losses of $300 million in 2018, with respect to our equity index put option contracts. These gains and losses were primarily due to changes in the equity index values. These contracts may not be unilaterally terminated or fully settled before the expiration dates and the ultimate amount of cash basis gains or losses on these contracts will not be determined until the contract expiration dates.

Our regulated utility subsidiaries may use forward purchases and sales, futures, swaps and options to manage a portion of their commodity price risks. Most of these net derivative contract assets or liabilities of our regulated utilities are probable of recovery through rates and are offset by regulatory liabilities or assets.

(15) Unpaid losses and loss adjustment expenses

Our liabilities for unpaid losses and loss adjustment expenses (also referred to as "claim liabilities") under property and casualty insurance and reinsurance contracts are based upon estimates of the ultimate claim costs associated with claim occurrences as of the balance sheet date and include estimates for incurred-but-not-reported ("IBNR") claims. A reconciliation of the changes in claim liabilities, excluding liabilities under retroactive reinsurance contracts (see Note 16), for each of the three years ending December 31, 2020 is as follows (in millions).

	2020	2019	2018
Balances at beginning of year:			
Gross liabilities	$ 73,019	$ 68,458	$ 61,122
Reinsurance recoverable on unpaid losses	(2,855)	(3,060)	(3,201)
Net liabilities	70,164	65,398	57,921
Incurred losses and loss adjustment expenses:			
Current accident year events	43,400	43,335	39,876
Prior accident years' events	(356)	(752)	(1,406)
Total	43,044	42,583	38,470
Paid losses and loss adjustment expenses:			
Current accident year events	(17,884)	(19,482)	(18,391)
Prior accident years' events	(18,862)	(17,642)	(15,452)
Total	(36,746)	(37,124)	(33,843)
Foreign currency translation adjustment	480	(23)	(331)
Business acquisition (disposition)	—	(670)	3,181
Balances at December 31:			
Net liabilities	76,942	70,164	65,398
Reinsurance recoverable on unpaid losses	2,912	2,855	3,060
Gross liabilities	$ 79,854	$ 73,019	$ 68,458

Incurred losses and loss adjustment expenses shown in the preceding table were recorded in earnings and related to insured events occurring in the current year ("current accident year") and events occurring in all prior years ("prior accident years"). Current accident year losses included approximately $950 million in 2020, $1.0 billion in 2019 and $1.6 billion in 2018 from significant catastrophe events occurring in each year. Current accident year losses in 2020 also reflected the effects of low private passenger automobile claims frequencies and increased loss estimates for certain commercial insurance and reinsurance business attributable to the COVID-19 pandemic.

We recorded net reductions of estimated ultimate liabilities for prior accident years of $356 million in 2020, $752 million in 2019 and $1,406 million in 2018, which produced corresponding reductions in incurred losses and loss adjustment expenses. These reductions, as percentages of the net liabilities at the beginning of each year, were 0.5% in 2020, 1.1% in 2019 and 2.4% in 2018.

(15) Unpaid losses and loss adjustment expenses *(Continued)*

Estimated ultimate liabilities for prior years' loss events related to primary insurance were reduced by $518 million in 2020, $457 million in 2019 and $937 million in 2018. The decrease in 2020 was primarily attributable to reductions for private passenger automobile, medical professional liability and workers' compensation claims, partly offset by increases for other casualty claims. The decrease in 2019 reflected reductions in medical professional liability and workers' compensation claims, partially offset by higher commercial auto and other liability claims. The decrease in 2018 was primarily due to reductions for workers' compensation, medical professional liability and private passenger automobile claims. Estimated ultimate liabilities for prior years' loss events related to property and casualty reinsurance increased $162 million in 2020 and were reduced $295 million in 2019 and $469 million in 2018. The increase in 2020 included increased claims estimates for legacy casualty exposures.

Estimated claim liabilities included amounts for environmental, asbestos and other latent injury exposures, net of reinsurance recoverable, of approximately $2.1 billion at December 31, 2020 and $1.7 billion at December 31, 2019. These liabilities are subject to change due to changes in the legal and regulatory environment. We are unable to reliably estimate additional losses or a range of losses that are reasonably possible for these claims.

Disaggregated information concerning our claims liabilities is provided below and in the pages that follow. The effects of businesses acquired or disposed during the period are reflected in the data presented on a retrospective basis. A reconciliation of the disaggregated net unpaid losses and allocated loss adjustment expenses (the latter referred to as "ALAE") of GEICO, Berkshire Hathaway Primary Group ("BH Primary") and Berkshire Hathaway Reinsurance Group ("BHRG") to our consolidated unpaid losses and loss adjustment expenses as of December 31, 2020 follows (in millions).

	GEICO Physical Damage	GEICO Auto Liability	BH Primary Medical Professional Liability	BH Primary Workers' Compensation and Other Casualty	BHRG Property	BHRG Casualty	Total
Unpaid losses and ALAE, net	$ 524	$ 18,755	$ 7,897	$ 11,294	$ 11,280	$ 22,890	$ 72,640
Reinsurance recoverable	—	1,109	49	621	181	864	2,824
Unpaid unallocated loss adjustment expenses							2,671
Other unpaid losses and loss adjustment expenses							1,719
Unpaid losses and loss adjustment expenses							$ 79,854

GEICO

GEICO's claim liabilities predominantly relate to various types of private passenger auto liability and physical damage claims. For such claims, we establish and evaluate unpaid claim liabilities using standard actuarial loss development methods and techniques. The actuarial methods utilize historical claims data, adjusted when deemed appropriate to reflect perceived changes in loss patterns. Claim liabilities include average, case, case development and IBNR estimates.

We establish average liabilities based on expected severities for newly reported physical damage and liability claims prior to establishing individual case reserves when insufficient time or information is available for specific claim estimates and for large volumes of minor physical damage claims that once reported are quickly settled. We establish case loss estimates for liability claims, including estimates for loss adjustment expenses, as the facts and merits of the claim are evaluated.

Estimates for liability coverages are more uncertain than for physical damage coverages, primarily due to the longer claim-tails, the greater chance of protracted litigation and the incompleteness of facts at the time the case estimate is first established. The "claim-tail" is the time period between the claim occurrence date and settlement date. Consequently, we establish additional case development liabilities, which are usually percentages of the case liabilities. For unreported claims, IBNR liabilities are estimated by projecting the ultimate number of claims expected (reported and unreported) for each significant coverage and deducting reported claims to produce estimated unreported claims. The product of the average cost per unreported claim and the number of unreported claims produces the IBNR liability estimate. We may record supplemental IBNR liabilities in certain situations when actuarial techniques are difficult to apply.

(15) Unpaid losses and loss adjustment expenses *(Continued)*

GEICO's incurred and paid losses and ALAE, net of reinsurance, are summarized by accident year below for physical damage and auto liability claims. IBNR and case development liabilities are as of December 31, 2020. Claim counts are established when accidents that may result in a liability are reported and are based on policy coverage. Each claim event may generate claims under multiple coverages, and thus may result in multiple counts. The "Cumulative Number of Reported Claims" includes the combined number of reported claims for all policy coverages and excludes projected IBNR claims. Dollars are in millions.

Physical Damage

Accident Year	Incurred Losses and ALAE through December 31, 2019*	2020	IBNR and Case Development Liabilities	Cumulative Number of Reported Claims (in thousands)
2019	$ 9,020	$ 8,920	$ 69	8,929
2020		8,603	296	7,794
	Incurred losses and ALAE	$ 17,523		

Accident Year	Cumulative Paid Losses and ALAE through December 31, 2019*	2020
2019	$ 8,678	$ 8,905
2020		8,118
	Paid losses and ALAE	17,023
	Net unpaid losses and ALAE for 2019 – 2020 accident years	500
	Net unpaid losses and ALAE for accident years before 2019	24
	Net unpaid losses and ALAE	$ 524

Auto Liability

Accident Year	Incurred Losses and ALAE through December 31, 2016*	2017*	2018*	2019*	2020	IBNR and Case Development Liabilities	Cumulative Number of Reported Claims (in thousands)
2016	$ 11,800	$ 12,184	$ 12,149	$ 12,178	$ 12,198	$ 222	2,451
2017		14,095	13,864	13,888	13,824	502	2,639
2018			15,383	15,226	14,985	1,163	2,702
2019				16,901	16,678	2,905	2,749
2020					14,637	4,482	1,945
				Incurred losses and ALAE	$ 72,322		

Accident Year	Cumulative Paid Losses and ALAE through December 31, 2016*	2017*	2018*	2019*	2020
2016	$ 5,069	$ 8,716	$ 10,330	$ 11,294	$ 11,718
2017		5,806	9,944	11,799	12,729
2018			6,218	10,772	12,658
2019				6,742	11,671
2020					5,395
				Paid losses and ALAE	54,171
				Net unpaid losses and ALAE for 2016 – 2020 accident years	18,151
				Net unpaid losses and ALAE for accident years before 2016	604
				Net unpaid losses and ALAE	$ 18,755

* *Unaudited required supplemental information*

(15) Unpaid losses and loss adjustment expenses *(Continued)*

BH Primary

BH Primary's liabilities for unpaid losses and loss adjustment expenses primarily derive from medical professional liability and workers' compensation and other casualty insurance, including commercial auto and general liability insurance. Incurred and paid losses and ALAE are summarized by accident year in the following tables, disaggregated by medical professional liability coverages and workers' compensation and other casualty coverages. IBNR and case development liabilities are as of December 31, 2020. The cumulative number of reported claims reflects the number of individual claimants and includes claims that ultimately resulted in no liability or payment. Dollars are in millions.

BH Primary Medical Professional Liability

We estimate the ultimate expected incurred losses and loss adjustment expenses for medical professional claim liabilities using a variety of commonly accepted actuarial methodologies, such as the paid and incurred development method and Bornhuetter-Ferguson based methods, as well as other techniques that consider insured loss exposures and historical and expected loss trends, among other factors. These methodologies produce loss estimates from which we determine our best estimate. In addition, we study developments in older accident years and adjust initial loss estimates to reflect recent development based upon claim age, coverage and litigation experience.

	Incurred Losses and ALAE through December 31,											
Accident Year	**2011***	**2012***	**2013***	**2014***	**2015***	**2016***	**2017***	**2018***	**2019***	**2020**	**IBNR and Case Development Liabilities**	**Cumulative Number of Reported Claims (in thousands)**
2011	$1,346	$1,334	$1,321	$1,262	$1,173	$1,115	$1,050	$1,004	$ 968	$ 972	$ 39	11
2012		1,336	1,306	1,277	1,223	1,168	1,078	1,035	998	988	53	11
2013			1,328	1,296	1,261	1,195	1,127	1,086	1,019	985	65	11
2014				1,370	1,375	1,305	1,246	1,218	1,127	1,061	129	11
2015					1,374	1,342	1,269	1,290	1,218	1,157	202	12
2016						1,392	1,416	1,414	1,394	1,341	286	14
2017							1,466	1,499	1,495	1,474	494	20
2018								1,602	1,650	1,659	780	22
2019									1,670	1,691	1,204	17
2020										1,704	1,529	13
										Incurred losses and ALAE	$13,032	

	Cumulative Paid Losses and ALAE through December 31,									
Accident Year	**2011***	**2012***	**2013***	**2014***	**2015***	**2016***	**2017***	**2018***	**2019***	**2020**
2011	$ 16	$ 82	$ 200	$ 356	$ 517	$ 632	$ 711	$ 767	$ 822	$ 842
2012		15	93	218	377	522	642	725	789	830
2013			15	90	219	368	518	635	743	793
2014				21	106	238	396	540	671	752
2015					23	108	218	382	543	663
2016						22	115	274	461	620
2017							27	128	300	457
2018								35	166	367
2019									39	160
2020										34
									Paid losses and ALAE	5,518
									Net unpaid losses and ALAE for 2011– 2020 accident years	7,514
									Net unpaid losses and ALAE for accident years before 2011	383
									Net unpaid losses and ALAE	$ 7,897

* *Unaudited required supplemental information*

(15) Unpaid losses and loss adjustment expenses *(Continued)*

BH Primary Workers' Compensation and Other Casualty

We periodically evaluate ultimate loss and loss adjustment expense estimates for the workers' compensation and other casualty claims using a combination of commonly accepted actuarial methodologies such as the Bornhuetter-Ferguson and chain-ladder approaches using paid and incurred loss data. Paid and incurred loss data is segregated and analyzed by state due to the different state regulatory frameworks that may impact certain factors, including the duration and amount of loss payments. We also separately study the various components of liabilities, such as employee lost wages, medical expenses and the costs of claims investigations and administration. We establish case liabilities for reported claims based upon the facts and circumstances of the claim. The excess of the ultimate projected losses, including the expected development of case estimates, and the case-basis liabilities is included in IBNR liabilities.

	Incurred Losses and ALAE through December 31,											
Accident Year	**2011***	**2012***	**2013***	**2014***	**2015***	**2016***	**2017***	**2018***	**2019***	**2020**	**IBNR and Case Development Liabilities**	**Cumulative Number of Reported Claims (in thousands)**
2011	$ 738	$ 675	$ 675	$ 624	$ 621	$ 618	$ 607	$ 596	$ 591	$ 576	$ 39	46
2012		873	850	837	791	780	762	750	736	718	53	53
2013			1,258	1,228	1,178	1,127	1,096	1,072	1,050	1,028	120	67
2014				1,743	1,638	1,614	1,548	1,482	1,497	1,477	190	90
2015					2,169	2,127	2,042	2,014	2,025	1,997	267	111
2016						2,511	2,422	2,359	2,325	2,365	470	115
2017							3,044	2,907	2,842	2,843	691	138
2018								3,544	3,412	3,480	1,152	160
2019									4,074	4,102	1,788	170
2020										4,421	2,987	120
										Incurred losses and ALAE	$23,007	

	Cumulative Paid Losses and ALAE through December 31,									
Accident Year	**2011***	**2012***	**2013***	**2014***	**2015***	**2016***	**2017***	**2018***	**2019***	**2020**
2011	$ 109	$ 220	$ 333	$ 403	$ 453	$ 481	$ 496	$ 505	$ 512	$ 519
2012		116	299	414	501	560	592	611	626	634
2013			177	422	609	725	793	835	858	874
2014				239	557	800	1,007	1,111	1,176	1,214
2015					289	700	1,017	1,289	1,488	1,570
2016						329	775	1,148	1,461	1,661
2017							441	1,003	1,434	1,771
2018								538	1,198	1,683
2019									682	1,478
2020										695
									Paid losses and ALAE	12,099
									Net unpaid losses and ALAE for 2011 – 2020 accident years	10,908
									Net unpaid losses and ALAE for accident years before 2011	386
									Net unpaid losses and ALAE	$11,294

* *Unaudited required supplemental information*

BHRG

We use a variety of methodologies to establish BHRG's estimates for property and casualty claims liabilities. We use certain methodologies, such as paid and incurred loss development techniques, incurred and paid loss Bornhuetter-Ferguson techniques and frequency and severity techniques, as well as ground-up techniques when appropriate.

Our claims liabilities are principally a function of reported losses from ceding companies, case development and IBNR liability estimates. Case loss estimates are reported under our contracts either individually or in bulk as provided under the terms of the contracts. We may independently evaluate case losses reported by the ceding company, and if deemed appropriate, we may establish case liabilities based on our estimates.

(15) Unpaid losses and loss adjustment expenses *(Continued)*

Estimated IBNR liabilities are affected by expected case loss emergence patterns and expected loss ratios, which are evaluated as groups of contracts with similar exposures or on a contract-by-contract basis. Estimated case and IBNR liabilities for major catastrophe events are generally based on a per-contract assessment of the ultimate cost associated with the individual loss event. Claim count data is not provided consistently by ceding companies under our contracts or is otherwise considered unreliable.

Incurred and paid losses and ALAE of BHRG are disaggregated based on losses that are expected to have shorter claim-tails (property) and losses expected to have longer claim-tails (casualty). Under certain contracts, the coverage can apply to multiple lines of business written by the ceding company, whether property, casualty or combined, and the ceding company may not report loss data by such lines consistently, if at all. In those instances, we allocated losses to property and casualty coverages based on internal estimates. BHRG's disaggregated incurred and paid losses and ALAE are summarized by accident year, net of reinsurance. IBNR and case development liabilities are as of December 31, 2020. Dollars are in millions.

BHRG Property

Incurred Losses and ALAE through December 31,

Accident Year	2011*	2012*	2013*	2014*	2015*	2016*	2017*	2018*	2019*	2020	IBNR and Case Development Liabilities
2011	$ 4,111	$ 4,095	$ 3,804	$ 3,711	$ 3,707	$ 3,672	$ 3,654	$ 3,638	$ 3,627	$ 3,616	$ 30
2012		3,153	2,846	2,644	2,403	2,351	2,348	2,329	2,315	2,305	35
2013			3,255	3,093	2,745	2,653	2,631	2,570	2,518	2,504	45
2014				2,648	2,436	2,322	2,178	2,123	2,050	2,021	48
2015					3,287	3,135	2,577	2,979	2,976	3,000	143
2016						3,293	3,923	3,646	3,614	3,616	218
2017							5,291	4,986	4,837	4,727	187
2018								4,426	4,524	4,397	678
2019									4,146	4,299	952
2020										5,858	3,129
									Incurred losses and ALAE	$ 36,343	

Cumulative Paid Losses and ALAE through December 31,

Accident Year	2011*	2012*	2013*	2014*	2015*	2016*	2017*	2018*	2019*	2020
2011	$ 609	$ 2,259	$ 2,917	$ 3,188	$ 3,304	$ 3,387	$ 3,429	$ 3,474	$ 3,493	$ 3,507
2012		262	1,232	1,813	1,950	2,040	2,117	2,135	2,181	2,199
2013			526	1,459	1,906	2,105	2,226	2,307	2,347	2,376
2014				467	1,249	1,574	1,713	1,779	1,829	1,858
2015					581	1,614	1,969	2,166	2,271	2,453
2016						709	1,811	2,208	2,670	2,923
2017							1,028	2,734	3,660	3,972
2018								915	2,341	2,868
2019									751	2,282
2020										960
									Paid losses and ALAE	25,398
									Net unpaid losses and ALAE for 2011 – 2020 accident years	10,945
									Net unpaid losses and ALAE for accident years before 2011	335
									Net unpaid losses and ALAE	$ 11,280

* *Unaudited required supplemental information*

(15) Unpaid losses and loss adjustment expenses *(Continued)*

BHRG Casualty

Incurred Losses and ALAE through December 31,

Accident Year	2011*	2012*	2013*	2014*	2015*	2016*	2017*	2018*	2019*	2020	IBNR and Case Development Liabilities
2011	$ 2,635	$ 2,726	$ 2,595	$ 2,536	$ 2,447	$ 2,354	$ 2,346	$ 2,307	$ 2,272	$ 2,255	$ 279
2012		2,820	3,002	2,837	2,899	2,827	2,712	2,645	2,588	2,581	317
2013			2,160	2,298	2,328	2,170	2,114	2,060	1,964	1,892	387
2014				1,900	2,099	2,068	2,030	1,944	1,980	1,970	537
2015					1,902	2,109	2,137	2,035	1,908	1,870	455
2016						1,928	2,138	2,047	2,003	1,922	555
2017							2,216	2,711	2,588	2,494	762
2018								2,948	3,585	3,509	1,183
2019									3,455	3,931	1,924
2020										3,883	2,754
										Incurred losses and ALAE	$ 26,307

Cumulative Paid Losses and ALAE through December 31,

Accident Year	2011*	2012*	2013*	2014*	2015*	2016*	2017*	2018*	2019*	2020
2011	$ 294	$ 824	$ 1,169	$ 1,412	$ 1,501	$ 1,595	$ 1,673	$ 1,713	$ 1,748	$ 1,776
2012		312	757	1,150	1,381	1,539	1,664	1,764	1,825	1,883
2013			294	530	818	947	1,052	1,155	1,215	1,273
2014				153	488	655	765	889	974	1,119
2015					199	500	725	846	938	1,029
2016						255	563	742	874	972
2017							233	574	830	1,282
2018								267	875	1,649
2019									356	906
2020										406
									Paid losses and ALAE	12,295
									Net unpaid losses and ALAE for 2011 – 2020 accident years	14,012
									Net unpaid losses and ALAE for accident years before 2011	8,878
									Net unpaid losses and ALAE	$ 22,890

* *Unaudited required supplemental information*

(15) Unpaid losses and loss adjustment expenses *(Continued)*

Required supplemental unaudited average historical claims duration information based on the net losses and ALAE incurred and paid accident year data in the preceding tables follows. The percentages show the average portions of net losses and ALAE paid by each succeeding year, with year 1 representing the current accident year.

Average Annual Percentage Payout of Incurred Losses by Age, Net of Reinsurance

In Year	1	2	3	4	5	6	7	8	9	10
GEICO Physical Damage	97%	2%								
GEICO Auto Liability	42%	29%	13%	7%	4%					
BH Primary Medical Professional Liability	2%	7%	12%	14%	14%	12%	9%	6%	5%	2%
BH Primary Workers' Compensation and Other Casualty	16%	21%	16%	13%	8%	4%	3%	2%	1%	1%
BHRG Property	19%	37%	17%	8%	4%	4%	1%	2%	1%	0%
BHRG Casualty	11%	16%	14%	9%	5%	5%	5%	2%	2%	1%

(16) Retroactive reinsurance contracts

Retroactive reinsurance policies provide indemnification of losses and loss adjustment expenses of short-duration insurance contracts with respect to underlying loss events that occurred prior to the contract inception date. Claims payments may commence immediately after the contract date or, when applicable, once a contractual retention amount has been reached. Reconciliations of the changes in estimated liabilities for retroactive reinsurance unpaid losses and loss adjustment expenses ("claim liabilities") and related deferred charge reinsurance assumed assets for each of the three years ended December 31, 2020 follow (in millions).

	2020		2019		2018	
	Unpaid losses and loss adjustment expenses	Deferred charges reinsurance assumed	Unpaid losses and loss adjustment expenses	Deferred charges reinsurance assumed	Unpaid losses and loss adjustment expenses	Deferred charges reinsurance assumed
Balances at beginning of year	$ 42,441	$ (13,747)	$ 41,834	$ (14,104)	$ 42,937	$ (15,278)
Incurred losses and loss adjustment expenses:						
Current year contracts	—	—	1,138	(453)	603	(86)
Prior years' contracts	(399)	1,306	378	810	(341)	1,260
Total	(399)	1,306	1,516	357	262	1,174
Paid losses and loss adjustment expenses	(1,076)	—	(909)	—	(1,365)	—
Balances at December 31	$ 40,966	$ (12,441)	$ 42,441	$ (13,747)	$ 41,834	$ (14,104)
Incurred losses and loss adjustment expenses, net of deferred charges	$ 907		$ 1,873		$ 1,436	

In the preceding table, classifications of incurred losses and loss adjustment expenses are based on the inception dates of the contracts. We do not believe that analysis of losses incurred and paid by accident year of the underlying event is relevant or meaningful given that our exposure to losses incepts when the contract incepts. Further, we believe the classifications of reported claims and case development liabilities have little or no practical analytical value.

Estimated ultimate claim liabilities included $17.7 billion at December 31, 2020 and $18.2 billion at December 31, 2019, with respect to an agreement with various subsidiaries of American International Group, Inc. (collectively, "AIG") to indemnify AIG for 80% of up to $25 billion of losses and allocated loss adjustment expenses in excess of $25 billion retained by AIG for certain commercial insurance loss events occurring prior to 2016. The related deferred charge assets were $5.4 billion at December 31, 2020 and $6.3 billion at December 31, 2019.

(16) Retroactive reinsurance contracts *(Continued)*

Incurred losses and loss adjustment expenses related to contracts written in prior years were $907 million in 2020, $1,188 million in 2019 and $919 million in 2018, which included recurring amortization of deferred charges and the effect of changes in the timing and amount of expected future loss payments.

In establishing retroactive reinsurance claim liabilities, we analyze historical aggregate loss payment patterns and project losses into the future under various probability-weighted scenarios. We expect the claim-tail to be very long for many contracts, with some lasting several decades. We monitor claim payment activity and review ceding company reports and other information concerning the underlying losses. We reassess and revise the expected timing and amounts of ultimate losses periodically or when significant events are revealed through our monitoring and review processes.

Our retroactive reinsurance claim liabilities include estimated liabilities for environmental, asbestos and other latent injury exposures of approximately $12.5 billion at December 31, 2020 and $12.9 billion at December 31, 2019. Retroactive reinsurance contracts are generally subject to aggregate policy limits and thus, our exposure to such claims under these contracts is likewise limited. We monitor evolving case law and its effect on environmental and other latent injury claims. Changing laws or government regulations, newly identified toxins, newly reported claims, new theories of liability, new contract interpretations and other factors could result in increases in these liabilities, which could be material to our results of operations. We are unable to reliably estimate the amount of additional net loss or the range of net loss that is reasonably possible.

(17) Notes payable and other borrowings

Notes payable and other borrowings are summarized below (in millions). The weighted average interest rates and maturity date ranges shown in the following tables are based on borrowings as of December 31, 2020.

	Weighted Average Interest Rate	December 31, 2020	December 31, 2019
Insurance and other:			
Berkshire Hathaway Inc. ("Berkshire"):			
U.S. Dollar denominated due 2021-2047	3.2%	$ 8,308	$ 8,324
Euro denominated due 2021-2035	1.0%	8,326	7,641
Japanese Yen denominated due 2023-2060	0.7%	6,031	3,938
Berkshire Hathaway Finance Corporation ("BHFC"):			
U.S. Dollar denominated due 2021-2050	3.7%	10,766	8,679
Great Britain Pound denominated due 2039-2059	2.5%	2,347	2,274
Other subsidiary borrowings due 2021-2045	4.2%	4,682	5,262
Short-term subsidiary borrowings	2.5%	1,062	1,472
		$ 41,522	$ 37,590

In March 2020, Berkshire repaid €1.0 billion of maturing senior notes and issued €1.0 billion of 0.0% senior notes due in 2025. In April 2020, Berkshire issued ¥195.5 billion (approximately $1.8 billion) of senior notes with maturity dates ranging from 2023 to 2060 and a weighted average interest rate of 1.07%.

Borrowings of BHFC, a wholly owned finance subsidiary of Berkshire, consist of senior unsecured notes used to fund manufactured housing loans originated or acquired and equipment held for lease of certain subsidiaries. BHFC borrowings are fully and unconditionally guaranteed by Berkshire. During 2020, BHFC repaid $900 million of maturing senior notes and issued $3.0 billion of senior notes consisting of $500 million of 1.85% notes due in 2030, $750 million of 1.45% notes due in 2030 and $1.75 billion of 2.85% notes due in 2050.

The carrying values of Berkshire and BHFC non-U.S. Dollar denominated senior notes (€6.85 billion, £1.75 billion and ¥625.5 billion par) reflect the applicable exchange rates as of the balance sheet dates. The effects of changes in foreign currency exchange rates during the period are recorded in earnings as a component of selling, general and administrative expenses. Changes in the exchange rates resulted in pre-tax losses of approximately $1.0 billion in 2020 and pre-tax gains of $192 million in 2019 and $366 million in 2018.

(17) Notes payable and other borrowings *(Continued)*

In addition to BHFC borrowings, Berkshire guaranteed approximately $1.2 billion of other subsidiary borrowings at December 31, 2020. Generally, Berkshire's guarantee of a subsidiary's debt obligation is an absolute, unconditional and irrevocable guarantee for the full and prompt payment when due of all payment obligations.

	Weighted Average Interest Rate	December 31, 2020	December 31, 2019
Railroad, utilities and energy:			
Berkshire Hathaway Energy Company ("BHE") and subsidiaries:			
BHE senior unsecured debt due 2021-2051	4.2%	$ 13,447	$ 8,581
Subsidiary and other debt due 2021-2064	4.1%	36,420	30,772
Short-term borrowings	1.8%	2,286	3,214
Burlington Northern Santa Fe ("BNSF") and subsidiaries due 2021-2097	4.6%	23,220	23,211
		$ 75,373	$ 65,778

BHE subsidiary debt represents amounts issued pursuant to separate financing agreements. Substantially all of the assets of certain BHE subsidiaries are, or may be, pledged or encumbered to support or otherwise secure debt. These borrowing arrangements generally contain various covenants, including covenants which pertain to leverage ratios, interest coverage ratios and/or debt service coverage ratios. In November 2020, BHE's subsidiary debt increased $5.6 billion for the debt assumed in connection with the Dominion pipeline business acquisition. See Note 2 to the Consolidated Financial Statements. During 2020, BHE and its subsidiaries also issued new term debt of approximately $7.6 billion with maturity dates ranging from 2025 to 2062 and a weighted average interest rate of 3.2% and repaid $3.2 billion of term debt and reduced short-term borrowings.

BNSF's borrowings are primarily senior unsecured debentures. During 2020, BNSF issued $575 million of 3.05% senior unsecured debentures due in 2051 and repaid debt of $570 million. As of December 31, 2020, BNSF, BHE and their subsidiaries were in compliance with all applicable debt covenants. Berkshire does not guarantee any debt, borrowings or lines of credit of BNSF, BHE or their subsidiaries.

As of December 31, 2020, our subsidiaries had unused lines of credit and commercial paper capacity aggregating approximately $9.3 billion to support short-term borrowing programs and provide additional liquidity. Such unused lines of credit included approximately $8.2 billion related to BHE and its subsidiaries.

Debt principal repayments expected during each of the next five years are as follows (in millions). Amounts in 2021 include short-term borrowings.

	2021	2022	2023	2024	2025
Insurance and other	$ 4,354	$ 1,593	$ 6,021	$ 2,343	$ 2,817
Railroad, utilities and energy	5,044	3,405	4,792	3,965	3,777
	$ 9,398	$ 4,998	$ 10,813	$ 6,308	$ 6,594

(18) Income taxes

The liabilities for income taxes reflected in our Consolidated Balance Sheets are as follows (in millions).

	December 31, 2020	December 31, 2019
Currently payable (receivable)	$ (276)	$ 24
Deferred	73,261	65,823
Other	1,113	952
	$ 74,098	$ 66,799

(18) Income taxes *(Continued)*

The tax effects of temporary differences that give rise to significant portions of deferred tax assets and deferred tax liabilities are shown below (in millions).

	December 31,	
	2020	2019
Deferred tax liabilities:		
Investments – unrealized appreciation and cost basis differences	$ 40,181	$ 32,134
Deferred charges reinsurance assumed	2,613	2,890
Property, plant and equipment and equipment held for lease	30,203	29,388
Goodwill and other intangible assets	6,753	7,293
Other	3,736	3,144
	83,486	74,849
Deferred tax assets:		
Unpaid losses and loss adjustment expenses	(1,135)	(1,086)
Unearned premiums	(900)	(853)
Accrued liabilities	(2,193)	(1,981)
Regulatory liabilities	(1,421)	(1,610)
Other	(4,576)	(3,496)
	(10,225)	(9,026)
Net deferred tax liability	$ 73,261	$ 65,823

We have not established deferred income taxes on accumulated undistributed earnings of certain foreign subsidiaries, which are expected to be reinvested indefinitely. Repatriation of all accumulated earnings of foreign subsidiaries would be impracticable to the extent that such earnings represent capital to support normal business operations. Generally, no U.S. federal income taxes will be imposed on future distributions of foreign earnings under current law. However, distributions to the U.S. or other foreign jurisdictions could be subject to withholding and other local taxes.

On December 22, 2017, legislation known as the Tax Cuts and Jobs Act of 2017 ("TCJA") was enacted. Among its provisions, the TCJA reduced the statutory U.S. Corporate income tax rate from 35% to 21% effective January 1, 2018 and provided for a one-time tax on certain accumulated undistributed post-1986 earnings of foreign subsidiaries. These effects were largely recorded in 2017 upon the enactment. In 2018, we reduced our estimate of the income taxes on the deemed repatriation of earnings of foreign subsidiaries and recognized additional deferred income tax rate change effects.

Income tax expense reflected in our Consolidated Statements of Earnings for each of the three years ending December 31, 2020 is as follows (in millions).

	2020	2019	2018
Federal	$ 10,596	$ 19,069	$ (1,613)
State	1,086	625	175
Foreign	758	1,210	1,117
	$ 12,440	$ 20,904	$ (321)
Current	$ 5,052	$ 5,818	$ 5,176
Deferred	7,388	15,086	(5,497)
	$ 12,440	$ 20,904	$ (321)

(18) Income taxes *(Continued)*

Income tax expense is reconciled to hypothetical amounts computed at the U.S. federal statutory rate for each of the three years ending December 31, 2020 in the table below (in millions).

	2020	2019	2018
Earnings before income taxes	$ 55,693	$ 102,696	$ 4,001
Hypothetical income tax expense computed at the U.S. federal statutory rate	$ 11,696	$ 21,566	$ 840
Dividends received deduction and tax-exempt interest	(448)	(433)	(393)
State income taxes, less U.S. federal income tax benefit	858	494	138
Foreign tax rate differences	13	(6)	271
U.S. income tax credits	(1,519)	(942)	(711)
Net benefit from the enactment of the TCJA	—	—	(302)
Goodwill impairments	1,977	20	21
Other differences, net	(137)	205	(185)
	$ 12,440	$ 20,904	$ (321)
Effective income tax rate	22.3%	20.4%	(8.0)%

We file income tax returns in the United States and in state, local and foreign jurisdictions. We have settled income tax liabilities with the U.S. federal taxing authority ("IRS") for tax years through 2011 and the tax years 2012 and 2013 remain open. The IRS is auditing Berkshire's consolidated U.S. federal income tax returns for the 2014 through 2016 tax years. We are also under audit or subject to audit with respect to income taxes in many state and foreign jurisdictions. It is reasonably possible that certain of these income tax examinations will be settled in 2021. We currently do not believe that the outcome of unresolved issues or claims will be material to our Consolidated Financial Statements.

At December 31, 2020 and 2019, net unrecognized tax benefits were $1,113 million and $952 million, respectively. Included in the balance at December 31, 2020, were $920 million of tax positions that, if recognized, would impact the effective tax rate. The remaining balance in net unrecognized tax benefits principally relates to tax positions where the ultimate recognition is highly certain but there is uncertainty about the timing of recognition. Because of the impact of deferred income tax accounting, these positions, when recognized, would not affect the annual effective income tax rate. We recorded income tax expense of $60 million in 2020 and $377 million in 2019 for uncertain tax positions related to investments by a subsidiary in certain tax equity investment funds that generated income tax benefits from 2015 through 2018. We now believe that it is more likely than not those income tax benefits are not valid. We do not expect any material increases to the estimated amount of unrecognized tax benefits during 2021.

(19) Dividend restrictions – Insurance subsidiaries

Payments of dividends by our insurance subsidiaries are restricted by insurance statutes and regulations. Without prior regulatory approval, our principal insurance subsidiaries may declare up to approximately $23 billion as ordinary dividends during 2021. Investments in fixed maturity and equity securities and short-term investments on deposit with U.S. state insurance authorities in accordance with state insurance regulations were approximately $5.5 billion at December 31, 2020 and $6.3 billion at December 31, 2019.

Combined shareholders' equity of U.S. based insurance subsidiaries determined pursuant to statutory accounting rules (Surplus as Regards Policyholders) was approximately $237 billion at December 31, 2020 and $216 billion at December 31, 2019. Statutory surplus differs from the corresponding amount based on GAAP, due to differences in accounting for certain assets and liabilities. For instance, deferred charges reinsurance assumed, deferred policy acquisition costs, unrealized gains on certain investments and related deferred income taxes are recognized for GAAP but not for statutory reporting purposes. In addition, the carrying values of certain assets, such as goodwill and non-insurance entities owned by our insurance subsidiaries, are not fully recognized for statutory reporting purposes.

(20) Fair value measurements

Our financial assets and liabilities are summarized below as of December 31, 2020 and December 31, 2019, with fair values shown according to the fair value hierarchy (in millions). The carrying values of cash and cash equivalents, U.S. Treasury Bills, receivables and accounts payable, accruals and other liabilities are considered to be reasonable estimates of their fair values.

	Carrying Value	Fair Value	Quoted Prices (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)
December 31, 2020					
Investments in fixed maturity securities:					
U.S. Treasury, U.S. government corporations and agencies	$ 3,403	$ 3,403	$ 3,358	$ 45	$ —
Foreign governments	11,338	11,338	9,259	2,079	—
Corporate bonds	5,191	5,191	—	5,191	—
Other	478	478	—	478	—
Investments in equity securities	281,170	281,170	271,848	38	9,284
Investment in Kraft Heinz common stock	13,336	11,280	11,280	—	—
Loans and finance receivables	19,201	20,554	—	2,692	17,862
Derivative contract assets *(1)*	270	270	1	72	197
Derivative contract liabilities:					
Railroad, utilities and energy *(1)*	121	121	6	96	19
Equity index put options	1,065	1,065	—	—	1,065
Notes payable and other borrowings:					
Insurance and other	41,522	46,676	—	46,665	11
Railroad, utilities and energy	75,373	92,593	—	92,593	—
December 31, 2019					
Investments in fixed maturity securities:					
U.S. Treasury, U.S. government corporations and agencies	$ 3,090	$ 3,090	$ 3,046	$ 44	$ —
Foreign governments	8,638	8,638	5,437	3,201	—
Corporate bonds	6,352	6,352	—	6,350	2
Other	605	605	—	605	—
Investments in equity securities	248,027	248,027	237,271	46	10,710
Investment in Kraft Heinz common stock	13,757	10,456	10,456	—	—
Loans and finance receivables	17,527	17,861	—	1,809	16,052
Derivative contract assets *(1)*	145	145	—	23	122
Derivative contract liabilities:					
Railroad, utilities and energy *(1)*	76	76	6	59	11
Equity index put options	968	968	—	—	968
Notes payable and other borrowings:					
Insurance and other	37,590	40,589	—	40,569	20
Railroad, utilities and energy	65,778	76,237	—	76,237	—

(1) Assets are included in other assets and liabilities are included in accounts payable, accruals and other liabilities.

(20) Fair value measurements *(Continued)*

The fair values of substantially all of our financial instruments were measured using market or income approaches. The hierarchy for measuring fair value consists of Levels 1 through 3, which are described below.

Level 1 – Inputs represent unadjusted quoted prices for identical assets or liabilities exchanged in active markets.

Level 2 – Inputs include directly or indirectly observable inputs (other than Level 1 inputs) such as quoted prices for similar assets or liabilities exchanged in active or inactive markets; quoted prices for identical assets or liabilities exchanged in inactive markets; other inputs that may be considered in fair value determinations of the assets or liabilities, such as interest rates and yield curves, volatilities, prepayment speeds, loss severities, credit risks and default rates; and inputs that are derived principally from or corroborated by observable market data by correlation or other means. Pricing evaluations generally reflect discounted expected future cash flows, which incorporate yield curves for instruments with similar characteristics, such as credit ratings, estimated durations and yields for other instruments of the issuer or entities in the same industry sector.

Level 3 – Inputs include unobservable inputs used in the measurement of assets and liabilities. Management is required to use its own assumptions regarding unobservable inputs because there is little, if any, market activity in the assets or liabilities and it may be unable to corroborate the related observable inputs. Unobservable inputs require management to make certain projections and assumptions about the information that would be used by market participants in valuing assets or liabilities.

Reconciliations of assets and liabilities measured and carried at fair value on a recurring basis with the use of significant unobservable inputs (Level 3) for each of the three years ending December 31, 2020 follow (in millions).

	Investments in equity and fixed maturity securities	Net derivative contract liabilities
Balance December 31, 2017	$ 6	$ (2,069)
Gains (losses) included in:		
Earnings	—	(118)
Other comprehensive income	—	2
Regulatory assets and liabilities	—	3
Acquisitions	2	3
Dispositions and settlements	(1)	(164)
Balance December 31, 2018	7	(2,343)
Gains (losses) included in:		
Earnings	404	1,972
Other comprehensive income	—	(1)
Regulatory assets and liabilities	—	(26)
Acquisitions	10,000	6
Dispositions and settlements	(4)	(465)
Balance December 31, 2019	10,407	(857)
Gains (losses) included in:		
Earnings	(1,426)	603
Other comprehensive income	—	—
Regulatory assets and liabilities	—	(17)
Acquisitions	—	5
Dispositions and settlements	(2)	(621)
Balance December 31, 2020	$ 8,979	$ (887)

(20) Fair value measurements *(Continued)*

We acquired investments in Occidental Cumulative Perpetual Preferred Stock ("Occidental Preferred") and Occidental common stock warrants in August 2019 at an aggregate cost of $10 billion. We currently consider the related fair value measurements to contain Level 3 inputs. See Note 4 for information regarding these investments.

Quantitative information as of December 31, 2020, with respect to assets and liabilities measured and carried at fair value on a recurring basis with the use of significant unobservable inputs (Level 3) follows (in millions).

	Fair Value	Principal Valuation Techniques	Unobservable Inputs	Weighted Average
Investments in equity securities:				
Preferred stock	$ 8,891	Discounted cash flow	Expected duration	9 years
			Discount for transferability restrictions and subordination	375 bps
Common stock warrants	86	Warrant pricing model	Expected duration	9 years
			Volatility	32%
Derivative contract liabilities	1,065	Option pricing model	Volatility	19%

Investments in equity securities in the preceding table include our investments in the Occidental Preferred and Occidental common stock warrants. These investments are subject to contractual restrictions on transferability and contain provisions that currently prevent us from economically hedging our investments. In applying discounted cash flow techniques in valuing the Occidental Preferred, we made assumptions regarding the expected duration of the investment. The Occidental Preferred is redeemable at Occidental's option beginning in 2029. We also made estimates regarding the impact of subordination, as the Occidental Preferred has a lower priority in liquidation than debt instruments. In valuing the Occidental common stock warrants, we used a warrant valuation model. While most of the inputs to the model are observable, we made assumptions regarding the expected duration and volatility of the warrants. The Occidental common stock warrants contractually expire on the one-year anniversary on which no Occidental Preferred remains outstanding.

Our equity index put option contracts are illiquid and contain contract terms that are not standard in derivatives markets. For example, we are not required to post collateral under most of our contracts. We determine the fair value of the equity index put option contract liabilities based on the Black-Scholes option valuation model.

(21) Accumulated other comprehensive income

A summary of the net changes in after-tax accumulated other comprehensive income attributable to Berkshire Hathaway shareholders for each of the three years ending December 31, 2020 follows (in millions).

	Unrealized appreciation of investments, net	Foreign currency translation	Defined benefit pension plans	Other	Accumulated other comprehensive income
Balance December 31, 2017	$ 62,093	$ (3,114)	$ (420)	$ 12	$ 58,571
Reclassifications to retained earnings upon adoption of new accounting standards	(61,340)	(65)	36	(6)	(61,375)
Other comprehensive income, net	(383)	(1,424)	(432)	28	(2,211)
Balance December 31, 2018	370	(4,603)	(816)	34	(5,015)
Other comprehensive income, net	111	257	(553)	(43)	(228)
Balance December 31, 2019	481	(4,346)	(1,369)	(9)	(5,243)
Other comprehensive income, net	55	1,264	(276)	(43)	1,000
Balance December 31, 2020	$ 536	$ (3,082)	$ (1,645)	$ (52)	$ (4,243)

(22) Common stock

Changes in Berkshire's issued, treasury and outstanding common stock during the three years ending December 31, 2020 are shown in the table below. In addition to our common stock, 1,000,000 shares of preferred stock are authorized, but none are issued.

	Class A, $5 Par Value (1,650,000 shares authorized)			Class B, $0.0033 Par Value (3,225,000,000 shares authorized)		
	Issued	Treasury	Outstanding	Issued	Treasury	Outstanding
Balance December 31, 2017	762,755	(11,680)	751,075	1,342,066,749	(1,409,762)	1,340,656,987
Conversions of Class A common stock to Class B common stock and exercises of replacement stock options	(20,542)	—	(20,542)	31,492,234	—	31,492,234
Treasury stock acquired	—	(1,217)	(1,217)	—	(4,729,147)	(4,729,147)
Balance December 31, 2018	742,213	(12,897)	729,316	1,373,558,983	(6,138,909)	1,367,420,074
Conversions of Class A common stock to Class B common stock and exercises of replacement stock options	(22,906)	—	(22,906)	34,624,869	—	34,624,869
Treasury stock acquired	—	(4,440)	(4,440)	—	(17,563,410)	(17,563,410)
Balance December 31, 2019	719,307	(17,337)	701,970	1,408,183,852	(23,702,319)	1,384,481,533
Conversions of Class A common stock to Class B common stock and exercises of replacement stock options	(40,784)	—	(40,784)	61,176,000	—	61,176,000
Treasury stock acquired	—	(17,255)	(17,255)	—	(95,614,062)	(95,614,062)
Balance December 31, 2020	678,523	(34,592)	643,931	1,469,359,852	(119,316,381)	1,350,043,471

Each Class A common share is entitled to one vote per share. Class B common stock possesses dividend and distribution rights equal to one-fifteen-hundredth (1/1,500) of such rights of Class A common stock. Each Class B common share possesses voting rights equivalent to one-ten-thousandth (1/10,000) of the voting rights of a Class A share. Unless otherwise required under Delaware General Corporation Law, Class A and Class B common shares vote as a single class. Each share of Class A common stock is convertible, at the option of the holder, into 1,500 shares of Class B common stock. Class B common stock is not convertible into Class A common stock. On an equivalent Class A common stock basis, there were 1,543,960 shares outstanding as of December 31, 2020 and 1,624,958 shares outstanding as of December 31, 2019.

Since we have two classes of common stock, we provide earnings per share data on the Consolidated Statements of Earnings for average equivalent Class A shares outstanding and average equivalent Class B shares outstanding. Class B shares are economically equivalent to one-fifteen-hundredth (1/1,500) of a Class A share. Average equivalent Class A shares outstanding represents average Class A shares outstanding plus one-fifteen-hundredth (1/1,500) of the average Class B shares outstanding. Average equivalent Class B shares outstanding represents average Class B shares outstanding plus 1,500 times average Class A shares outstanding.

Berkshire's common stock repurchase program, as amended, permits Berkshire to repurchase shares any time that Warren Buffett, Berkshire's Chairman of the Board and Chief Executive Officer, and Charlie Munger, Vice Chairman of the Board, believe that the repurchase price is below Berkshire's intrinsic value, conservatively determined. The program continues to allow share repurchases in the open market or through privately negotiated transactions and does not specify a maximum number of shares to be repurchased. However, repurchases will not be made if they would reduce the total value of Berkshire's consolidated cash, cash equivalents and U.S. Treasury Bill holdings below $20 billion. The repurchase program does not obligate Berkshire to repurchase any specific dollar amount or number of Class A or Class B shares and there is no expiration date to the program.

(23) Revenues from contracts with customers

We recognize revenue when a good or service is transferred to a customer. A good or service is transferred when or as the customer obtains control of that good or service. Revenues are based on the consideration we expect to receive in connection with our promises to deliver goods and services to our customers.

(23) Revenues from contracts with customers *(Continued)*

The following tables summarize customer contract revenues disaggregated by reportable segment and the source of the revenue for each of the three years ended December 31, 2020 (in millions). Other revenues included in consolidated revenues were primarily insurance premiums earned, interest, dividend and other investment income and leasing revenues, which are not considered to be revenues from contracts with customers under GAAP.

2020	Manufacturing	McLane Company	Service and retailing	BNSF	Berkshire Hathaway Energy	Insurance, Corporate and other	Total
Manufactured products:							
Industrial and commercial products	$ 20,772	$ —	$ 192	$ —	$ —	$ —	$ 20,964
Building products	15,943	—	—	—	—	—	15,943
Consumer products	14,757	—	—	—	—	—	14,757
Grocery and convenience store distribution	—	30,795	—	—	—	—	30,795
Food and beverage distribution	—	15,368	—	—	—	—	15,368
Auto sales	—	—	8,258	—	—	—	8,258
Other retail and wholesale distribution	2,452	—	12,470	—	—	—	14,922
Service	1,456	584	3,332	20,693	4,595	—	30,660
Electricity and natural gas	—	—	—	—	15,066	—	15,066
Total	55,380	46,747	24,252	20,693	19,661	—	166,733
Other revenues	3,598	93	3,859	57	1,353	69,817	78,777
	$ 58,978	$ 46,840	$ 28,111	$ 20,750	$ 21,014	$ 69,817	$245,510

2019							
Manufactured products:							
Industrial and commercial products	$ 25,311	$ —	$ 184	$ —	$ —	$ —	$ 25,495
Building products	15,620	—	—	—	—	—	15,620
Consumer products	14,120	—	—	—	—	—	14,120
Grocery and convenience store distribution	—	33,057	—	—	—	—	33,057
Food and beverage distribution	—	16,767	—	—	—	—	16,767
Auto sales	—	—	8,481	—	—	—	8,481
Other retail and wholesale distribution	2,299	—	12,213	—	—	—	14,512
Service	1,642	539	4,062	23,302	4,096	—	33,641
Electricity and natural gas	—	—	—	—	14,819	—	14,819
Total	58,992	50,363	24,940	23,302	18,915	—	176,512
Other revenues	3,632	95	4,459	55	1,181	68,682	78,104
	$ 62,624	$ 50,458	$ 29,399	$ 23,357	$ 20,096	$ 68,682	$254,616

2018							
Manufactured products:							
Industrial and commercial products	$ 25,707	$ —	$ 204	$ —	$ —	$ —	$ 25,911
Building products	14,323	—	—	—	—	—	14,323
Consumer products	14,790	—	—	—	—	—	14,790
Grocery and convenience store distribution	—	33,518	—	—	—	—	33,518
Food and beverage distribution	—	16,309	—	—	—	—	16,309
Auto sales	—	—	8,181	—	—	—	8,181
Other retail and wholesale distribution	2,091	—	12,067	—	—	—	14,158
Service	1,519	84	4,100	23,652	3,949	—	33,304
Electricity and natural gas	—	—	—	—	14,951	—	14,951
Total	58,430	49,911	24,552	23,652	18,900	—	175,445
Other revenues	3,340	76	4,297	51	1,070	63,558	72,392
	$ 61,770	$ 49,987	$ 28,849	$ 23,703	$ 19,970	$ 63,558	$247,837

(23) Revenues from contracts with customers *(Continued)*

A summary of the transaction price allocated to the significant unsatisfied remaining performance obligations relating to contracts with expected durations in excess of one year as of December 31, 2020 and the timing of when the performance obligations are expected to be satisfied follows (in millions).

	Less than 12 months	Greater than 12 months	Total
Electricity and natural gas	$ 3,210	$ 22,088	$ 25,298
Other sales and service contracts	1,228	2,382	3,610

(24) Pension plans

Certain of our subsidiaries sponsor defined benefit pension plans. Benefits under the plans are generally based on years of service and compensation or fixed benefit rates. Plan sponsors may make contributions to the plans to meet regulatory requirements and may also make discretionary contributions. The components of our net periodic pension expense for each of the three years ending December 31, 2020 follow (in millions).

	2020	2019	2018
Service cost	$ 235	$ 224	$ 271
Interest cost	510	618	593
Expected return on plan assets	(955)	(936)	(988)
Amortization of actuarial losses and other	171	26	188
Net periodic pension expense	$ (39)	$ (68)	$ 64

The projected benefit obligation ("PBO") is the actuarial present value of benefits earned based upon service and compensation prior to the valuation date and, if applicable, includes assumptions regarding future compensation levels. Benefit obligations under qualified U.S. defined benefit pension plans are funded through assets held in trusts. Pension obligations under certain non-U.S. plans and non-qualified U.S. plans are unfunded and the aggregate PBOs of such plans were approximately $1.6 billion and $1.3 billion as of December 31, 2020 and 2019, respectively. The cost of pension plans covering employees of certain regulated subsidiaries of BHE are generally recoverable through the regulated rate making process.

The funded status at year end 2020 and 2019 and reconciliations of the changes in PBOs and plan assets related to BHE's pension plans and all other pension plans for each of the two years ending December 31, 2020 follow (in millions).

	2020			2019		
	BHE	Other	Total	BHE	Other	Total
Benefit obligations						
PBO beginning of year	$ 4,898	$ 13,808	$ 18,706	$ 4,551	$ 12,371	$ 16,922
Service cost	33	202	235	32	192	224
Interest cost	133	377	510	161	457	618
Benefits paid	(285)	(709)	(994)	(257)	(776)	(1,033)
Settlements	(63)	(12)	(75)	(121)	(46)	(167)
Actuarial (gains) or losses and other	566	1,481	2,047	532	1,610	2,142
PBO end of year	$ 5,282	$ 15,147	$ 20,429	$ 4,898	$ 13,808	$ 18,706
Plan assets						
Plan assets beginning of year	$ 4,808	$ 11,688	$ 16,496	$ 4,385	$ 10,574	$ 14,959
Employer contributions	69	127	196	68	131	199
Benefits paid	(285)	(709)	(994)	(257)	(776)	(1,033)
Actual return on plan assets	554	1,820	2,374	650	1,764	2,414
Settlements	(63)	(12)	(75)	(121)	(46)	(167)
Other	75	(134)	(59)	83	41	124
Plan assets end of year	$ 5,158	$ 12,780	$ 17,938	$ 4,808	$ 11,688	$ 16,496
Funded status – net liability	$ 124	$ 2,367	$ 2,491	$ 90	$ 2,120	$ 2,210

(24) Pension plans *(Continued)*

The funded status reflected in assets was $1,351 million and in liabilities was $3,842 million at December 31, 2020. The funded status included in assets was $857 million and in liabilities was $3,067 million at December 31, 2019.

The accumulated benefit obligation ("ABO") is the actuarial present value of benefits earned based on service and compensation prior to the valuation date. The ABO was $19.4 billion at December 31, 2020 and $17.5 billion at December 31, 2019. Information for plans with PBOs and ABOs in excess of plan assets as of December 31, 2020 and 2019 follows (in millions)

	2020	2019
PBOs	$ 12,775	$ 12,625
Plan assets	9,018	9,627
ABOs	10,875	10,617
Plan assets	7,820	8,367

Weighted average assumptions used in determining PBOs and net periodic pension expense follow.

	2020	2019	2018
Discount rate applicable to pension benefit obligations	2.3%	3.1%	3.9%
Expected long-term rate of return on plan assets	6.2	6.4	6.4
Rate of compensation increase	2.6	2.5	2.6
Discount rate applicable to net periodic pension expense	3.1	4.0	3.4

Pension benefit payments expected over the next ten years are as follows (in millions): 2021 – $1,105; 2022 – $1,031; 2023 – $1,034; 2024 – $1,037; 2025 – $1,040; and 2026 to 2030 – $5,119. Sponsoring subsidiaries expect to contribute $202 million to the plans in 2021.

Fair value measurements of plan assets as of December 31, 2020 and 2019 follow (in millions).

	Fair Value				Investment funds and partnerships at net asset value
	Total	Level 1	Level 2	Level 3	
December 31, 2020					
Cash and cash equivalents	$ 383	$ 243	$ 140	$ —	$ —
Equity securities	11,383	10,123	851	409	—
Fixed maturity securities	3,173	2,214	926	33	—
Investment funds and other	2,999	198	398	56	2,347
	$ 17,938	$ 12,778	$ 2,315	$ 498	$ 2,347
December 31, 2019					
Cash and cash equivalents	$ 412	$ 309	$ 103	$ —	$ —
Equity securities	11,105	9,860	836	409	—
Fixed maturity securities	2,328	1,593	704	31	—
Investment funds and other	2,651	143	358	40	2,110
	$ 16,496	$ 11,905	$ 2,001	$ 480	$ 2,110

See Note 20 for a discussion of the three levels of fair value measurements. Plan assets are generally invested with the long-term objective of producing earnings to adequately cover expected benefit obligations, while assuming a prudent level of risk. Allocations may change due to changing market conditions and investment opportunities. The expected rates of return on plan assets reflect subjective assessments of expected long-term investment returns. Generally, past investment returns are not given significant consideration when establishing assumptions for expected long-term rates of return on plan assets. Actual experience will differ from the assumed rates of return.

(24) Pension plans *(Continued)*

A reconciliation of the pre-tax accumulated other comprehensive income (loss) related to defined benefit pension plans for each of the two years ending December 31, 2020 follows (in millions).

	2020	2019
Balance beginning of year	$ (1,896)	$ (1,184)
Amount included in net periodic pension expense	141	94
Actuarial gains (losses) and other	(496)	(806)
Balance end of year	$ (2,251)	$ (1,896)

Several of our subsidiaries also sponsor defined contribution retirement plans, such as 401(k) or profit-sharing plans. Employee contributions are subject to regulatory limitations and the specific plan provisions. Several plans provide for employer matching contributions up to levels specified in the plans and provide for additional discretionary contributions as determined by management. Our defined contribution plan expense was approximately $1.4 billion in 2020, $1.2 billion in 2019 and $1.0 billion in 2018.

(25) Contingencies and Commitments

We are parties in a variety of legal actions that routinely arise out of the normal course of business, including legal actions seeking to establish liability directly through insurance contracts or indirectly through reinsurance contracts issued by Berkshire subsidiaries. Plaintiffs occasionally seek punitive or exemplary damages. We do not believe that such normal and routine litigation will have a material effect on our financial condition or results of operations. Berkshire and certain of its subsidiaries are also involved in other kinds of legal actions, some of which assert or may assert claims or seek to impose fines and penalties. We believe that any liability that may arise as a result of other pending legal actions will not have a material effect on our consolidated financial condition or results of operations.

Our subsidiaries regularly make commitments in the ordinary course of business to purchase goods and services used in their businesses. As of December 31, 2020, estimated future payments under such arrangements were as follows: $14.6 billion in 2021, $4.5 billion in 2022, $3.4 billion in 2023, $2.8 billion in 2024, $3.1 billion in 2025 and $20.0 billion after 2025. The most significant of these relate to our railroad, utilities and energy businesses and our shared aircraft ownership and leasing business.

Pursuant to the terms of agreements with noncontrolling shareholders in our less than wholly-owned subsidiaries, we may be obligated to acquire their equity interests. If we had acquired all outstanding noncontrolling interests as of December 31, 2020, we estimate the cost would have been approximately $6.3 billion. However, the timing and the amount of any such future payments that might be required are contingent on future actions of the noncontrolling owners.

(26) Supplemental cash flow information

A summary of supplemental cash flow information for each of the three years ending December 31, 2020 is presented in the following table (in millions).

	2020	2019	2018
Cash paid during the period for:			
Income taxes	$ 5,001	$ 5,415	$ 4,354
Interest:			
Insurance and other	1,001	1,011	1,111
Railroad, utilities and energy	3,006	2,879	2,867
Non-cash investing and financing activities:			
Liabilities assumed in connection with business acquisitions	6,981	766	3,735
Operating lease liabilities arising from obtaining right-of-use assets	729	782	—

(27) Business segment data

Our operating businesses include a large and diverse group of insurance, manufacturing, service and retailing businesses. We organize our reportable business segments in a manner that reflects how management views those business activities. Certain businesses are grouped together for segment reporting based upon similar products or product lines, marketing, selling and distribution characteristics, even though those business units are operated under separate local management.

The tabular information that follows shows data of reportable segments reconciled to amounts reflected in our Consolidated Financial Statements. Intersegment transactions are not eliminated from segment results when management considers those transactions in assessing the results of the respective segments. Furthermore, our management does not consider investment and derivative gains/losses, impairments or amortization of certain business acquisition accounting adjustments related to Berkshire's business acquisitions or certain other corporate income and expense items in assessing the financial performance of operating units. Collectively, these items are included in reconciliations of segment amounts to consolidated amounts.

Berkshire's operating segments are as follows.

Business Identity	Business Activity
Insurance:	
GEICO	Underwriting private passenger automobile insurance mainly by direct response methods
Berkshire Hathaway Primary Group	Underwriting multiple lines of property and casualty insurance policies for primarily commercial accounts
Berkshire Hathaway Reinsurance Group	Underwriting excess-of-loss, quota-share and facultative reinsurance worldwide
BNSF	Operation of one of the largest railroad systems in North America
Berkshire Hathaway Energy	Regulated electric and gas utility, including power generation and distribution activities and real estate brokerage activities
Manufacturing	Manufacturers of numerous products including industrial, consumer and building products, including home building and related financial services
McLane Company	Wholesale distribution of groceries and non-food items
Service and retailing	Providers of numerous services including shared aircraft ownership programs, aviation pilot training, electronic components distribution, various retailing businesses, including automobile dealerships and trailer and furniture leasing

(27) Business segment data *(Continued)*

A disaggregation of our consolidated data for each of the three most recent years is presented as follows (in millions).

	Revenues			Earnings before income taxes		
	2020	2019	2018	2020	2019	2018
Operating Businesses						
Insurance:						
Underwriting:						
GEICO	$ 35,093	$ 35,572	$ 33,363	$ 3,428	$ 1,506	$ 2,449
Berkshire Hathaway Primary Group	9,615	9,165	8,111	110	383	670
Berkshire Hathaway Reinsurance Group	18,693	16,341	15,944	(2,700)	(1,472)	(1,109)
Insurance underwriting	63,401	61,078	57,418	838	417	2,010
Investment income	5,960	6,615	5,518	5,949	6,600	5,503
Total insurance	69,361	67,693	62,936	6,787	7,017	7,513
BNSF	20,869	23,515	23,855	6,792	7,250	6,863
Berkshire Hathaway Energy	21,031	20,114	19,987	2,479	2,618	2,472
Manufacturing	59,079	62,730	61,883	8,010	9,522	9,366
McLane Company	46,840	50,458	49,987	251	288	246
Service and retailing	28,178	29,487	28,939	2,628	2,555	2,696
	245,358	253,997	247,587	26,947	29,250	29,156
Reconciliation to consolidated amount						
Investment and derivative gains/losses	—	—	—	40,746	72,607	(22,455)
Interest expense, not allocated to segments	—	—	—	(483)	(416)	(458)
Equity method investments	—	—	—	726	1,176	(2,167)
Goodwill and intangible asset impairments	—	—	—	(10,671)	(96)	(382)
Corporate, eliminations and other	152	619	250	(1,572)	175	307
	$ 245,510	$ 254,616	$ 247,837	$ 55,693	$ 102,696	$ 4,001

	Interest expense			Income tax expense		
	2020	2019	2018	2020	2019	2018
Operating Businesses						
Insurance	$ —	$ —	$ —	$ 1,089	$ 1,166	$ 1,374
BNSF	1,037	1,070	1,041	1,631	1,769	1,644
Berkshire Hathaway Energy	1,941	1,835	1,777	(1,010)	(526)	(452)
Manufacturing	737	752	690	1,795	2,253	2,188
McLane Company	—	—	15	71	71	59
Service and retailing	61	86	91	669	603	634
	3,776	3,743	3,614	4,245	5,336	5,447
Reconciliation to consolidated amount						
Investment and derivative gains/losses	—	—	—	8,855	15,159	(4,673)
Interest expense, not allocated to segments	483	416	458	(102)	(88)	(96)
Equity method investments	—	—	—	57	148	(753)
Corporate, eliminations and other	(176)	(198)	(219)	(615)	349	(246)
	$ 4,083	$ 3,961	$ 3,853	$ 12,440	$ 20,904	$ (321)

(27) Business segment data *(Continued)*

	Capital expenditures			Depreciation of tangible assets		
	2020	2019	2018	2020	2019	2018
Operating Businesses						
Insurance	$ 50	$ 108	$ 130	$ 74	$ 82	$ 79
BNSF	3,063	3,608	3,187	2,423	2,350	2,268
Berkshire Hathaway Energy	6,765	7,364	6,241	3,376	2,947	2,830
Manufacturing	2,133	2,981	3,116	2,026	1,951	1,890
McLane Company	98	158	276	204	225	204
Service and retailing	903	1,760	1,587	1,216	1,192	1,115
	$ 13,012	$ 15,979	$ 14,537	$ 9,319	$ 8,747	$ 8,386

	Goodwill at year-end			Identifiable assets at year-end		
	2020	2019	2018	2020	2019	2018
Operating Businesses						
Insurance	$ 15,224	$ 15,289	$ 15,289	$ 399,169	$ 364,550	$ 289,746
BNSF	14,851	14,851	14,851	73,809	73,699	70,242
Berkshire Hathaway Energy	11,763	9,979	9,851	109,286	88,651	80,543
Manufacturing	25,512	34,800	34,019	104,318	104,437	99,912
McLane Company	232	734	734	6,771	6,872	6,243
Service and retailing	6,152	6,229	6,281	26,173	26,494	24,724
	$ 73,734	$ 81,882	$ 81,025	719,526	664,703	571,410
Reconciliation to consolidated amount						
Corporate and other				80,469	71,144	55,359
Goodwill				73,734	81,882	81,025
				$ 873,729	$ 817,729	$ 707,794

Property/casualty and life/health insurance premiums written and earned are summarized below (in millions).

	Property/Casualty			Life/Health		
	2020	2019	2018	2020	2019	2018
Premiums Written:						
Direct	$ 47,838	$ 47,578	$ 44,513	$ 510	$ 839	$ 1,111
Assumed	11,533	10,214	8,970	5,960	5,046	5,540
Ceded	(898)	(821)	(869)	(42)	(45)	(49)
	$ 58,473	$ 56,971	$ 52,614	$ 6,428	$ 5,840	$ 6,602
Premiums Earned:						
Direct	$ 46,418	$ 46,540	$ 43,095	$ 510	$ 839	$ 1,111
Assumed	11,449	9,643	8,649	5,973	4,952	5,438
Ceded	(907)	(851)	(825)	(42)	(45)	(50)
	$ 56,960	$ 55,332	$ 50,919	$ 6,441	$ 5,746	$ 6,499

(27) Business segment data *(Continued)*

Insurance premiums written by geographic region (based upon the domicile of the insured or reinsured) are summarized below (in millions).

	Property/Casualty			Life/Health		
	2020	**2019**	**2018**	**2020**	**2019**	**2018**
United States	$ 50,250	$ 50,529	$ 46,146	$ 2,820	$ 2,553	$ 3,598
Western Europe	3,751	2,535	2,157	1,120	908	939
Asia Pacific	3,410	3,114	3,726	1,652	1,582	1,361
All other	1,062	793	585	836	797	704
	$ 58,473	$ 56,971	$ 52,614	$ 6,428	$ 5,840	$ 6,602

Consolidated sales, service and leasing revenues were $132.3 billion in 2020, $140.8 billion in 2019 and $139.1 billion in 2018. Sales, service and leasing revenues attributable to the United States were 86% in 2020, 85% in 2019 and 84% in 2018 of such amounts. The remainder of sales, service and leasing revenues were primarily in Europe, Canada and the Asia Pacific. Railroad, utilities and energy revenues were $41.8 billion in 2020, $43.5 billion in 2019 and $43.7 billion in 2018. In each of the three years, approximately 96% of such revenues were attributable to the United States. At December 31, 2020, approximately 89% of our consolidated net property, plant and equipment and equipment held for lease was located in the United States with the remainder primarily in Canada and the United Kingdom.

(28) Quarterly data

A summary of revenues and net earnings by quarter for each of the last two years follows. This information is unaudited. Amounts are in millions, except per share amounts.

	1st Quarter	**2nd Quarter**	**3rd Quarter**	**4th Quarter**
2020				
Revenues	$ 61,265	$ 56,840	$ 63,024	$ 64,381
Net earnings (loss) attributable to Berkshire shareholders *	(49,746)	26,295	30,137	35,835
Net earnings (loss) attributable to Berkshire shareholders per equivalent Class A common share	(30,653)	16,314	18,994	23,015
2019				
Revenues	$ 60,678	$ 63,598	$ 64,972	$ 65,368
Net earnings attributable to Berkshire shareholders *	21,661	14,073	16,524	29,159
Net earnings attributable to Berkshire shareholders per equivalent Class A common share	13,209	8,608	10,119	17,909

* *Includes after-tax investment and derivative gains/losses as follows:*

	1st Quarter	***2nd Quarter***	***3rd Quarter***	***4th Quarter***
2020	*$ (55,617)*	*$ 31,645*	*$ 24,737*	*$ 30,826*
2019	*16,106*	*7,934*	*8,666*	*24,739*

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

None

Item 9A. Controls and Procedures

At the end of the period covered by this Annual Report on Form 10-K, the Corporation carried out an evaluation, under the supervision and with the participation of the Corporation's management, including the Chairman (Chief Executive Officer) and the Senior Vice President (Chief Financial Officer), of the effectiveness of the design and operation of the Corporation's disclosure controls and procedures pursuant to Exchange Act Rule 13a-15. Based upon that evaluation, the Chairman (Chief Executive Officer) and the Senior Vice President (Chief Financial Officer) concluded that the Corporation's disclosure controls and procedures are effective in timely alerting them to material information relating to the Corporation (including its consolidated subsidiaries) required to be included in the Corporation's periodic SEC filings. The report called for by Item 308(a) of Regulation S-K is incorporated herein by reference to Management's Report on Internal Control Over Financial Reporting, included on page K-66 of this report. The attestation report called for by Item 308(b) of Regulation S-K is incorporated herein by reference to Report of Independent Registered Public Accounting Firm, included on page K-67 of this report. There has been no change in the Corporation's internal control over financial reporting during the quarter ended December 31, 2020 that has materially affected, or is reasonably likely to materially affect, the Corporation's internal control over financial reporting.

Item 9B. Other Information

None

Part III

Except for the information set forth under the caption "Executive Officers of the Registrant" in Part I hereof, information required by this Part (Items 10, 11, 12, 13 and 14) is incorporated by reference from the Registrant's definitive proxy statement, filed pursuant to Regulation 14A, for the Annual Meeting of Shareholders of the Registrant to be held on May 1, 2021, which meeting will involve the election of directors.

Part IV

Item 15. Exhibits and Financial Statement Schedules

(a)1. *Financial Statements*

The following Consolidated Financial Statements, as well as the Report of Independent Registered Public Accounting Firm, are included in Part II Item 8 of this report:

	PAGE
Report of Independent Registered Public Accounting Firm	K-67
Consolidated Balance Sheets—	
December 31, 2020 and December 31, 2019	K-70
Consolidated Statements of Earnings—	
Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-72
Consolidated Statements of Comprehensive Income—	
Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-73
Consolidated Statements of Changes in Shareholders' Equity—	
Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-73
Consolidated Statements of Cash Flows—	
Years Ended December 31, 2020, December 31, 2019, and December 31, 2018	K-74
Notes to Consolidated Financial Statements	K-75
2. *Financial Statement Schedule*	
Report of Independent Registered Public Accounting Firm	K-117
Schedule I—Parent Company Condensed Financial Information	
Balance Sheets as of December 31, 2020 and 2019, Statements of Earnings and Comprehensive Income and Cash Flows for the years ended December 31, 2020, December 31, 2019 and December 31, 2018 and Note to Condensed Financial Information	K-118

Other schedules are omitted because they are not required, information therein is not applicable, or is reflected in the Consolidated Financial Statements or notes thereto.

(b) *Exhibits*

See the "Exhibit Index" at page K-120.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Shareholders and the Board of Directors of
Berkshire Hathaway Inc.
Omaha, Nebraska

Opinion on the Financial Statement Schedule

We have audited the consolidated financial statements of Berkshire Hathaway Inc. and subsidiaries (the "Company") as of December 31, 2020 and 2019, and for each of the three years in the period ended December 31, 2020, and the Company's internal control over financial reporting as of December 31, 2020, and have issued our report thereon dated February 27, 2021; such consolidated financial statements and report are included elsewhere in this Form 10-K. Our audits also included the financial statement schedule of the Company listed in the Index at Item 15. This financial statement schedule is the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statement schedule based on our audits. In our opinion, such financial statement schedule, when considered in relation to the financial statements taken as a whole, presents fairly, in all material respects, the information set forth therein.

/s/ Deloitte & Touche LLP
Omaha, Nebraska
February 27, 2021

BERKSHIRE HATHAWAY INC. (Parent Company)
Condensed Financial Information
(Dollars in millions)
Schedule I
Balance Sheets

	December 31,	
	2020	**2019**
Assets:		
Cash and cash equivalents	$ 12,329	$ 15,004
Short-term investments in U.S. Treasury Bills	29,773	25,514
Investments in and advances to/from consolidated subsidiaries	411,826	392,162
Investment in The Kraft Heinz Company	13,336	13,757
Other assets	108	131
	$ 467,372	$ 446,568
Liabilities and Shareholders' Equity:		
Accounts payable, accrued interest and other liabilities	$ 369	$ 320
Income taxes, principally deferred	1,174	1,554
Notes payable and other borrowings	22,665	19,903
	24,208	21,777
Berkshire Hathaway shareholders' equity	443,164	424,791
	$ 467,372	$ 446,568

Statements of Earnings and Comprehensive Income

	Year ended December 31,		
	2020	**2019**	**2018**
Income items:			
From consolidated subsidiaries:			
Dividends and distributions	$ 26,110	$ 15,603	$ 9,658
Undistributed earnings (losses)	17,402	65,237	(3,952)
	43,512	80,840	5,706
Investment gains (losses)	(24)	(125)	(4)
Equity in net earnings (losses) of The Kraft Heinz Company	95	493	(2,730)
Other income	328	780	649
	43,911	81,988	3,621
Cost and expense items:			
General and administrative	194	122	216
Interest expense	489	591	601
Foreign exchange (gains) losses on non-U.S. Dollar denominated debt	970	(193)	(366)
Income tax expense (benefit)	(263)	51	(851)
	1,390	571	(400)
Net earnings attributable to Berkshire Hathaway shareholders	42,521	81,417	4,021
Other comprehensive income attributable to Berkshire Hathaway shareholders	1,000	(228)	(2,211)
Comprehensive income attributable to Berkshire Hathaway shareholders	$ 43,521	$ 81,189	$ 1,810

See Note to Condensed Financial Information

BERKSHIRE HATHAWAY INC. (Parent Company)
Condensed Financial Information
(Dollars in millions)
Schedule I (continued)
Statements of Cash Flows

	Year ended December 31,		
	2020	2019	2018
Cash flows from operating activities:			
Net earnings attributable to Berkshire Hathaway shareholders	$ 42,521	$ 81,417	$ 4,021
Adjustments to reconcile net earnings to cash flows from operating activities:			
Investment gains/losses	24	125	4
Undistributed earnings of consolidated subsidiaries	(17,402)	(65,237)	3,952
Non-cash dividends from subsidiaries	(8,296)	—	—
Income taxes payable	(72)	(56)	(972)
Other	1,100	(693)	3,062
Net cash flows from operating activities	17,875	15,556	10,067
Cash flows from investing activities:			
Investments in and advances to/from consolidated subsidiaries, net	(1,947)	60	460
Purchases of U.S. Treasury Bills	(54,715)	(40,107)	(29,740)
Sales and maturities of U.S. Treasury Bills	59,035	36,943	21,442
Other	11	737	—
Net cash flows from investing activities	2,384	(2,367)	(7,838)
Cash flows from financing activities:			
Proceeds from borrowings	2,923	3,967	17
Repayments of borrowings	(1,151)	(758)	(1,563)
Acquisition of treasury stock	(24,706)	(4,850)	(1,346)
Other	—	19	61
Net cash flows from financing activities	(22,934)	(1,622)	(2,831)
Increase (decrease) in cash and cash equivalents	(2,675)	11,567	(602)
Cash and cash equivalents at beginning of year	15,004	3,437	4,039
Cash and cash equivalents at end of year	$ 12,329	$ 15,004	$ 3,437
Other cash flow information:			
Income taxes paid	$ 3,391	$ 3,531	$ 2,790
Interest paid	359	364	388

Note to Condensed Financial Information

Berkshire currently owns 26.6% of the outstanding shares of The Kraft Heinz Company ("Kraft Heinz") common stock, which is accounted for pursuant to the equity method. See Note 5 to the accompanying Consolidated Financial Statements for additional information regarding this investment.

In 2020, the Parent Company repaid €1.0 billion of maturing senior notes and issued €1.0 billion of 0.0% senior notes due in 2025 and ¥195.5 billion (approximately $1.8 billion) of senior notes with maturity dates ranging from 2023 to 2060 with a weighted average interest rate of 1.07%. As of December 31, 2020, the Parent Company's non-U.S. Dollar denominated borrowings included €6.85 billion and ¥625.5 billion par value senior notes. The gains and losses from the periodic remeasurement of these non-U.S. Dollar denominated notes due to changes in foreign currency exchange rates are included in earnings. See Note 17 to the accompanying Consolidated Financial Statements for additional information.

Parent Company debt maturities over the next five years are as follows: 2021—$2,172 million; 2022—$600 million; 2023—$4,633 million; 2024—$2,272 million and 2025—$1,801 million. The Parent Company guarantees certain debt of subsidiaries, which in the aggregate, approximated $14.4 billion at December 31, 2020 and included $13.1 billion of debt issued by Berkshire Hathaway Finance Corporation. Such guarantees are an absolute, unconditional and irrevocable guarantee for the full and prompt payment when due of all present and future payment obligations. The Parent Company has also provided guarantees in connection with equity index put option contracts and certain retroactive reinsurance contracts issued by subsidiaries. The amounts of subsidiary payments under these contracts, if any, is contingent upon the outcome of future events.

EXHIBIT INDEX

Exhibit No.	
2(i)	Agreement and Plan of Merger dated as of June 19, 1998 between Berkshire and General Re Corporation. Incorporated by reference to Annex I to Registration Statement No. 333-61129 filed on Form S-4.
2(ii)	Agreement and Plan of Merger dated as of November 2, 2009 by and among Berkshire, R Acquisition Company, LLC and BNSF. Incorporated by reference to Annex A to Registration Statement No. 333-163343 on Form S-4.
2(iii)	Agreement and Plan of Merger dated August 8, 2015, by and among Berkshire, NW Merger Sub Inc. and Precision Castparts Corporation ("PCC") Incorporated by reference to Exhibit 2.1 to PCC's Current Report on Form 8-K filed on August 10, 2015 (SEC File No. 001-10348)
3(i)	Restated Certificate of Incorporation Incorporated by reference to Exhibit 3(i) to Form 10-K filed on March 2, 2015.
3(ii)	By-Laws Incorporated by reference to Exhibit 3(ii) to Form 8-K filed on May 4, 2016.
4.1	Indenture, dated as of December 22, 2003, between Berkshire Hathaway Finance Corporation, Berkshire Hathaway Inc. and The Bank of New York Mellon Trust Company, N.A. (as successor to J.P. Morgan Trust Company, National Association), as trustee. Incorporated by reference to Exhibit 4.1 on Form S-4 of Berkshire Hathaway Finance Corporation and Berkshire Hathaway Inc. filed on February 4, 2004. SEC File No. 333-112486
4.2	Indenture, dated as of February 1, 2010, among Berkshire Hathaway Inc., Berkshire Hathaway Finance Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee. Incorporated by reference to Exhibit 4.1 to Berkshire's Registration Statement on Form S-3 filed on February 1, 2010. SEC File No. 333-164611
4.3	Indenture, dated as of January 26, 2016, by and among Berkshire Hathaway Inc., Berkshire Hathaway Finance Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee. Incorporated by reference to Exhibit 4.1 to Berkshire's Registration Statement on Form S-3 filed on January 26, 2016. SEC File No. 333-209122
4.4	Indenture, dated as of December 1, 1995, between BNSF and The First National Bank of Chicago, as trustee. Incorporated by reference to Exhibit 4 on Form S-3 of BNSF filed on February 8, 1999.
4.5	Indenture, dated as of October 4, 2002, by and between MidAmerican Energy Holdings Company and The Bank of New York, Trustee. Incorporated by reference to Exhibit 4.1 to the Berkshire Hathaway Energy Company Registration Statement No. 333-101699 dated December 6, 2002.
	Other instruments defining the rights of holders of long-term debt of Registrant and its subsidiaries are not being filed since the total amount of securities authorized by all other such instruments does not exceed 10% of the total assets of the Registrant and its subsidiaries on a consolidated basis as of December 31, 2020. The Registrant hereby agrees to furnish to the Commission upon request a copy of any such debt instrument to which it is a party.
10.1	Equity Commitment Letter of Berkshire Hathaway Inc. with Hawk Acquisition Holding Corporation dated February 13, 2013. Incorporated by reference to Exhibit 10.1 on Form 8-K of Berkshire Hathaway Inc. filed on February 14, 2013.
14	Code of Ethics Berkshire's Code of Business Conduct and Ethics is posted on its Internet website at *www.berkshirehathaway.com*
21	Subsidiaries of Registrant
23	Consent of Independent Registered Public Accounting Firm
31.1	Rule 13a—14(a)/15d-14(a) Certification
31.2	Rule 13a—14(a)/15d-14(a) Certification
32.1	Section 1350 Certification
32.2	Section 1350 Certification

Exhibit No.	
95	Mine Safety Disclosures
101	The following financial information from Berkshire Hathaway Inc.'s Annual Report on Form 10-K for the year ended December 31, 2020, formatted in iXBRL (Inline Extensible Business Reporting Language) includes: (i) the Cover Page (ii) the Consolidated Balance Sheets, (iii) the Consolidated Statements of Earnings, (iv) the Consolidated Statements of Comprehensive Income, (v) the Consolidated Statements of Changes in Shareholders' Equity, (vi) the Consolidated Statements of Cash Flows, and (vii) the Notes to Consolidated Financial Statements and Schedule I, tagged in summary and detail.
104	Cover Page Interactive Data File (formatted as iXBRL and contained in Exhibit 101)

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

BERKSHIRE HATHAWAY INC.

Date: February 27, 2021

/S/ MARC D. HAMBURG

Marc D. Hamburg
Senior Vice President and
Principal Financial Officer

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the Registrant and in the capacities and on the dates indicated.

/S/ WARREN E. BUFFETT **Warren E. Buffett**	Chairman of the Board of Directors—Chief Executive Officer	February 27, 2021 Date
/S/ GREGORY E. ABEL **Gregory E. Abel**	Director—Vice Chairman—Non Insurance Operations	February 27, 2021 Date
/S/ HOWARD G. BUFFETT **Howard G. Buffett**	Director	February 27, 2021 Date
/S/ STEPHEN B. BURKE **Stephen B. Burke**	Director	February 27, 2021 Date
/S/ KENNETH I. CHENAULT **Kenneth I. Chenault**	Director	February 27, 2021 Date
/S/ SUSAN L. DECKER **Susan L. Decker**	Director	February 27, 2021 Date
/S/ DAVID S. GOTTESMAN **David S. Gottesman**	Director	February 27, 2021 Date
/S/ CHARLOTTE GUYMAN **Charlotte Guyman**	Director	February 27, 2021 Date
/S/ AJIT JAIN **Ajit Jain**	Director—Vice Chairman—Insurance Operations	February 27, 2021 Date
/S/ CHARLES T. MUNGER **Charles T. Munger**	Director—Vice Chairman	February 27, 2021 Date
/S/ THOMAS S. MURPHY **Thomas S. Murphy**	Director	February 27, 2021 Date
/S/ RONALD L. OLSON **Ronald L. Olson**	Director	February 27, 2021 Date
/S/ WALTER SCOTT, JR. **Walter Scott, Jr.**	Director	February 27, 2021 Date
/S/ MERYL B. WITMER **Meryl B. Witmer**	Director	February 27, 2021 Date
/S/ MARC D. HAMBURG **Marc D. Hamburg**	Senior Vice President—Principal Financial Officer	February 27, 2021 Date

/S/ DANIEL J. JAKSICH	Vice President—Principal Accounting Officer	February 27, 2021
Daniel J. Jaksich		Date